

Health Co-Benefits and Low-Income Residential Weatherization in the United States

Bruce Tonn
Erin Rose
Michaela Marincic

Three³, Inc.
520 W. Summit Hill Drive, Suite 1101
Knoxville, TN 37902
USA

May 14, 2026

Three³

Executive Summary	4
1.0 Introduction.....	6
2.0 Energy Poverty.....	8
2.1 Overview.....	8
2.2. Multiple dimensions of poverty	9
3.0 Weatherization Programs in the United States: Federal and Utility-based	11
3.1 Federal Weatherization Program.....	11
3.2 Utility Weatherization Programs.....	14
4.0 Why Evaluate the HHR Co-Benefits of Weatherization.....	15
5.0 How Weatherization Can Yield Health Impacts: A Theory of Change	16
6.0 Approaches to Measuring Changes in Co-Benefits Attributable to Weatherization	20
7.0 Evaluations of the Co-Benefits of Weatherization.....	21
7.1 U.S. Department of Energy’s (DOE) Weatherization Assistance Program (WAP)	21
7.2 Knoxville Extreme Energy Makeover Project (KEEM)	21
7.3 Low-Income Multifamily Buildings (LIMF) Co-Benefits Evaluation	22
7.4 Home Uplift Program (HUP).....	23
8.0 Health and Other Household-Related (HHR) Co-Benefits of Weatherization	24
8.1 Demographics of Four Evaluations’ Main Respondents	24
8.2 Descriptive Statistics – Treatment Groups.....	25
8.3 Patterns of Co-Benefit Results.....	31
8.4 Cascading Co-Benefits.....	34
8.5 Future Research on the Health Co-Benefits of Weatherization	41
9.0 Monetization of Co-Benefits of Weatherization	46
9.1 Monetization Overview	46
9.2 Approaches to Estimating Monetization Values for NEIs	48
9.3 Monetization Results	49
9.4 Future Research – Monetizing the Benefits of Weatherization	51
10.0 Weatherization and Related Policy Issues	53
10.1 National Weatherization Policy Map	53
10.2 Weatherization Policy Issues: Detailed Discussion	54
10.2.1 Weatherization – Federal Perspective	55
10.2.2 Weatherization – Utility Perspective.....	58
10.2.3 Weatherization – Renters and Multifamily Issues	60
10.2.4 Weatherization Workforce Issues.....	60

10.2 Weatherization – Healthcare	61
10.3 Weatherization – Climate Change.....	63
11. Summary and Concluding Thoughts.....	65
Acknowledgments.....	66
Appendix A. History of Weatherization NEI Research	67
References.....	69

Executive Summary

This report is designed to be a comprehensive resource on low-income weatherization and health and other household related co-benefits. Weatherization programs save energy in low-income homes and affordable multifamily buildings. Energy savings can lead to reductions in the emissions of greenhouse gases. Weatherization programs can also produce important health co-benefits for occupants and a wide range of additional non-energy impacts (NEIs). Weatherization programs can play important roles in climate change mitigation and adaptation plans.

Weatherization is a term of art used in the United States (U.S.) for programs that install energy efficiency measures in income-eligible homes for free. The U.S. federal government and both investor owned and public utilities are the main sources of weatherization funds. Comprehensive weatherization programs install energy efficiency measures such as air sealing, insulation, heating and cooling systems repair and replacement, and door and window repair and replacement. Community action agencies and private sector contractors deliver weatherization services.

Energy efficiency programs are rigorously evaluated in the U.S. In addition to estimating energy savings, evaluations regularly address the NEIs of these programs. Energy program evaluations are important for policy makers, program administrators, and implementers to help them understand the full complement of impacts of their programs. The focus of this report is the health and household-related NEIs---which are also referred to as co-benefits herein---of low-income weatherization programs.

A theory of change is presented in Section 5 that explains why weatherization can produce health and household-related co-benefits. Basically, weatherization programs install measures in homes that change the physical structure of homes in ways that make them more comfortable to live in. Measures such as air sealing and insulation can keep warm air in homes in the winter and cool air in homes in the summer. These measures can also keep pests and outside allergens from entering homes to the benefit of those with chronic respiratory problems. Additionally, these measures can also reduce the intrusion of outdoor noise, which can result in better rest and sleep. More thermally sound homes can reduce symptoms of arthritis and the need for medical attention for individuals who get too hot or cold in their homes. Weatherization reduces household energy costs, which can allow households to better afford food and prescriptions. Better health can lead to fewer missed days of work, which can also improve finances of low-income households.

Hypotheses derivable from the theory of change were explored in four major weatherization program evaluations documented herein. These evaluations focused on the federal Weatherization Assistance Program (WAP), a weatherization program implemented in Knoxville, Tennessee known as Knoxville Extreme Energy Makeover (KEEM), a regional weatherization program offered by the Tennessee Valley Authority (TVA) known as Home Uplift, and weatherization programs in the Northeastern and Midwestern regions of the U.S. that focus on affordable multifamily buildings. The appendices also contain summaries of more limited evaluations that address the health co-benefits of an air conditioner replacement program in Phoenix, Arizona and changes in temperature and humidity measured pre- and post-weatherization in Home Uplift homes.

The evaluations relied upon surveys administered pre- and post-weatherization to collect data on health and household-related co-benefits. The survey results consistently show that occupants believe that their homes are more comfortable, and less drafty, dusty and noisy post-weatherization. Instances of pest infestation are reduced. Households less frequently keep their homes at unhealthy temperatures. Respondents report improved physical and mental health and rest and sleep. Headaches and arthritis symptoms are reduced as are medical interventions for thermal stress. The results are mixed with respect to asthma, COPD and other respiratory conditions such as colds, flu, and bronchitis. Financially, households report that utility bills are less hard to pay. There are fewer tradeoffs between paying utility bills and affording food and prescriptions. Use of short-term, high-interest predatory loans is also reduced. Installed measures that reduce the chances of home fires, carbon monoxide poisoning, and thermal stress can also save lives.

The impacts of weatherization upon these and other health and household-related co-benefits are not always statistically significant across the four major evaluations. However, additional analyses indicate that patterns of changes in NEIs from pre- to post- across NEIs studies move in directions consistent with the theory of change. Furthermore, a systems analysis of relationships between home conditions, chronic health, acute health, household financial, and life satisfaction variables shows that all of the categories of variables are highly statistically interrelated. The modeling also suggests that improvements in general and chronic health drive improvements in the other four variables. It is also clear that weatherization can have a cascading impact on the other four variables. The evaluations also monetized a select number of NEIs. Across the board, the combined discounted present values for even a select set of NEIs exceed weatherization job costs over a ten-year period. The value of a saved life (VSL) had the most influence upon the aggregated monetizations.

Future research on health and household-related co-benefits of weatherization can follow many paths. Data can be collected for many more health issues, such as diabetes, kidney disease, PTSD and ADHD. Additional evidence is needed to ascertain the persistence of these co-benefits or even if the benefits may increase over time through positive feedback effects through the household system. It is possible that more direct measurements of health pre-and post-weatherization are needed (e.g., collecting spirometer data from those with respiratory conditions). Big data approaches could be used to assess whether over the course of decades the weatherized housing stock itself can be found to reduce medical costs. With respect to monetization, future research can provide better insights into how households' budgets and spending decisions change with the infusion of energy and other cost savings. It would also be helpful to have forecasts for categories of major cost reductions, such as medical costs.

In addition to future research, there are many important policy issues that relate to weatherization. The national weatherization policy map is complicated and multifaceted. Policy issues run the gamut from being very detailed to weatherization program administration (e.g., what measures can be installed in homes, can monetized co-benefits be included in economic effectiveness calculations) to broad issues around the collaboration of weatherization and the healthcare sector. There numerous important policy issues surrounding the blending of federal and utility funded weatherization, healthy homes, and housing programs.

1.0 Introduction

A dollar spent on *weatherizing* low-income homes and affordable multifamily buildings is a dollar very well spent. Weatherization is a term-of-art used in the United States (U.S.) to describe programs that install for free comprehensive sets of energy efficiency measures in income eligible homes. Frequently installed measures include air sealing, insulation, and heating system repair and replacement. Comprehensive energy retrofit programs targeting homes of low-income households are implemented across the Global North and can be found in the Global South where countries aim to reduce fuel poverty and mitigate climate change through improved energy efficiency and improve housing conditions (Howden-Chapman et al. 2007; IEA 2014; Warmer Homes 2020). In the U.S., the main weatherization programs are administered by the federal U.S. Department of Energy and investor-owned and public electricity and natural gas utilities.

Weatherization saves energy and reduces household energy bills (Tonn, Rose and Hawkins 2018). Weatherization also yields an impressive range of co-benefits (Schweitzer and Tonn, 2003).¹ These include reductions in the emissions of greenhouse gases (GHGs) and other air pollutants (Carroll and Bausch, 2015), the production of jobs (Burton and Bray 2014; Brown et al. 1993), and improvements in home conditions (Skumatz and Gardner, 2005) and national security (Riggert et al. 1999).

This piece focuses on the broad range of health and other household-related (HHR) co-benefits of low-income weatherization. Here, the health benefits accrue to household members from improvements in indoor environmental conditions and improvements in household budgets that allow increased expenditures related to health (e.g., being more able to afford prescriptions). We acknowledge that weatherization can improve public health through the reduction in emissions of air pollutants and GHGs, but this report is not addressing public health benefits. As documented below, research strongly indicates that weatherization can improve general health of household members, reduce household financial burdens, and improve life satisfaction. The research results are weaker or mixed with respect to reducing thermal stress and ameliorating respiratory conditions.

Weatherization programs are one of many types of social programs whose primary aim is to reduce poverty. Weatherization programs save energy which in turn can reduce household energy costs. Energy poverty and energy burden are all too common in the U.S. and elsewhere in the world, as discussed in Section 2. Section 3 provides a detailed description of the weatherization process. Section 4 addresses more directly why evaluation of the HHR co-benefits can be valuable to weatherization program administrators, and to the healthcare and public health sectors.

Section 5 presents a theory of change to explain why weatherization could yield HHR co-benefits. The theory of change provides a framework to understand why weatherization could reduce asthma and arthritis symptoms, for example, and therefore help researchers develop plans to collect data that could support or refute hypotheses about the HHR co-benefits of

¹ Co-benefits are also referred to as non-energy benefits or non-energy impacts of weatherization.

weatherization. Section 6 sets out various approaches that can be used to collect data to estimate HHR co-benefits.

Several rigorous HHR co-benefits evaluations of weatherization programs have been conducted. The four most comprehensive evaluations completed thus far and what data were collected are described in Section 7.

Results from these four evaluations are presented in Section 8. Demographic results are presented first to paint a picture of typical weatherization recipients. Descriptive statistics related to changes in home conditions, occupant health, household finances and life satisfaction are presented in Section 8.2. The following subsection presents results that show that almost all measures of co-benefits improve post-weatherization even if all of the results are not statistically significant. Section 8.4 documents a five-component household model that shows how the impacts of weatherization can have a cascading impact on health, finances and life satisfaction.

Section 9 presents our approach to monetizing HHR co-benefits and the results of four in-depth monetization exercises, two that make use of pre-post survey results and two that assess changes in direct medical costs. Section 10 addresses policy issues surrounding weatherization specifically and the HHR co-benefits more generally. It is hoped that insights into weatherization programs and policy-related issues in the U.S. can be generalizable to other contexts across the globe. Appendix A presents a short history of weatherization co-benefits research in the U.S.

2.0 Energy Poverty

This section presents a discussion of energy poverty. The first subsection presents an overview of the literature around the concepts of energy poverty and energy burden. The second subsection expands upon these concepts to show how energy poverty is just one of the multiple dimensions of poverty.

2.1 Overview

Access to affordable and reliable energy can be seen as a basic human right (Demski et al. 2019). Violating this human right is fundamentally a socio-political injustice (Gillard et al. 2017). Similar to general poverty, energy poverty is normally associated with household income. Different from general poverty, energy poverty is measured by energy burden, which is a function of energy consumption divided by household income, expressed as a percentage. In the U.S., low-income households typically have an energy burden three times greater than non-low-income households (Drehobl and Ross, 2016). An energy burden exceeding 10% is generally considered an indication of energy poverty (Daniel, 2019). A recent study added two additional categories, “energy stressed” (4–7%) and “energy burdened” (7–10%), to capture the range of financial hardship relating to energy (Cook and Shah, 2018; Brown et al. 2020). Others put forward a 6% cut off for energy poverty based on the notion that housing costs should not exceed 30% of income, with no more than 20% of housing costs allocated for energy (Colton, 2011); according to this logic, energy poverty would not be influenced by cost-of-living factors other than housing prices.

Notably, the U.K. government previously defined energy poverty as having an energy burden exceeding 10% (Boardman, 2010) but revised its definition following a 2012 report investigating the causes and characteristics of domestic fuel poverty (Hills, 2012). The report concluded that energy burden had both merits and flaws, including in the latter its sensitivity to energy prices and issues with accurately calculating income; the author asserts, “The trends it reports do not well reflect changes in the underlying problems, and its definition can encompass households that clearly are not poor.” The new definition does not mention energy burden and instead specifies two metrics: 1) a household’s “required fuel costs” must be above the national median, and 2) if the household spent this amount on fuel, their remaining income would put them below the poverty line.² Others in the U.K. and Europe more broadly have continued to refine methods of measuring energy poverty and its effects without relying solely on energy burden (Dubois and Meier 2016; Thomson et al. 2017a; Grey et al. 2017; Thomson et al. 2017b; Bridgeman et al. 2012).

As indicated in part by this shift in policy, the simple definition of energy poverty as energy burden obscures many other important aspects of energy poverty (Mould and Baker, 2017). A singular energy burden estimate may obscure whether households are compressing their energy needs to meet what they can afford, are able to satisfy their energy needs, or maybe even wasting energy, though few low-income households fall into the latter category (Papada and

² Department for Business, Energy & Industrial Strategy. Fuel Poverty Statistics. GOV.UK. Last updated June 25, 2020. <https://www.gov.uk/government/collections/fuel-poverty-statistics>.

Kaliampakos, 2020). This singular metric also hides how serious the consequences of energy poverty could be if households are not able to heat their homes above unsafe temperatures (Meyer et al. 2018). For example, a study in Spain found that 11% of homes could not heat their homes to adequate temperatures (Llorca et al. 2020). Also hidden are the fear and stigma members of energy vulnerable households may feel (Longhurst and Hargreaves, 2019) and the relationships between energy use and a minimum standard of living (Walker et al. 2016).

Energy poverty is a serious issue in the United States (Power, 2019; Hernandez, 2013) and internationally (Liddel and Morris, 2010; Belaid, 2018; Howden-Chapman, 2012; Ezratty et al. 2009; Boardman, 2010). The U.S. Department of Energy estimates that over 30 million households in the U.S. are substantially burdened by energy costs (EIA, 2018). The number of EU households estimated to suffer energy poverty is 50 million (Filcak and Zivcic, 2017).

Boardman (1991) and others (e.g., Campbell, 1993) argue that energy poverty should be viewed as distinct from general poverty because energy poverty can be solved through capital investments in homes and in the energy infrastructure. Indeed, weatherization programs do invest in the installation of energy efficiency and conservation measures that increase the performance of homes, with the average lifetime of such measures approaching 20 or more years. Additionally, these programs can be straightforwardly given to utility companies and governmental energy efficiency programs to implement. On the other hand, this view has siloed energy poverty programs from other efforts to reduce poverty. It can also be argued that this view has artificially constrained societal investments in improving low-income housing that could reduce energy burdens while also improving health and reducing material deprivation. Making this case to the health care and public health sectors in the U.S. is a major motivation driving recent HHR research (Capps et al. 2019; Hayes et al. 2020).

2.2. Multiple dimensions of poverty

Similar to energy poverty, general poverty is conventionally defined in terms of income. Income thresholds are set such that households falling below the thresholds are deemed to lack enough income to support an acceptable quality of life. Such thresholds are used by most poverty programs in the U.S. to determine eligibility for social programs, including weatherization programs. While thresholds are administratively manageable to implement, their development and use is not without complexity and controversy. For example, should the thresholds be set to express absolute or relative poverty, and should thresholds reflect other underlying features of poverty (Jensen, 2009)? Additionally, strict income-threshold definitions of poverty also hide the non-income components of poverty (Forster et al. 2004).

Amartya Sen (1983) defines poverty as the deprivation of a person's capabilities to live the life they have reason to value. This powerful definition expresses the various aspects of poverty experienced by the poor in the Global South, encompassing lack of clean water and sanitation, threats from a wide-ranging set of deadly diseases, and lack of basic freedoms. Poverty in the Global North may not share all of the same features, but it is multidimensional nevertheless (Townsend et al. 1988).

A multidimensional understanding of poverty needs to move beyond monetary indicators to encompass non-monetary indicators (OECD, 2008); Shen et al. 2023; Chen, 2022). For example, Glassman presents a seven-scale multidimensional poverty index that has these components: health, education, standard of living, consumption, economic security, housing quality, and neighborhood quality (Glassman, 2017). Schenck-Fontaine and Panico (2019) propose these three dimensions of economic hardship: income poverty, material deprivation and subjective financial stress.

Drawing from our survey questions, our case studies, and our listening sessions with weatherization recipients across the U.S., the concepts of hardship and material deprivation come closest to capturing what we see in the data and have experienced in the real world. Townsend (1979) used the term deprivation to describe the broader notion of ‘inability of living a decent life’. Material deprivation or hardships have been variously defined as having these components:

- bill-paying hardship, food hardship, health care hardship (Campbell and Pearlman, 2019);
- food insecurity, inadequate housing, unmet medical needs, utility cutoffs, and financial insecurity (Neckerman et al. 2016);
- not having access to a car, not being able to pay utility bills, and not being able to pay rent in the past year (Jabre et al. 2020); and
- income deprivation; employment deprivation; health deprivation and disability; education, skills and training deprivation; barriers to housing and services; crime; and living environment deprivation domain (Marchand et al. 2019).

Each of these frameworks has something to offer with respect to understanding the potential impacts of weatherization on material deprivation. The multidimensional approach to poverty that comes closest to our theory of change with respect to weatherization presented in Section 3.1 was developed in New York City to study wellbeing. It is called the “Poverty Tracker” (Neckerman et al. 2016). This model has these three major components:

- Income Poverty;
- Material Hardship (including indicators of food insecurity, housing hardship, unmet medical needs, utility cutoffs, and financial insecurity); and
- Adult Health Problems (which can drain family time and resources).

3.0 Weatherization Programs in the United States: Federal and Utility-based

3.1 Federal Weatherization Program

DOE's Weatherization Assistance Program (WAP) was created by Congress in 1976 under Title IV of the Energy Conservation and Production Act. The purpose and scope of the Program as currently stated in the Code of Federal Regulations (CFR) 10CRF 440.1 is "to increase the energy efficiency of dwellings owned or occupied by low-income persons, reduce their total residential energy expenditures, and improve their health and safety, especially low-income persons who are particularly vulnerable such as the elderly, persons with disabilities, families with children, high residential energy users, and households with high energy burden." (Code of Federal Regulations, 2011). The U.S. Congress authorizes funding for WAP on an annual basis.

To implement WAP, DOE provides grants to Grantees (i.e., states, territories, District of Columbia, a small number of Tribes), and the states provide grants to their Subgrantees (e.g., local weatherization agencies) to do the actual weatherization work. There are about 56 Grantees and 900 Subgrantees. A national weatherization network has evolved over the years. The National Association of State Community Services Programs (NASCSPP) is the professional organization that provides support for the Grantees. NASCSPP maintains the Weatherization Assistance Program Technical Assistance Center, which provides up-to-date information about WAP, such as formal program notices issued by DOE, as well as training and technical assistance information.³ The National Community Action Foundation (NCAF) is the professional home for the traditional weatherization Subgrantees that operate as Community Action Agencies.⁴

WAP supports comprehensive weatherization. WAP funds can be used to weatherize a broad range of building types: single family detached and attached homes, manufactured (i.e., mobile) homes, and small and large affordable multifamily buildings. Weatherization is conducted across the entire U.S. and in all climate zones (e.g., very cold to hot and dry). WAP supports the installation of a wide range of energy conservation measures into homes and buildings. Among the most important are air sealing, insulation, and heating and air conditioning repair and replacement.

Figure 1 depicts the weatherization process followed by a local weatherization agency.⁵ The process begins when a household applies to the local WAP program. Many households are referred to WAP by the energy assistance programs run by the community action agencies that also administer the local WAP programs. Other households hear about their local WAP program by word-of-mouth through social networks (Rose et al. 2015a). Indeed, our visits to local weatherization agencies and weatherization recipients across the U.S. indicates that the recipients are very appreciative of the services and that the agencies are held in high regard. As a result, most local weatherization agencies do not have to market their programs. In fact, the many

³ <https://nascsp.org>

⁴ <http://www.ncaf.org>

⁵ Tonn et al. 2014c provides about a dozen case studies of high performing and innovative weatherization programs across the U.S.

agencies have waiting lists for weatherization services that extended many months and, in some cases, years into the future.

The local weatherization agencies are responsible for determining whether households are eligible for weatherization. Currently, households whose income is at 200% or less than the U.S. federal poverty rate are income eligible to receive WAP services. Homes that were recently weatherized are not eligible for re-weatherization. Households that expect to move within a year or two are not eligible to have their homes weatherized. Renters and households living in multifamily properties may have difficulty in getting their homes or apartments weatherized. Property owners need to provide their approval to have their properties weatherized. Conversely, property owners may need to agree not to raise rents for a few years as a type of match to DOE's weatherization funds. Owners of large affordable multifamily buildings may also be asked to co-fund the purchase and installation of weatherization measures. Because of these barriers, and the fact that different skills are needed to weatherize multifamily buildings versus single family homes and mobile homes, many local weatherization agencies simply do not weatherize the former. These issues are frequent barriers that renters face in getting their homes weatherized.

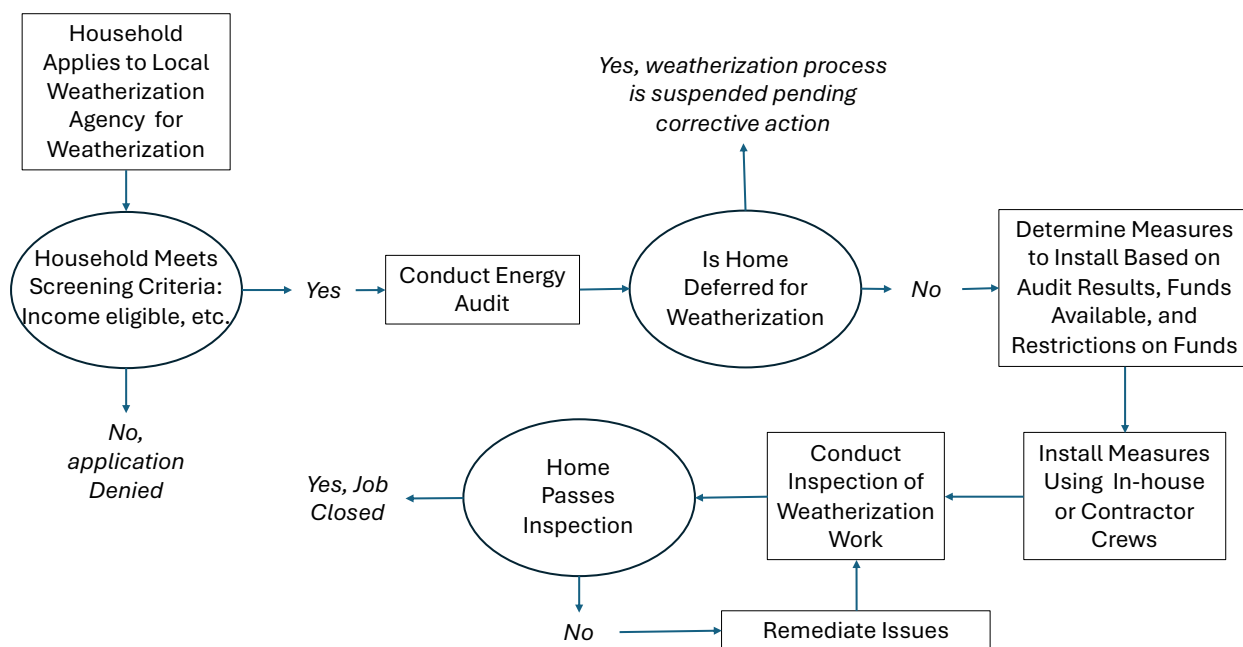


Figure 1. Local Weatherization Agency Weatherization Services Process

Homes that pass the eligibility screening criteria and are not caught up on the barriers just mentioned are placed in a queue to have home energy audits. Prior to weatherization, all homes receive an energy audit to determine what measures should be installed in each specific home. Energy auditors thoroughly inspect and test homes to determine what measures could save energy in the homes. Auditors employ a range of diagnostic tests to support their assessments. The most common diagnostic measure is the blower door test. Essentially, a large fan is attached to an outside door and the home is depressurized. The test indicates how airtight a home is and

can also indicate the sources of major air leaks throughout the home. Other common diagnostics include carbon monoxide measurements in flues of space and water heating systems.

The auditors input these data into DOE approved computerized audit tools that will determine which measures are cost effective to install in the homes. Auditors also use priority lists, which are lists of measures that are generally known to be cost-effective without having to conduct computerized audits for every home. To be deemed cost-effective, each installed measure needs to pass a Savings-to-Investment Ratio (SIR) test, where the present value of the energy cost savings over the life of the measure (e.g., 20 years) needs to exceed the present value of its cost (i.e., $SIR \geq 1.0$). Summing up the costs of cost-effective measures provides an estimate for the total job costs. Lastly, the local programs are allowed to invest a small amount of money to deal with health and safety issues found in homes (typically around 15% of funds invested in a weatherized unit).

It is common for agencies to encounter homes that are in such poor physical condition structurally that weatherization would have virtually no impact on energy consumption or simply cannot be installed. It is also common to encounter homes whose conditions pose health and safety risks to agency staff. In the cases where the agency does not have the resources to help the households with rectifying these types of problems, weatherization is deferred. The homes can re-enter the weatherization queue once the household has addressed the reason(s) for the deferral. It should be noted, though, that very few homes have the resources to address deferral issues and, therefore, never re-enter the weatherization queue (Rose et al. 2015b).

For homes not being deferred, the next step in the process is to create a job order that details exactly what measures will be installed in the home. The job order also prices out the job cost. Not all measures that pass the SIR test could be included in the job order if the total job costs greatly exceed the average amount of money that an agency is allowed to spend on each home. However, agencies are often able to blend weatherization funds provided from other funders to increase the overall investments in homes. For example, states can allocate funds from the Department of Health and Human Services (HHS) Low-Income Home Energy Assistance Program (LIHEAP) to their weatherization Grantee and then to the states Subgrantees to increase weatherization resources. LIHEAP funds are not subject to the SIR constraint and are often used for heating system replacements that do not meet SIR. Utility weatherization programs may provide funds directly to local weatherization agencies.

After the job order is complete, the agency implements its weatherization model. There are two dominant models. In the first, agencies use in-house crews to do the weatherization work. In the other, agencies hire private-sector contractors.⁶ Most agencies that use in-house crews contract out for heating and cooling system repair and replacement work. Most agencies that use contractors use in-house staff to conduct the audits. After weatherization work is completed, the agencies are required to inspect all weatherized homes. To address conflict-of-interest issues, whenever possible inspectors typically refrain from inspecting homes they had initially audited. It should be noted that state weatherization staff and DOE Project Officers also inspect a sample of weatherized homes. Homes that do not pass inspection are passed back to the local

⁶ It should be noted that DOE funds weatherization training centers across the country.
<https://nascsp.org/wap/waptac/weatherization-training-centers/>

weatherization agency for remediation. When remediation is successful, the weatherization job is officially closed.

WAP is the leading comprehensive weatherization program in the U.S. It sets the standard of technical excellence for the weatherization industry. WAP, when implemented by local community action agencies⁷, can refer households to many other social services in addition to energy assistance. Frequently offered programs include Meals on Wheels and services for elders. Often, household needs for these additional programs only become apparent when energy auditors visit the households' homes. To our knowledge, the WAP model, trusted community-based local community action agencies funded through federal and state grants, is unique among weatherization programs. We believe it is a model that should receive serious consideration worldwide.

3.2 Utility Weatherization Programs

Investor-owned utilities (IOUs) may run their own weatherization programs. These programs are funded by customer (i.e., ratepayer) revenues. Regulated IOUs must receive approval from their state's Public Utility Commission (PUCs) to include costs for their weatherization programs in their consumers' energy rates. Typically, energy efficiency programs administered by IOUs have to meet total resource cost (TRC) tests for cost effectiveness. TRC tests essentially pit energy efficiency programs against other sources of energy that utilities can procure (e.g., electricity from natural gas and coal plants, renewable energy sources). Energy efficiency is viewed as a source of energy for IOU customers and is very frequently cost competitive with other energy resources.

Unfortunately, low-income energy efficiency programs are often not cost competitive. To defend the funding of comprehensive weatherization programs, IOUs will often argue that they have a 'social contract' to provide for the public good as regulated entities. In many cases, IOUs will only offer reduced weatherization programs that install only highly cost-effective measures, such as LED lights. We are finding that for many utilities it is important to monetize non-energy impacts of low-income weatherization to increase estimates of their cost effectiveness. Section 5 below addresses the non-energy impacts of weatherization.

IOUs may contract with local weatherization programs to implement their programs. As noted above, in many cases agencies can blend WAP and utility dollars to maximize investments in homes. IOUs will also contract with home energy retrofit contractors to administer their programs. Utility programs that use contractors may have difficulties in recruiting homes because of a lack of referrals from energy assistance programs and lack of word-of-mouth recommendations. Additionally, contractors are not as well positioned to refer low-income households to other social service programs as most low-weatherization agencies are.

⁷ Link to CAAs

4.0 Why Evaluate the HHR Co-Benefits of Weatherization

There are two major reasons to evaluate all co-benefits of weatherization. One, it is good practice to evaluate all of the outcomes of weatherization programs, not just the intended energy demand and household cost reduction outcomes. At the federal level, a fuller appreciation of the non-energy impacts could shore up support for Congressional funding and support collaborative interagency initiatives---e.g. between housing, indoor air quality, and residential energy efficiency programs. With respect to electric and natural gas utilities in the U.S., demonstrating the co-benefits of weatherization programs can help them meet their social contract as regulated utilities. Monetary estimates of the values of co-benefits can be incorporated into rate cases to support using customer revenues for these programs. This is becoming particularly important in the U.S. because the success that utilities have had in installing LED lights in homes has left them with fewer cost-effective energy efficiency measures to install in low-income homes (Kravatz 2018).

A second major reason to evaluate the HHR co-benefits of weatherization is to inform the healthcare and public health sectors about the benefits of weatherization. A link between weatherization and improvements in health can be made through the concept of social determinants of health (SDOH). This is a concept that has arisen recently in the literature and has been taken up by healthcare and public health sectors⁸ in the U.S., due in part to the realization individuals' life expectancy varies tremendously simply based upon the neighborhood where they live (Glasgow Center for Population Health, 2013). This insight led to the conclusion that social and economic conditions can significantly impact health (Braveman and Gottlieb, 2014). The Healthy People Program of the U.S. Department of Health and Human Services defines social determinants of health as “conditions in the environments in which people are born, live, learn, work, play, worship, and age that affect a wide range of health, functioning, and quality-of-life outcomes and risks.”⁹ It can be argued that weatherization directly addresses SDOH through investments that improve the physical conditions of homes.

Demonstrating the links between weatherization and social determinants is critical for several reasons. Research on the social determinants impacts of weatherization is vital to broadening the perspectives of low-income weatherization program administrators beyond energy burden to poverty writ large. Another important reason for demonstrating the link between weatherization and health in the U.S. context is that healthcare's financial resources dwarf those available to energy efficiency programs in general and low-income weatherization programs in particular. In the U.S., low-income weatherization is severely underfunded and waiting lists for weatherization are long (Tonn et al., 2015). It may take a year or more to get one's home weatherized. If it can be further demonstrated that healthcare costs could be reduced through investments in weatherization, then it could be a win-win policy solution for the energy efficiency and healthcare sectors.

⁸ In the U.S., public health agencies work with their communities to improve general health and prevent the spread of disease. The healthcare sector is responsible for providing medical services to individuals on an as needed basis.

⁹ <https://www.healthypeople.gov/2020/topics-objectives/topic/social-determinants-of-health>

5.0 How Weatherization Can Yield Health Impacts: A Theory of Change

Figure 2 presents our theory of change and explains how weatherization can positively influence home conditions, health, and household finances. The model begins with physical changes to homes through the installation of energy efficiency measures. These measures have significant lifetimes---a decade or more for a new furnace and over twenty years for insulation, for example. The installation of these energy efficiency measures results in energy cost savings to households. In addition, measures installed to reduce water heating energy demand by reducing water consumption (e.g., low-flow showerheads and faucets) can also reduce water costs. When these cost savings allow households to better pay their utility bills on time and in full, they may also face fewer costs associated with arrearages and utility disconnections.

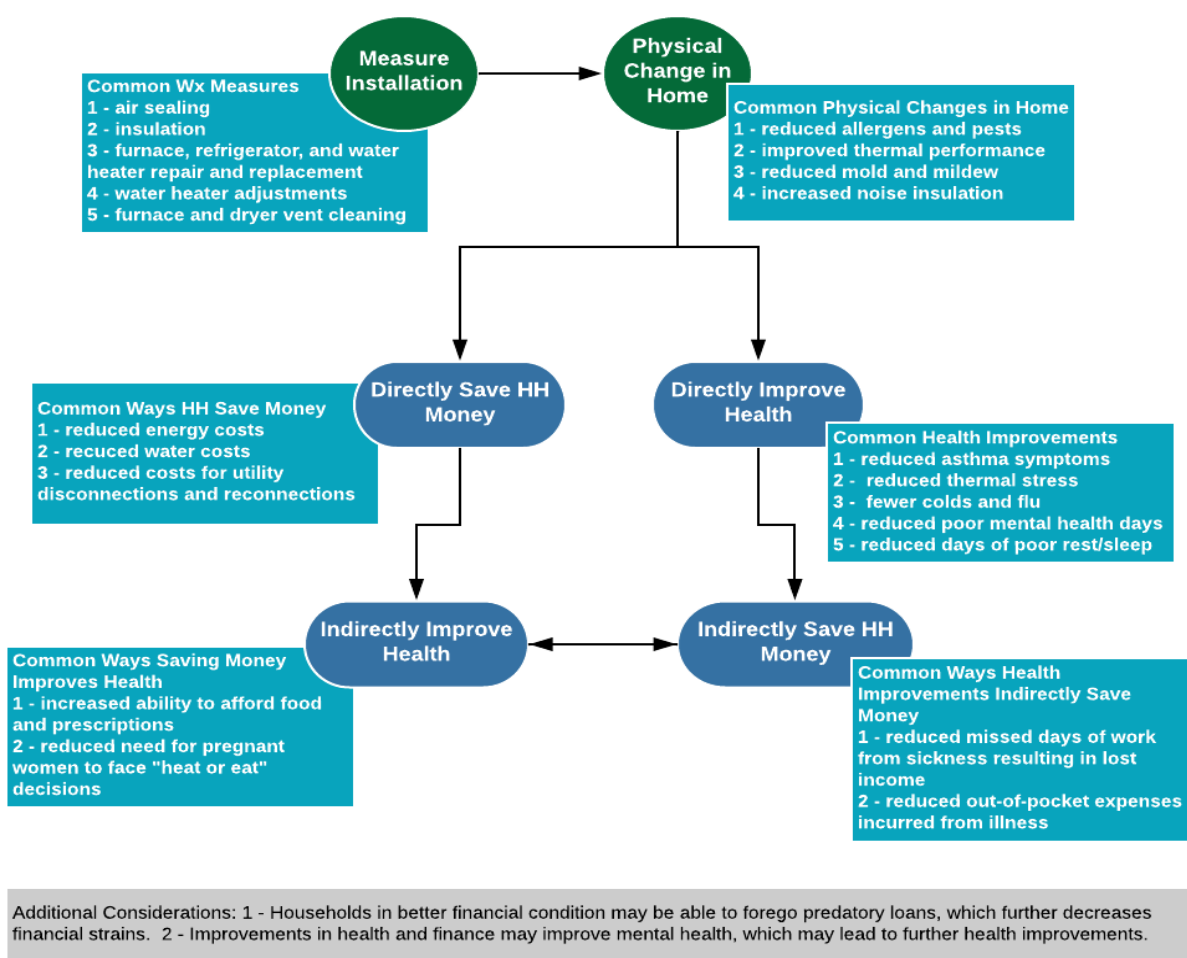


Figure 2. How Weatherization Can Yield Health and other Household Benefits

Weatherization can directly impact human health by improving indoor environmental quality (Doll et al., 2016; Pigg et al., 2014; Norris et al., 2013; Francisco et al. 2017). Air sealing, repair and replacement of combustion appliances, improvements in ventilation (Underhill et al., 2020; Engvall et al., 2003), and the cleaning of furnace and dryer filters are among the weatherization

measures that can improve IAQ (Doll et al., 2016; Pigg et al., 2014; Norris et al., 2013; Francisco et al., 2017; Wilson et al., 2020). Air sealing can keep pests and outdoor allergens out of homes (Breysse et al., 2014; De Souza et al., 2019; Rose et al., 2015; Sommerville et al., 2000). Improvements in IAQ can then lead to improvements in chronic respiratory conditions, which include asthma (Howden-Chapman et al., 2007; Breysse et al., 2014; De Souza et al., 2019; Rose et al., 2015; Sommerville et al., 2000) and COPD (Breysse et al., 2014; Colton et al., 2015; Wang and Rotondo, 2020; Fisk et al., 2010; Tohn et al., 2016). Improved IAQ can also reduce migraines and chronic headaches (Tietjen et al., 2012; Lukeso et al., 2016; Marc et al., 2018; Norbäck et al., 1990).

Insulation, air sealing, and HVAC system repair and replacement can keep homes warmer in the winter and cooler in the summer, which can greatly improve the health of occupants (Haverinen-Shoughnessy et al., 2018; Ahrentzen et al., 2016; Alam et al., 2018; Poortinga et al., 2018; Guo et al., 2016; Lee et al., 2020; Chard and Walker, 2016; Nicholls and Strengers, 2018; Wilson et al., 2014) and reduce instances of thermal stress that may require medical intervention (Haverinen-Shoughnessy et al., 2018; Ahrentzen et al., 2016; Alam et al., 2018; Poortinga et al., 2018; Tonn et al., 2014b). Insulation, air sealing and furnace replacement can also make homes more resilient to extreme weather events. Preventing homes from being too cold or too hot can also reduce the incidence of low-birth-weight babies (Frank et al., 2006; Chersich et al., 2020; He et al., 2016; Sun et al., 2019).

Improvements in home thermal conditions can improve rest and sleep (Wilson et al., 2014; Nicol, 2019; Okamoto-Mizuno and Mizuno, 2012; Fujii et al., 2015). Improvements in thermal comfort can reduce symptoms related to osteoporosis and rheumatoid arthritis. More comfortable indoor temperatures could also promote more physical activity (Lindemann et al., 2017; van Kasteren et al., 2019), which could also help reduce obesity (Wilson et al., 2014) and arthritis symptoms. Moderated thermal temperatures can also reduce instances of dehydration (Wilby et al., 2021; Kenny et al., 2019; Silva et al., 2020) and, further, reduce the incidence of kidney disease (Johnson et al., 2019; Barraclough et al., 2017; Nerbass et al., 2019). The latter, along with decreased stress and increased physical activity, could then lead to reductions in uncontrolled diabetes and cholesterol levels (Misra and Fitch, 2020; Shah et al., 2010; Kratz, 2005; Medline Plus, N.D).

Wall insulation, primarily, as well as ceiling and attic insulation, new windows, and air sealing, can reduce noise levels in homes (Tonn et al., 2014b). A direct result can be improvements in rest and sleep (Halonon et al., 2012; Öhrström et al., 2006; Basner and McGuire., 2018). We hypothesize that reductions in noise might also benefit those who suffer from post-traumatic stress disorder (PTSD) (Nuamah et al., 2021), autism spectrum disorders (Shireen et al., 2017; Mamashli et al., 2017; Landon et al., 2016), and hearing impairment (Mayes, 2021; Alsina-Pages et al., 2021).

Improvements in IAQ, indoor comfort, and reductions in the infiltration of outdoor noise can improve rest and sleep, which can then lead to improvements in mental health (Phillip et al., 2006; Kyle et al., 2013; Kyle et al., 2010; Leger et al., 2001; Schubert et al., 2002; Vermeulen et al., 2021). Improvements in general health and physical activities can also lead to improvements in mental health (Ohrnberger et al., 2017; Kim et al., 2012; Giannakopoulos et al., 2009).

Improvements in mental health can then reduce the impacts of chronic conditions such as hypertension (Unger et al., 2020; Gerin et al., 2012; Yano et al., 2016) and ischemic heart disease (Sanchis-Soler et al., 2020; Steptoe et al., 2011; Blumenthal et al., 2005).

Burns to occupants can be reduced by reducing water heater temperatures, done ostensibly to reduce the amount of energy needed to heat water (Simon et al., 1993). Furnace replacement and minor home repairs to electrical wiring and equipment can reduce home fires (Tonn et al., 2014b). Installation of CO monitors can reduce CO poisoning events (Tonn et al., 2014b). In every one of these cases, improved health can lead to reductions in medical costs, including visits to doctors' offices and emergency departments and hospitalizations (Tonn et al., 2014b).

Improved household financial situations can indirectly improve health by reducing psychosocial stressors, exemplified by reduced bad days of rest and sleep and fewer households having to trade off buying food or prescriptions to pay energy bills (Tonn, 2014b). Improved health of households' primary wage earners and dependents can result in fewer missed days of work (Tonn et al., 2014b). This is vitally important because almost three-quarters of lower-wage workers in the U.S. do not enjoy the benefits of paid sick days. Improved financial situations could also allow households to avoid costly short-term predatory loans (e.g., payday loans, pawnshop transactions) (Tonn et al., 2014b).

Research shows that low-income households whose single-family homes have been weatherized have improved household finances. This is primarily due to reductions in energy (Tonn et al., 2018) and water costs (Shrubsole et al., 2014; Tonn et al., 2014a; IEA, 2014). Fewer missed days of work due to illness, fewer interest payments associated with short-term, high-interest loans, fewer utility disconnection and reconnection fees, and reduced out-of-pocket costs associated with treatment of chronic health conditions can also improve household financial situations (Tonn, 2014b). Improved household finances can synergistically reduce mental stress, as has been shown with decreases in energy burden (Hernandez et al., 2016). It also has been found that households are better able to afford necessary prescriptions and nutritious food post weatherization (Tonn et al., 2014b).

It is argued here that these benefits are most achievable through comprehensive weatherization programs. For example, it takes a combination of measures to reduce asthma triggers, not simply the cleaning of furnace filters. Similarly, it takes several measures to improve home comfort and reduce dangerous incidents of thermal stress. Reduction in fire risk can be seen as an accumulation of risk reductions through furnace replacement, insulation, and minor electrical repairs. It is only the rare case where a single measure can be mapped directly to a health benefit, such as the linkage of CO monitors to reduced CO poisoning.

The framework of results represented by these five categories is generic, in that it can be applied to SF and mobile homes and MF buildings. However, most of the research in the literature is based on studies of SF homes. Only one previous study addressed the health impacts of weatherizing multifamily buildings (Wilson et al. 2020), though it should be noted that the study also co-mingled single-family homes and small multifamily buildings (i.e., MF buildings with less than five units), was not limited to low-income properties, and had a much smaller sample size.

We do not believe that the results of previous studies on the health benefits attributable to weatherizing low-income SF homes are transferrable to the MF sector for several reasons. First, SF homes and MF buildings behave differently from a building science perspective. The latter have shared walls, common areas, central heating, and oftentimes shared ventilation systems. Stack effects can be more pronounced in high-rise buildings than in single floor SF homes. Second, because of physical differences between SF homes and MF buildings, the frequency of weatherization measures installed is different between these home types. For example, SF homes more often receive air sealing and insulation measures while MF buildings are more likely to receive major HVAC replacements and new windows. For MF, insulation and air sealing measures are often installed only in common areas rather than at a unit-level. Lastly, the demographics of occupants differ between the two housing types. Residents of affordable MF buildings are older and more racially diverse (Tonn et al. 2021a). Thus, evaluating the health benefits potentially accruable from weatherizing affordable MF buildings separate from SF health benefits is advisable.

There are numerous differences between the SF and MF contexts that could impact measurable health benefits of weatherization. Here are several hypothesized differences:

- 1) IAQ impacts will be smaller – This is because less work is typically done in MF units that could improve IAQ. For example, air sealing and insulation are less frequently installed in medium size and large MF buildings. Many MF buildings have central heating and common laundry areas, which eliminates the potentially positive contributions to IAQ of cleaning in-home/unit heating system filters that are commonly done as part of the weatherization of SF homes.
- 2) Thermal comfort impacts will be larger – This is because in the medium and large-sized MF buildings old and energy inefficient heating systems are replaced with new systems that are less likely to breakdown and are operated by building-wide energy management systems. Compared to the residents of SF homes, tenants living in MF buildings typically have less control over the temperatures in their units until new systems are installed that have in-unit energy management controls.
- 3) Noise reductions will be larger – Inefficient single pane windows are more often replaced with double pane windows during large MF weatherization projects. Window replacement in the SF sector is relatively rare because they are deemed less effective at reducing energy use than other measures. Another factor is that the majority of the MF buildings in the study are located in urban areas that have more noise pollution. Synergistic impacts on mental health will be larger – This is because thermal comfort and noise levels will be improved.
- 4) Synergistic impacts on household financial health will be smaller – This is mainly because utility costs are included in the rents of many MF buildings. Thus, many MF households will not experience reductions in their energy costs nor avoid the costs of utility shut-offs and reconnection service fees. Also, MF households in our sample will not recoup the same level of lost wages from fewer missed days of work as their counterparts in the SF weatherization sector because more of the head of households are retired in the MF sector.

6.0 Approaches to Measuring Changes in Co-Benefits Attributable to Weatherization

There are several approaches to measuring changes in HHR co-benefit variables. Here are three for consideration:

Surveys – The approach is to survey weatherization participants pre- and post-weatherization. The survey must include questions that can measure changes that are monetizable. It is also recommended that questions be included that could provide additional evidence that the monetizable changes can be attributed to weatherization. To take asthma as an example, questions could focus on emergency department and hospital visits. Additional questions could focus on dust levels and infestations of pests. If the latter improve, then explanations are available for reductions in medical care needed for asthma post-weatherization. We typically also include several questions related to general health that can help make the case that weatherization can improve health. It is recommended that a control group be surveyed at the same times as the treatment group. In this way differences-in-differences statistical tests can be used to determine whether the measured changes in NEI variables are statistically significant.¹⁰

Direct measurements – Let us assume that it is hypothesized that weatherization could reduce high blood pressure, due in part to reduced stresses on household budgets and improved indoor comfort. Instead of posing a survey question about blood pressure, the blood pressure of occupants could be measured directly. Instead of asking about improvements in sleep, occupants could be asked to wear sleep monitors. Other methods of direct measurement include breathing and blood tests and extensively monitoring food intake and use of prescriptions. Overall, direct measurement approaches can be more accurate than surveys but may be more expensive and maybe more intrusive from the point of view of participants.

Special Studies – In some cases, it may not be practical or possible to measure changes within the scope of an NEI evaluation. For example, it may take years to collect enough data to estimate NEIs attributable to the installation of CO monitors. This is because CO poisoning is a rare event, compared to, for example, asthma emergency department visits. In these instances, the recommendation is to search the literature for special studies that present NEI-relevant impacts that can be attributable to the installation of individual or combinations of weatherization measures. In the case of CO, there is only a small number of papers that estimate the reduction in CO poisoning from the installation of CO monitors (Yoon et al., 1998; Krnzelok and Roth, 1996). We also used this approach to estimate the fire prevention NEIs (Tonn et al., 2014b).

¹⁰ For legal, programmatic, and ethical reasons, we have found that it is rarely feasible to implement randomized control trials in our energy program evaluation research and therefore rely on quasi-experimental designs. This is common in the energy research and social science space.

7.0 Evaluations of the Co-Benefits of Weatherization

Assessments of the HHR co-benefits of weatherization are, essentially, evaluations of existing weatherization programs. In other words, data such as those described above are collected in regard to specific weatherization programs. This section describes four weatherization program evaluations where HHR co-benefits were assessed, with selected results from those evaluations reported on in Sections 8 and 9. The first evaluation focused on the national Weatherization Assistance Program, the second on a local weatherization program called the Knoxville Extreme Energy Makeover, the third actually encompassed affordable multifamily buildings located in two regions of the U.S., and the fourth focused on the Home Uplift Program offered by the Tennessee Valley Authority (TVA).

7.1 U.S. Department of Energy's (DOE) Weatherization Assistance Program (WAP)

WAP is a U.S. federal program administered by the U.S. Department of Energy. As described in more depth in above, WAP provides grants to states and territories, who then provide grants to local weatherization agencies to deliver free comprehensive weatherization services to income-eligible households. WAP operates in all areas of the country. WAP weatherizes single family (SF) and mobile homes (MH), and affordable multifamily buildings (MF).

The most recent comprehensive national evaluations of WAP were conducted between 2009 and 2015 (Tonn et al., 2014a; Tonn et al., 2015b). The evaluations were funded by the U.S. Department of Energy and led by Oak Ridge National Laboratory.¹¹ As part of these evaluations, a national survey of WAP recipients was conducted. The treatment group respondents were randomly selected from lists of about to be weatherized SF and MH homes provided by a randomly selected set of local weatherization agencies. A comparison group of previously weatherized homes was similarly selected. A sample of control homes was not developed. A phone survey was administered in 2011/2012 pre- and post-weatherization for the treatment group and two times to the comparison group at similar times. The survey contained questions about home conditions, health, household budgets, and demographics.¹² The sample sizes for the treatment and comparison group were 665 and 803, respectively. This was the first national survey of WAP recipients ever conducted.

7.2 Knoxville Extreme Energy Makeover Project (KEEM)

The Knoxville Extreme Energy Makeover Project (KEEM) was a weatherization program conducted in Knoxville, Tennessee, USA, from 2016 to 2018 that was funded by the TVA. Because TVA is an electric utility, KEEM focused only on homes heated with electricity. Further, KEEM only enrolled single-family, owner-occupied homes. KEEM was designed to reduce

¹¹ Three³ researchers, Bruce Tonn and Erin Rose, led the evaluations when they worked at Oak Ridge National Laboratory.

¹² We note that the WAP resident survey was constructed using questions drawn from several existing federal questionnaires, such as those administered by the Centers for Disease Control, the Energy Information Administration, and the Bureau of Labor Statistics. With only a few exceptions, the WAP survey was reused in the KEEM, LIMF and HUP evaluations in order to foster cross-program HHR co-benefits insights.

homes' electricity use by 25%. The homes were weatherized by the local community action agency, City of Knoxville-Knox County Community Action Committee (CAC). During the early stages of the project, homes were heavily recruited from historically under-represented minority communities in Knoxville.

KEEM's weatherization approach differed from WAP in two significant ways. First, a savings-to-investment-ratio was not applied to measures installed. Instead, measures were identified to best reach the 25% reduction in electricity use for each home. Second, KEEM replaced more heating and air conditioning systems than would a conventional WAP program.

The Robert Wood Johnson Foundation provided a grant to Three³ to assess the HHR co-benefits of KEEM. A phone survey was administered to all homes that KEEM weatherized between 2016 and 2017. Contact information was provided by CAC. Surveys were administered after homes received energy audits but before homes were weatherized and then again one year later. The Center for Applied Research and Evaluation (CARE), University of Tennessee, Knoxville, administered the phone survey. The control group sample was drawn from CAC's Knoxville waiting list for their KEEM, WAP, and local-utility funded programs. A random sample of KEEM homes weatherized one year previously was also surveyed to create a comparison group. The sample sizes for the comparison, treatment and control groups were 196, 98, and 154, respectively.

7.3 Low-Income Multifamily Buildings (LIMF) Co-Benefits Evaluation

The JPB Foundation and the Energy Efficiency Program Administrator in the Commonwealth of Massachusetts provided funds to Three³ and partners Slipstream and NMR to estimate the HHR and resilience co-benefits attributable to weatherizing low-income MF buildings (LIMF) located in the Northeast and Midwestern regions of the United States. The MF buildings in our sample had 5 or more units. We worked with state and local weatherization programs, utilities that offer MF weatherization, non-profit low-income and affordable housing advocates, and non-profit and commercial owners of affordable MF buildings to construct a convenience sample of treatment, control, and comparison buildings.

Treatment, control, and comparison group households were first surveyed between 2018 and 2019. The project team asked property managers and property owners for permission to enter buildings to hang survey packets on residents' doorknobs and in the appropriate language if known. Each clear bag contained a cover letter, a paper survey, and a pre-paid return envelope. Survey packets were hung on every door in a visited building, except for the largest MF buildings, where surveys were hung on every other or every third door, though in some buildings project staff personally handed the surveys to primary respondents and were able to wait in the building until the surveys were completed. Over 7,700 surveys were distributed. Potential participants were also given the option of completing the survey over the phone or via the web, though most chose to fill out the paper survey. Based on recommendations provided by property managers and owners, paper surveys were translated into Spanish, Mandarin and Russian. Respondents mailed completed paper surveys to CARE for processing; CARE also administered the phone and web versions. The sample sizes for the number of units for the comparison, treatment, and control homes were 653, 432, and 836, respectively.

7.4 Home Uplift Program (HUP)

The Home Uplift Program (HUP) is a low-income weatherization program funded by TVA and local power companies in its jurisdiction. It is the successor to TVA's Extreme Energy Makeover initiative. The five major cities in TVA's region participated in the first round of funding of HUP: Knoxville, Nashville, Chattanooga, and Memphis, Tennessee, and Huntsville, Alabama. There were also a small number of rural communities in Mississippi and Kentucky that participated. HUP weatherized SF homes and a small number of MH. Most were owner-occupied. The small number of homes from Huntsville were owned by its public housing program.

TVA contracted with Three³ to estimate the HHR co-benefits attributable to HUP. Similar to KEEM, CARE administered the phone surveys just after homes received their energy audits but prior to weatherization and again one year later. All HUP homes that received energy audits between 2018 and 2019 were contacted by CARE. The control sample was drawn from waiting lists, mailing lists, and lists of income-eligible homes that received housing and energy assistance. The sample sizes for the treatment and control groups were 701 and 300, respectively.

8.0 Health and Other Household-Related (HHR) Co-Benefits of Weatherization

This section presents selected results from the HHR co-benefits evaluations of WAP, KEEM, LIMF, and HUP. The next section presents demographic statistics for the treatment groups from each evaluation. Section 8.2 presents descriptive statistics on home conditions, general health, household finances, respiratory health, health issues most impacted by home temperatures, and life satisfaction. The results presented in Section 8.3 illustrate that most HHR metrics changed in ways to benefit households post-weatherization even though statistical significance was not always reached. Section 8.4 presents a five-component household model that shows how the beneficial impacts of weatherization can cascade through the household.

8.1 Demographics of Four Evaluations' Main Respondents

Table 1 presents key demographic characteristics of the main respondents to the resident surveys administered as part of the WAP, KEEM, LIMF and HUP HHR co-benefits evaluations. The respondents are overwhelmingly female and approaching retirement age. That the majority of respondents are female is not too surprising as previous research suggests that females tend to know more about their households than others and then also tend to volunteer to answer survey questions on behalf of their households (Mullin and Tonn, 1993).

Table 1. Demographics of Evaluations' Main Respondents – Treatment Groups

Demographic Characteristic	WAP (n=665)	KEEM (n=98)	LIMF (n=417)	HUP (n=701)
Respondent Gender: Female (%)	73%	76%	69%	81%
Respondent Age	57	54	58	62
Average Household Size	2.38	2.04	1.8	1.8
Black or African American (%)	18%	29%	24%	77%
Respondent Employed (%)	34%	34%	27%	26%
Respondent Retired (%)	31%	20%	46%	44%
Respondent Married (%)	32%	25%	10%	15%
Respondent Education: GED or less (%)	60%	42%	62%	52%

The relatively high age of respondents can be explained as the result of several factors. One, respondents living in SF and MH tend to own their homes and are trying to age in place. Those who find themselves income-eligible for weatherization see the reduction of energy costs as an important goal. Analyses of the WAP data suggest that households whose respondents are close to retirement age suffer more economic issues and health problems than retired households,

which adds to the motivation of these households to seek weatherization (Tonn et al., 2021a). Additionally, younger respondents typically do not own their own homes, and the weatherization of SF and MH rental units lags the weatherization of owner-occupied units. Lower rates of employment, higher rates of retirement, and smaller household sizes follow from the relatively higher ages of the respondents.

The percentage of main respondents self-reporting as Black or African American is much higher for HUP because TVA prioritized weatherizing homes in historically minority communities. The much lower for WAP is the result of two factors: the majority of homes weatherized by WAP funds are owner-occupied and WAP funds weatherization in rural areas across the U.S. These factors suggest that a high majority of WAP homes would not be occupied by Black households.

The demographics of the LIMF sample were somewhat surprising. Almost all of the respondents in this sample were renters. It was expected that the average age of the respondents would be lower and that a higher percentage would self-report as Black. Instead, this sample has higher rates of retirement and a large number of single person households. Lastly, with respect to education, it is not surprising that lower levels of education seem to be correlated with weatherization income-eligibility, but it is somewhat surprising that between 40% and 58% percentage of the main respondents did self-report as having some sort of post-secondary school education.

In summary, the households of weatherization recipients in the U.S. tend to be older, with households with respondents' ages just shy of retirement having the greatest economic needs and worse health pre-weatherization.

8.2 Descriptive Statistics – Treatment Groups

This section presents key findings about the treatment groups' homes, main respondents' health, and households finances and life satisfaction pre- and post-weatherization. The statistics were produced using survey data collected during the four evaluations. As noted above, the same survey questions were used across all four evaluations. The results presented in this section are drawn from these publications: (Tonn et al., 2014b; Tonn et al., 2021b, Tonn et al., 2023a, and Rose et al., 2023).

This section begins with a summary of changes to homes' environmental conditions pre- to post-weatherization (Table 2). This is the logical place to start because according to the theory of change presented above, all the HHR co-benefits are driven by weatherization investments in homes.

The most consistent findings are that weatherization reduces drafts, dustiness, mold, mildew smells, and the intrusion of outdoor noise. The first two can be attributed to the air sealing of homes. Mold and mildew can be reduced to improvements in ventilation which are often undertaken in homes specifically to reduce mold and to even out the benefits of heating and cooling throughout the house and to properly vent combustion appliance emissions. The fourth can be attributed to the installation of insulation, air sealing, and repairs to broken windows.

The results with respect to infestations of insects and rodents are mixed. It is not clear how weatherization could lead to increases of infestations. This could be topic for future research.

Table 2. Housing Conditions – Pre-Post Weatherization for Treatment Groups

Physical Condition of Home	WAP	Δ	KEEM	Δ	LIMF	Δ	HUP	Δ
How often home too drafty? (all to some of the time)	71%	-32%	86%	-42%	60%	-14%	76%	-38%
How often home too dusty? (all to some of the time)	NA	NA	91%	-14%	71%	-18%	77%	-21%
Outdoor noise? (a great deal to some)	69%	-15%	70%	-12%	67%	-7%	72%	-15%
How infested is home with cockroaches, other insects, and spiders? (extremely to somewhat)	25%	-10%	28%	-20	26%	-3%	20%	-6%
How infested is home with mice? (extremely to somewhat)	10%	-3%	33%	-13%	15%	+2%	11%	-6%
Frequent mildew odor or musty smell? (%yes)	30%	-9%	39%	-13%	NA	NA	N/A	
How often have observed standing water in home? (always to sometimes)	20%	-7%	NA	NA	11%	0%	11%	-4%
Have seen mold in home? (%yes)	28%	-9%	42%	-17%	31%	-2%	30%	-11%

Among the strongest findings is that weatherization promotes general health. A particularly strong finding is that substantially fewer households keep their homes at unhealthy temperatures post-weatherization (Table 3). This result is fully expected because tighter and more insulated homes that have more effective and efficient heating and cooling systems will reduce the changes that homes will become too hot or too cold.

In part because homes are more comfortable thermally, respondents report fewer days of not good physical health and rest/sleep post-weatherization. Improvements in rest/sleep can also be attributed to a reduction in the infiltration of noise from outside. These results are strong across the four evaluations. The WAP, KEEM and HUP evaluations show substantial improvements in mental health, except for the evaluation that focused on MF housing. In this last case, the number of bad mental health days increased as small amount post-weatherization.

Table 3. Weatherization Impacts on General Health – Pre-Post Treatment Groups

Health Indicators	WAP	Δ	KEEM	Δ	LIMF	Δ	HUP	Δ
Home at unhealthy temperature (almost every month to some months)	18%	-13%	55%	-41%	50%	-11	21%	-15%
Number of days previous month physical health not good	10.3	-4.9	12.7	-4.9	NA	NA	8.6	-2.0
Number of days previous month mental health not good	7.1	-3.4	10.6	-4.2	7.06	+1.18	5.5	-0.7
Number of days rest/sleep last month	11.2	-5.1	15.0	-2.7	7.95	-2.1	10.6	-1.9

Table 4 presents results related to the impacts that weatherization may have upon household finances. The number of households that find it hard to pay energy bills decreases substantially post-weatherization for the programs that focus on single family homes. That almost three-fourths of households report it being hard or very hard to pay energy bills suggests that energy costs were a major motivation to seek weatherization. It should be noted though that even after weatherization, about one-half of homes still reported it being hard or very hard to pay energy bills. Weatherization by itself is not a standalone solution to energy poverty.

At this point it should be noted that the affordable multifamily context differs substantially from the SF context. First, in high majority of cases, it is the property owners that seek out weatherization programs, not the tenants. Second, many tenants' energy bills are included in their rents. The tenants may have difficulty in paying their rents, in part because utility bills for their buildings are high, but we did not ask this question in our MF survey. This also means that questions about tradeoffs with respect to paying energy bills versus paying for food and prescriptions are not relevant to these MF tenants.

Still, a fair portion of MF respondents do directly pay their energy bills. A much lower percentage of households report this financial problem. One can speculate that energy bills are lower for apartment dwellers.

Table 4. Impacts of Weatherization on Household Financial Health

Survey Item/Group	WAP	Δ	KEEM	Δ	LIMF	Δ	HUP	Δ
It is hard or very hard to pay energy bills (% yes)	75%	-26%	73%	-22%	37%*	0%	68%	-17%
Received a disconnect notice past year (% yes)	39%	-9%	47%	-8%	27%*	+3%	26%	-8%
Did not buy food to pay energy bills (% yes)	33%	-5%	37%	-8%	19%*	+5%	33%	-9%
Did not pay energy bills to buy food (% yes)	27%	-8%	19%	-1%	16%*	-4%	16%	-5%
Received food assistance (e.g., WIC) (% yes)	56%	-6%	54%	+2%	54%	+9%	36%	-1%
Did not pay energy bill to fill prescriptions (% yes)	12%	-1%	14%	-4 %	3%*	+3%	12%	-4%
Did not fill prescriptions to pay energy bills (% yes)	28%	-9%	35%	-8%	8%*	-2%	25%	-8%
Could not afford prescriptions (% yes)	33%	-11%	43%	-11%	14%	+2%	28%	-7%
Used one or more short-term, high-interest loans to pay utility bills (% yes)	19%	-7%	17%	+2	6%*	-2%	20%	-8%
Missed Days of Work**	7.67	-.63	4.7	+1.7	NA	NA	9.7	-3.5

* Only includes households whose utility costs are not embedded in their rent.

** Only included employed primary wage earners

Table 5 presents survey results with respect to respiratory issues. The theory of change suggests that weatherization should have major benefits for asthma sufferers from reducing the intrusion of outdoor asthma triggers and dust in homes. The results are mixed with respect to asthma. Symptoms are not guaranteed to lessen. Needs for emergency department and hospital visits do not uniformly decrease post-weatherization for the treatment group. The results for COPD, severe headaches, flu, and sinus infections are more robust.

Table 5. Respiratory Issues

Health Indicators	WAP	Δ	KEEM	Δ	LIMF	Δ	HU	Δ
Asthma rates	14%	--	15%	--	16%	--	15%	--
How long since last asthma symptoms (% less than 3 months)?	55%	+3%	53%	-3%	56%	+4%	65%	+13%
Asthma – ED visit past year (% yes)*	10%	-5%	13%	-3%	22%	0%	4.3%	+1.3%
Asthma -- Visited Hospital past year (% yes)*	14%	-6%	0%	0%	10%	0%	4.3%	-4.3%
Severe headaches (% yes)	20%	-4%	36%	-20%	29%	0%	31%	-10%
Past year, someone have flu (% yes)	22%	-4%	22%	-13%	N/A	NA	N/A	N/A
Past year, someone have sinus infections (% yes)	37%	-3%	59%	-13%	N/A	NA	N/A	N/A
Have COPD (% yes)	NA	NA	16%	--	11%	--	17%	--
Medical attention COPD (% yes)*	NA	NA	67%	-5%	21%	-10%	22%	-5%

* Amongst respondents who reported still having asthma or reporting having COPD.

Table 6 presents results related to thermal stress and arthritis. Both are related to the thermal comfort of homes. Thankfully, there were not many reported instances of occupants being too cold or too hot in their homes that required medical attention. However, needs for medical attention decreased post-weatherization in all cases. Please note that these results do not take into account the ways that weatherization can prevent instances of thermal stress before HVAC systems become too inefficient or inoperable.

Table 6 indicates that arthritis is very common among weatherization recipients. This is not surprising given the age demographics of the respondents. The results from the MF sector suggest that weatherization can reduce arthritis symptoms and medical costs. Arthritis needs to be included in more SF studies moving forward.

Table 6. Thermal stress and arthritis

Health Indicators	WAP	Δ	KEEM	Δ	LIMF	Δ	HUP	Δ
Medical attention needed past year for being too cold in home (% yes)	3.2%	-1.7%	5%	-2.5%	2.7%	-1%	2.9%	-2.6%
Medical attention needed past year for being too hot in home (% yes)	2.4%	-0.9%	1.3%	0%	0.5%	-0.5%	1.4%	-1.3%
Arthritis Rate	NA	NA	NA	NA	50%		60%	
Medical attention for arthritis (%)	NA	NA	NA	NA	NA	NA	NA	NA
Had arthritis symptoms last three months (%)	NA	NA	NA	NA	79.7%	-1.3%	NA	NA
Visited doctor for arthritis last year (%)	NA	NA	NA	NA	33.7%	-13.7%	NA	NA
Urgent care treatment last year (%)	NA	NA	NA	NA	9.8%	-4.9%	NA	NA
Visited emergency department for arthritis last year (%)	NA	NA	NA	NA	7.8	-3.9%	NA	NA
Visited hospital for arthritis last year (%)	NA	NA	NA	NA	6.9%	-3.0%	NA	NA

We visited a large number of local weatherization agencies and the homes of weatherization recipients as part of our evaluation research. We were inspired by the number of letters of heart felt gratitude that the agencies received from weatherization recipients. We attended meetings of weatherization recipients that featured emotional testimonials about the life changing impacts of weatherization. Based upon these experiences, we added questions to the LIMF and HUP residential surveys about changes in life satisfaction that could be attributed to weatherization. Survey results presented in Table 7 support hypotheses that weatherization is linked to increasing satisfaction with life, health and standard of living.

Table 7. Life Satisfaction

Life Satisfaction Indicators	WAP	Δ	KEEM	Δ	LIMF	Δ	HUP	Δ
Your life in general (very satisfied)	NA	NA	NA	NA	37%	+5%	60%	+7%
Your health (very satisfied)	NA	NA	NA	NA	25%	+4%	35%	+12%
Your present standard of living (very satisfied)	NA	NA	NA	NA	29%	+3%	43%	+12
Your neighborhood (Very satisfied)	NA	NA	NA	NA	27%	-2%	NA	NA

8.3 Patterns of Co-Benefit Results

The results presented above in Section 8.2 follow a conventional variable-by-variable format. In this section, the perspective is altered to focus on patterns of all statistical results visualized at one time. This section also presents visualizations of statistical results using different combinations of samples. Results presented in this section come from the KEEM and LIMF studies.

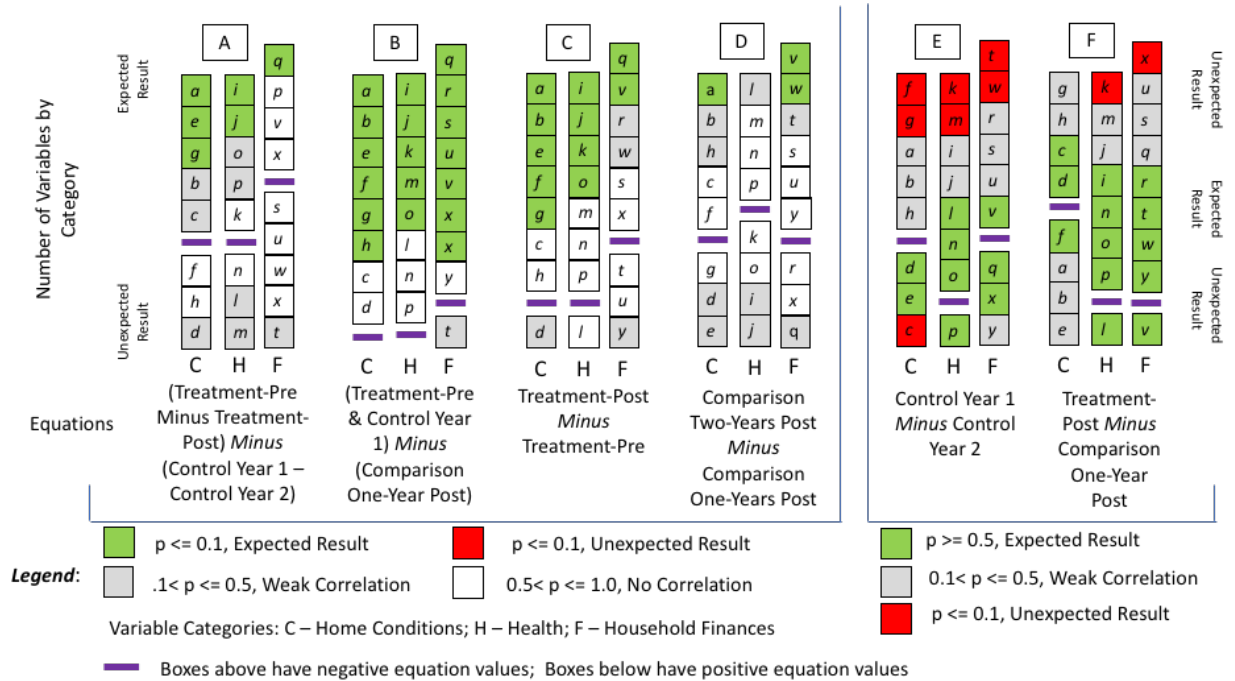
Both studies' research designs include comparison and control groups. For the KEEM study, survey data were collected pre- and post-weatherization for treatment group homes and control group homes. Survey data were also collected from an already weatherized group, the comparison group, twice as well (i.e., one year post and two years post weatherization).

Figure 3 presents a graphical summary with six panels of results. The equation used to estimate the results in each panel is presented underneath three stacked vertical bars. Each box in each bar represents the result of applying the equation to one of the variables in the categories of variables also noted at the bottom of the bar. Boxes appearing above the purple bar have negative values as calculated by the equations. Boxes appearing below the purple bar have positive values as calculated by the equations. Boxes further from the purple bar have higher levels of statistical significance. For example, Panel A's equation is the DID quasi-experimental design equation. The eight boxes in the leftmost column pertain to the home conditions category of variables. For five variables, the equation provided negative values and for three variables positive values.

For panels A, B, C, and D, negative values indicate expected results (i.e., the bad house conditions, health issues, and household financial issues are reduced). Positive values suggest that those problems became worse. The boxes filled with green and red are expected and unexpected results, respectively, both that are statistically significant with a p-value less than 0.1. The p-values for those boxes filled in grey are between 0.1 and 0.5, while the boxes with no fill color have p-values greater than 0.5. The first vertical bar in Panel A indicates that three home condition results with respect to the quasi-experimental design case were statistically significant in the expected direction (i.e., the three green boxes), two variables with negative equation

results had p values between 0.1 and 0.5, two variables with p values above 0.5 had positive equation results, and one variable with a positive equation result had a p value between 0.1 and 0.5. Visually, Panel A suggests that weatherization yielded only a few statistically significant non-energy impacts and several results moving in unexpected directions, though none of those were statistically significant.

Comparing the number of green boxes between Panels A and B provides a strong visual picture of how many more statistically significant results emerged from the cross-sectional analysis than the quasi-experimental design analysis. Only one result moved in an unexpected direction. Panel C provides a similar picture if one only uses the Treatment-pre and Treatment-post results. How closely the Treatment group matches the Comparison group one-year post weatherization is seen in Panel F. One can also see in Panel E how unexpected changes in the Control group from year 1 to year 2 contributed to the underwhelming quasi-experimental design results (Note: the expected results for Panels E and F were to have non-statistically significant differences between the means. Thus, the green boxes cluster near the purple bars and the red boxes are at either end of the columns.) Panel D visually confirms that for the most part, the non-energy impacts attributable to weatherization did not change much for the Comparison group from one-year post to two-years post. There is only weak confirmation that these impacts may improve from one year to the next due to positive feedback effects.



Visually, the strongest consistent signal in the data across Panels A, B, and C is that weatherization has the most impact on home conditions, followed by health and then household finances. However, if one were to only focus on Panel B, then weatherization about equally and strongly impacts all three categories of variables.

An important question that is beyond the scope of this report to resolve is which of the Panels A or B one should view as the most reliable and valid set of results. The research design underlying the former is more rigorous though suffers from small sample sizes and inexplicable changes in the control group. The research design underlying the latter is less rigorous but benefits from large sample sizes. Results presented in Panels C and F also support the conclusion drawn from Panel B that weatherization conducted by KEEM had a positive impact on 18 of the 25 non-energy impacts addressed herein.

Patterns of results from the LIMF study tell a similar story. There are two major observations to be made about the summary of results presented in Figure 4. First, the preponderance of results is consistent with the theory of change framework. Seventy percent of the cross-sectional analysis results were consistent with the framework and seventeen percent or less were inconsistent. The results for the treatment group were similar. This broad view of the results provides confidence that weatherization does correlate with improvements in health, broadly speaking.

Second, almost one half of the control group results changed substantially from period 1 to period 2. There does not seem to be a pattern as to whether control group apartment conditions improved or did not improve pre- to post. The high variability of the control group results was also seen in a study on the health impacts of weatherization conducted in the KEEM study just discussed. Unexpectedly, the variability of the control group results did not impact the percentage of differences-in-differences results that moved in the expected direction.

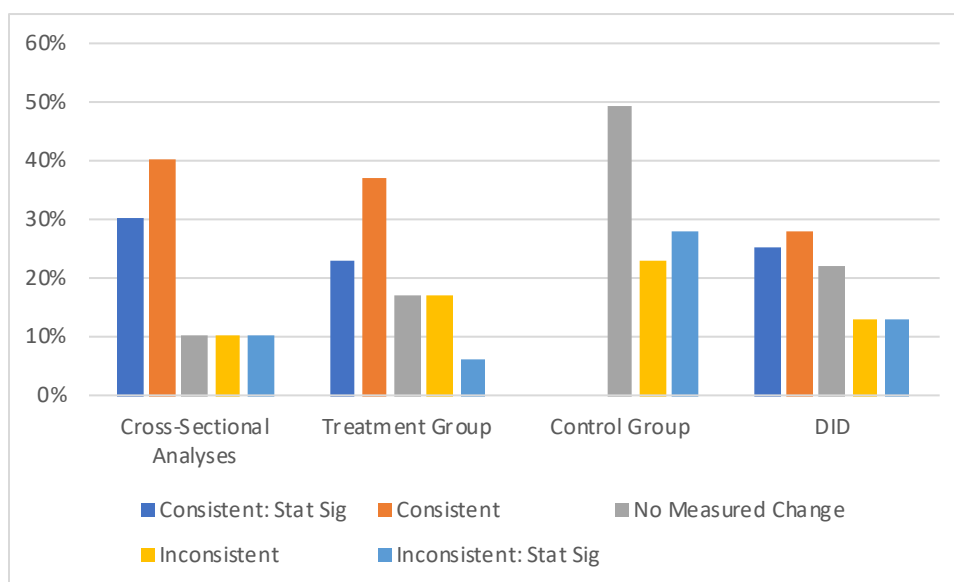


Figure 4. Visual Comparison of Significance of LIMF Statistical Results Across Samples

Figure 5 plots statistical support for the impacts of weatherization on categories of variables. As expected, the most statistical support is provided about the positive impacts of weatherization on home conditions, followed by thermal stress and arthritis, two variables impacted by temperature conditions in the home. There was medium statistical support for general health. Into the low statistical support group fell cholesterol, high blood pressure, spoiled food and missed days of work. As expected, there was very low to no statistical support for the two respiratory issues contained in the study, asthma and COPD, as well as for dental health, trips & falls, and diabetes.

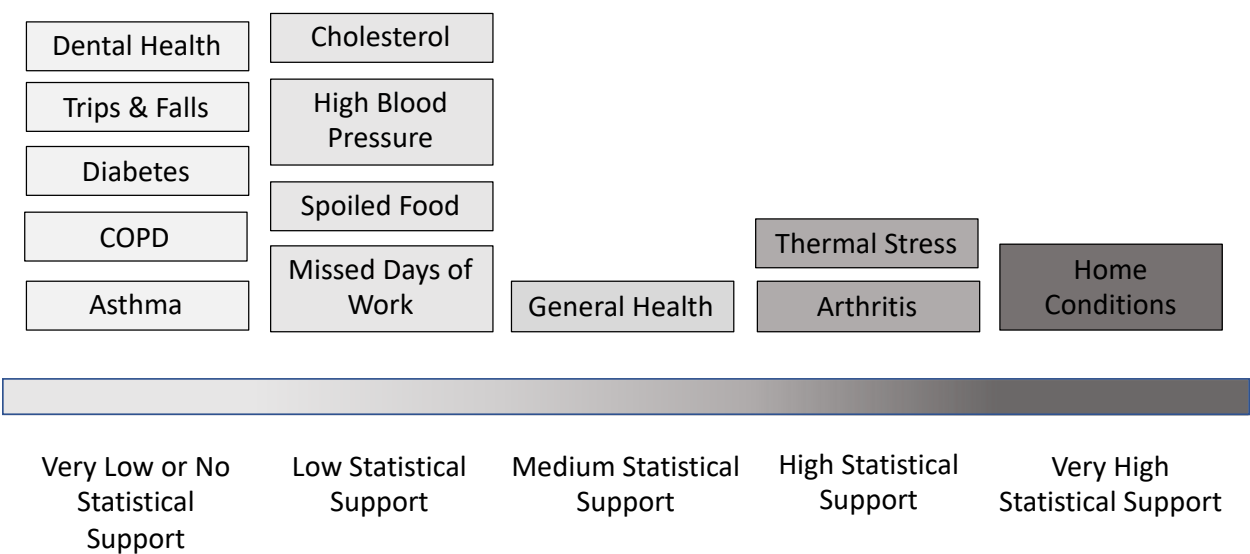


Figure 5. Visualization of LIMF Statistical Results by Variable Categories

In summary, the visualizations suggest that the preponderance of results from the KEEM and LIMF studies are consistent with the theory of change discussed above. Cross-sectional analyses have the strongest statistical support, followed by analyses using only the treatment groups. Some volatility across various variables with respect to the control group dampened the difference-in-differences results.

8.4 Cascading Co-Benefits¹³

Numerous studies report on the relationships between poor housing conditions and the quality of life of occupants, and how improvements in housing conditions can improve quality of life. For example, recent research has continued to link poor housing and poor mental health and to find that improvements in housing can improve mental health and well-being (Perreault et al., 2022; Sawyer et al., 2022; Garnham et al., 2022; Singh et al., 2020; Daniel et al., 2021; Singh et al., 2019; Park and Kim, 2022; Dotsikas et al., 2022; Tang et al., 2022; Baker et el., 2020). Kang (2021) found that housing instability adversely impacts mental health. Rolfe et al. (2020) found that good property quality and satisfactory neighborhood conditions are positively correlated with measures of health and wellbeing.

¹³ The results presented in this section are drawn from Tonn et al., (2023c).

Many studies have reported on how a very specific type of housing improvement, weatherization, can produce a wide range of non-energy impacts (NEIs) (Brown et al., 2020; Shrubsole et al., 2014; Skumatz, 2014). These types of studies also show that weatherization can positively impact the quality of life of occupants. For example, Willand et al. (2015) explain how residential energy efficiency interventions can improve health. Thomson and Thomas (2015) developed logic diagrams to track causal pathways, supported by research, from improvements in energy efficiency and housing conditions to a range of impacts, including:

- Improved general health (Howden-Chapman et al., 2007; Howden-Chapman et al., 2008; Braubach et al., 2008; Platt et al., 2007; Osman et al., 2010).
- Improved mental health (Shortt and Rugkasa, 2007; Barnes, 2003; Phillips et al., 2005)
- Improved social well-being (Bullen et al., 2008; Gibson et al., 2011; Ellaway et al., 2000).

Our previous research supports conclusions that weatherization can improve general health, mental health, and social well-being (Tonn et al., 2014b; Tonn et al., 2021b; Tonn et al., 2023a; Rose et al., 2022). Additional research indicates that weatherization can:

- Improve indoor air quality (IAQ) by preventing the intrusion outdoor air pollutants and pests through air sealing measures and improved ventilation (Doll et al., 2016; Francisco et al., 2017; Norris et al., 2013; Pigg et al., 2014; Wilson et al., 2020; Breysse et al., 2014; De Souza et al., 2019; Rose et al., 2015; Pigg, 2014; Sommerville et al., 2000).
- Reduce thermal stress by keeping homes from being too cold or too hot through a combination of new insulation, air sealing, and improvements to heating and cooling systems (Ahrentzen et al., 2016; Haverinen-Shaughnessy et al., 2018; Alam et al., 2018; Poortinga et al., 2018; Guo et al., 2016; Lee et al., 2020; Chard and Walker, 2016; Nicholls and Strengers, 2018; Wilson et al., 2014; Frank et al., 2006; Chersich et al., 2020; He et al., 2016; Sun et al., 2019).
- Reduce uncontrolled asthma and headaches through the installation of all of the energy efficiency measures just mentioned (Breysse et al., 2014; De Souza et al., 2019; Rose et al., 2015; Pigg, 2014; Sommerville et al., 2000; Howden-Chapman et al., 2007; Colton et al., 2015; Wang and Rotondo, 2020; Fisk et al., 2010; Tohn et al., 2016).
- Reduce noise, which can then improve mental health and rest and sleep (Tonn et al., 2014b; Halonen et al., 2012; Ohrstrom et al., 2006; Basner and McGuire, 2018; Phillip et al., 2006; Kyle et al., 2013; Kyle et al., 2010; Leger et al., 2001; Schubert et al., 2002; Vermeulen et al., 2021; Nicol, 2019; Okamoto-Mizuno and Mizuno, 2012; Fujii et al., 2015).
- Reduce energy burdens (Tonn et al., 2018) and water costs (Shrubsole et al., 2014; IEA, 2014), and lessen financial stresses (Tonn et al., 2014b; Tonn et al., 2021b; Tonn et al., 2023a; Hernandez et al., 2016).

One observation that can be made about this previous research is that the studies focus on exploring how housing conditions can impact various aspects of quality of life. The research does not explore how those aspects of quality of life may actually impact housing conditions and, in turn, each other. For example, one could hypothesize that bad mental or physical health could prevent occupants from engaging in activities that could keep their homes in good condition. Declining home conditions could then have a negative feedback effect upon mental and physical

health. Improvements in general health and physical activities, which could be fostered by weatherization, can lead to improvements in mental health (Ohrnberger et al., 2017; Kim et al., 2012; Giannakopoulos et al., 2009). Improvements in mental health can then reduce the impacts of chronic conditions such as hypertension (Unger et al., 2020; Gerin et al., 2012; Yano et al., 2016) and ischemic heart disease (Sanchis-Soler et al., 2020; Steptoe et al., 2011; Blumenthal et al., 2005).

Previous research also does not often explore energy poverty from a comprehensive systems perspective. While previous research indicates that energy poverty is linked to bad mental and physical health, one could also hypothesize that poor household finances in general could prevent investments to maintain good home conditions. Stressful household financial situations could also adversely impact mental and physical health, which could also work to reduce human capital investments in home maintenance (Boch et al., 2020). Households headed by elders, for example, have been found to only invest their housing as a last resort (Wu et al., 2022). This is quite relevant to the administration of weatherization programs because the majority weatherization recipients are elderly or near-elderly. Additional research has shown that poor mental health can exacerbate household financial problems (i.e., by making households more likely to be victims of scams and poor financial judgements and management (Rowe, 2021; Lee, 2020).

Lastly, it could be argued that previous research may have focused too much on the statistical significance of individual variables and not enough on broader patterns of results. For example, in our own previous research, one study may find that there is a statistically significant reduction in asthma-related emergency department visits (Tonn et al., 2014b) but another may not (Tonn et al., 2023a). We also note that all of the studies find that aspects of general health may improve post-weatherization, but not all of the results may be statistically significant. Examinations of changes in variables related to home conditions, health, and financial issues found that over 60% of the measured changes in the treatment group were in the expected direction and only 20% in an unexpected direction and 70% of the variables used in comparison group analyses changed in the expected direction (Tonn et al., 2021b; Tonn et al., 2023a). The systems approach used in this research to aggregate categories of variables into composite variables may be able to better express overall positive statistical outcomes seen in our data.

Most of the papers cited above present results of the impacts of home conditions and changes in home conditions on one topic of interest, with most papers focusing on physical health, followed by mental health, IAQ, financial problems, general health, and well-being. Only a few papers address multiple topics, which include some combination of mental health, well-being, and general health. Very few papers address how aspects of household health and well-being may impact home conditions. We found no previously published papers that take a systems approach to explore how housing conditions may be situated in a household-centric system.

Figure 7 presents the Spearman's correlations amongst the aggregate variables, as set out in Figure 6. *A priori*, we anticipated that the signs for the correlations for these four variables – _home conditions, general health problems, financial problems, and life events would be positive and for these four variables to the life satisfaction variable would be negative. These expectations were met for the full sample pre-weatherization, the full sample post-weatherization, and for the

treatment group both pre- and post-weatherization. All of the correlations are statistically significant at the $p < .001$ level and many of the correlation coefficients have high effect sizes. These results strongly support the contentions that home conditions are part of a larger household system and that all components of the system are strongly interrelated. That all of the correlations would be highly statistically significant and have such large coefficient values was certainly surprising. One more observation about the correlations results is worth noting at this point. This is that the correlations involving life events are lower in magnitude than the correlations involving the other variables. We hypothesize that life event as used herein – i.e., instances where serious health issues require hospitalization and emergency department visits – are more random and acute features of life than regular and chronic conditions of life. Even though the frequency of the life events is higher than we expected, the frequencies are still relatively low. For several of the variables, such as those related to thermal stress hot and cold, outdoor conditions that could create the conditions for the life events may not arise each and every year. The model results presented immediately below also suggest that life events can best be viewed as being relatively random and less tied to the other components of the household system.

<i>Home Conditions (yes, no)</i>	<i>Financial Problems (yes, no)</i>	<i>General Health Problems (yes, no)</i>	<i>Life Satisfaction (1 = not satisfied to 5 = satisfied)</i>	<i>Life Events (yes, no)</i>
Home Kept at Unsafe Temperature	Household Has Problems Paying Medical Bills	Respondent Physical Health Bad Last 30 Days	Respondent Satisfied with Life in General	Respondent Required COPD-Related Medical Intervention
Home too drafty	Household Cannot Afford Prescriptions	Respondent Sleep/Rest Bad Last 30 Days	Respondent Satisfied with Health	Respondent Required Asthma-Related Medical Intervention
Home too dusty	Household Did Not Pay Prescriptions to Pay Energy Bill	Respondent Mental Health Bad Last 30 Days	Respondent Satisfied with Standard of Living	Household Member Required Medical Intervention from Being Too Cold in Home
Outside Odors Infiltrate Home	Household Did Not Pay Energy to Pay Prescriptions	Respondent Has Headaches		Household Member Required Medical Intervention from Being Hot Cold in Home
Outdoor Noise Bothersome	Hard for Household to Pay Energy Bills	Respondent Had Asthma Symptoms Last 3 Months		Household Member Required Medical Intervention Due to Fire in Home
Mold Seen in Home	Household Did Not Pay Energy Bills to Pay Other Utilities	Respondent Had COPD Symptoms Requiring Doctors Visit		Household Member Required Medical Intervention Due to Food Poisoning Related to Malfunctioning Refrigerator
Standing Water in Home	Household Did Not Pay Other Utilities to Pay for Energy Bills			Household Member Required Medical Attention Due to Trip/Fall in Home
Insect Infestation in Home	Household Received Utility Disconnect Notice			Household Member Required Medical Intervention Due to CO Poisoning in Home
Mice Infestation in Home	Household Did Not Pay for Food to Pay Energy Bills			Respondent Required Medical Intervention for Migraines
	Household Did Not Pay Energy Bills to Pay for Food			Household Member Required Medical Intervention Due to Burns from Hot Water in Home
	Household Received Food Assistance			
	Household Used Short-term Predatory Loan			
	Household Received Energy Assistance			

Figure 6. Definitions of Five Aggregate Variables

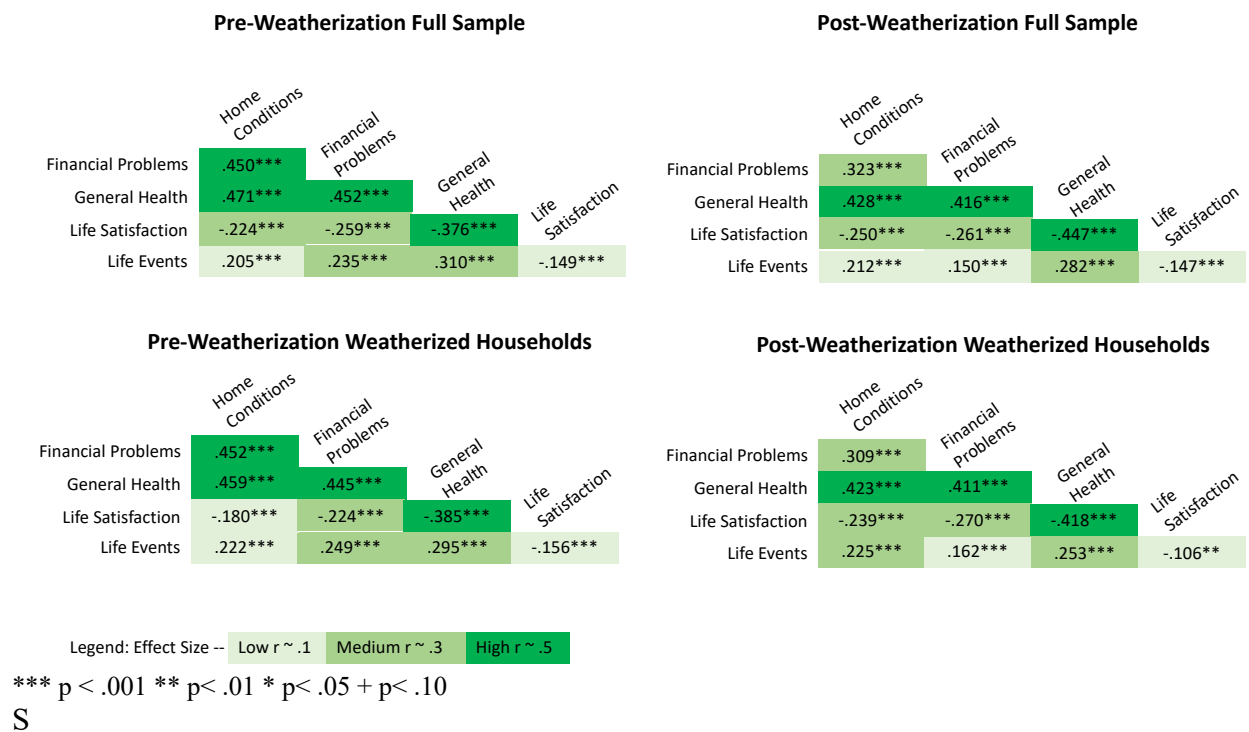


Figure 7. Aggregate Variable Correlations

To explore relationships between home conditions and the other four aggregate variables, a five-equation simultaneous equation model was estimated using three-stage least squares (3SLS). The five equations specified each have one of the five post-weatherization aggregate variables as its dependent variable. Also included was a lagged variable (i.e., the pre-weatherization value of the dependent variable). Over the course of the model development exercise, each endogenous variable was included in each equation. The endogenous variables were dropped from the equations when the p values were highly insignificant. Most of the beta-coefficients for the endogenous variables that remain in the model are statistically significant, though a couple of the p-values are over 0.20.

A treatment dummy variable was tested in each of five equations, though only remain in three of the five equations. A robust set of exogenous variables was also tested in all of the equations during the model development process in ways that did not cause the model to be unable to be estimated. These variables include age, gender, education level, and race of the respondent; having health insurance, household size, smoking allowed in home, and household income. Allow exogenous variables with p-values greater than 0.20 were dropped from the model.

The final model presented in Table 8 was judged to be reasonable and defensible. The R^2 for each equation is highly statistically significant. The high magnitudes of the R^2 's is most likely due to the presence of the lagged variables in the equations, which were all highly significant. An examination of the endogenous variables that remain in the model indicate that general health is the central component of the model. In other words, general health shows up in other equations as an endogenous variable more than the other variables and more endogenous variables impact

this variable as a dependent variable than other variables. These observations can be more clearly understood through an examination of Fig. 8. This means that improvements in general health can cascade more strongly through the model than might improvements in the other endogenous variables.

Table 8. Composite Variable Five Equation 3SLS Model (N=581)

Endogenous/Exogenous Variables (Post)	R² (p value)	Beta Coefficient	p value
<i>Eq. 1 Home Conditions Post-Wx</i>	.30 (.000)		
Home Conditions Pre-Wx		.31	.000
General Health Problems Post-Wx		.39	.000
Life Events Post-Wx		-.57	.301
Treatment Group Home		-.30	.012
Household Income		-.04	.068
Constant		.33	.056
<i>Eq. 2 Financial Problems Post-Wx</i>	.52 (.000)		
Financial Problems Pre-Wx		.48	.000
General Health Problems Post-Wx		.63	.000
Respondent Age		-.02	.003
Household Income		-.12	.000
Constant		2.06	.000
<i>Eq. 3 General Health Problems Post-Wx</i>	.53 (.000)		
General Health Problems Pre-Wx		.43	.000
Financial Problems Post-Wx		.08	.010
Life Events Post-Wx		.48	.246
Life Satisfaction Post-Wx		-.15	.026
Treatment Group Home		-.27	.010
Respondent Age		-.005	.139
Smoking Allowed in Home		-.20	.043
Constant		3.1	.002
<i>Eq 4. Life Satisfaction Post-Wx</i>	.32 (.000)		
Life Satisfaction Pre-Wx		.25	.000
General Health Problems Post-Wx		-.64	.000
Respondent Age		-.011	.055
Treatment Group Home		.42	.017
Household Income		-.04	.144
Constant		11.4	.000
<i>Eq. 5 Life Events Post-Wx</i>	.17 (.000)		
Life Events Pre-Wx		.17	.000
General Health Problems Post-Wx		.06	.000
Have Health Insurance		-.14	.011
Household Size		-.02	.055
Household Income		-.008	.181
Constant		.186	.006

Also, more clearly seen through Fig. 8 is the fact that home conditions are dependent upon other model components. For example, as general health problems increase, so do the number of problems with the home. One can surmise this is because general health problems can prevent household members from maintaining one's home. On the other hand, weatherization not only improves home conditions but also improves general health, which can then have positive feedback upon home conditions. Weatherization is also linked to improve life satisfaction, which improves general health and then has a positive impact on home conditions.

A striking finding from the model results is the relative isolation of financial problems in the model. Though financial problems are highly correlated with all of the other aggregate variables, in the model this variable only interacts with the general health endogenous variable. One could hypothesize that general health mediates the impact of financial problems on home conditions (e.g., better financial situations improve general health which then is used as the foundation to improve home conditions, life satisfaction and life events).

Age and income were the most influential demographic variables. As the age of the respondent increases, financial problems and general health problems decline. This finding is consistent with results from previous research that suggests that the health and home conditions of pre-elders (i.e., individuals 55–64 years old) are worse than those who have reached retirement age (Tonn et al., 2021a). Households with higher incomes (though still low enough to qualify weatherization and to be included in the control group) had fewer financial and home condition issues and suffered fewer serious life events.

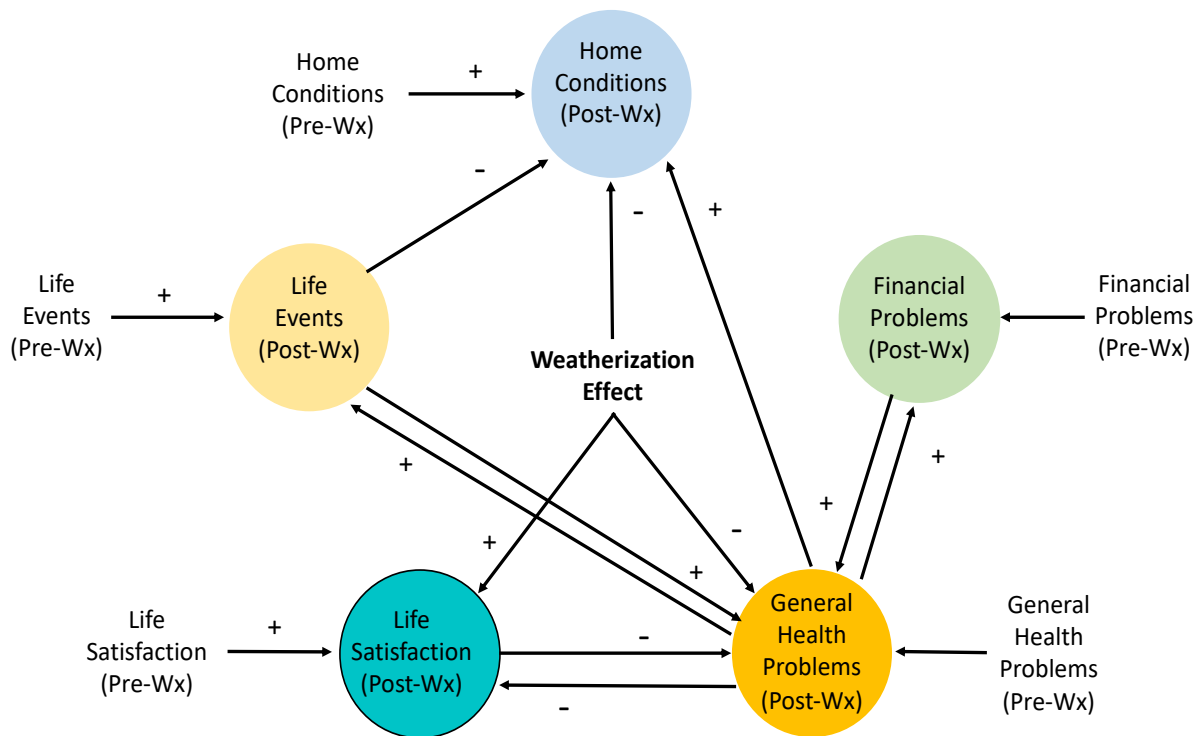


Figure 8. Three-Stage least Squares Systems Diagram

8.5 Future Research on the Health Co-Benefits of Weatherization

There is no lack of future research opportunities on the health co-benefits of weatherization. Nine broad areas are identified in this section. These include the exploration of many additional potential health co-benefits, the persistence of the health co-benefits, and how combining non-energy conservation measures with weatherization measures may have positive synergistic impacts on human health. Enthusiasm for pursuing these and other research areas needs to be tempered by the challenges with in-field data collection.

Explore Additional Health Co-Benefits – The health co-benefits studies presented above address general health and specific issues most straightforwardly assumed to be impacted by weatherization, respiratory and thermal stress. Numerous additional health issues could be the focus of future research. These issues include:

- a. *Kidney disease* – This disease could be exacerbated by dehydration caused by living in homes that are too cold.
- b. *Oral health* – Individuals with lower incomes may not have the resources to pay for adequate dental care. Weatherization and its other positive impacts on household budgets may foster improved dental care, which in turn could improve oral health.
- c. *Diabetes* – Similarly, individuals with lower incomes may not be able to buy more nutritious food but may be able to do so post-weatherization. Additionally, one could hypothesize that those living in more temperature comfortable homes may be more active and move around more in their homes, which contribute to ameliorating the impacts of diabetes.
- d. *Mental health* – As noted above, one of the strongest health co-benefits results is that the number of bad mental health days decreases post-weatherization. However, we lack insights into how this benefit may translate into fewer out-of-pocket and medical insurance costs related to mental health. Additionally, more research is needed to link potential improvements in mental health to reductions in maladies caused by poor mental health, such as being more susceptible to scams of all sorts and being cognitively unable to pay bills on time.
- e. *Low-birth weight* – The seminal study by Frank et al. (2006) showed how timely energy assistance can help pregnant women with limited incomes overcome the heat or eat dilemma that can result in low-birth weight babies. It is also possible that weatherization could also help resolve this painful tradeoff for pregnant women. This is a difficult issue to study because there are very few women of childbearing age that are respondents to health co-benefits surveys. One path forward would be to attempt to gain access to large databases of health records, merge these with large historical databases of weatherized homes, and then use machine learning techniques to identify relationships between weatherization and the reduction of low-birth weight babies.
- f. *Post-traumatic stress disorder (PTSD) and attention deficit/hyperactivity disorder (ADHD)* – Weatherization can reduce the intrusion of outdoor noise through air sealing and insulation. Quieter homes may benefit those with PTSD and ADHD.
- g. *Longer-term health impacts* – Most of the health co-benefits studies thus far can be immediately impacted by weatherization or impacted within a year or so. For

example, making homes more temperature comfortable can immediately impact the incidences of thermal stress. Reducing the infiltration of outdoor noise through insulation can have an immediate impact upon sleep. These impacts can be directly measured through pre-post surveys. However, weatherization could impact heart disease by reducing stress and improving sleep, but this impact could take years to emerge. Weatherization could have differential impacts upon cancer, maybe reducing the incidence through better foods but increase risks to cancer from increased levels of radon, both of which may take a decade or more to emerge.

Persistence of Health Co-benefits – Lifetimes of conventional weatherization measures are generally assumed to be about 20 years. Most analyses assume that annual energy savings attributable to the installed measures will be the same over this period of time. An important research question is whether one can assume that the health co-benefits of weatherization also accrue for a couple of decades or only for shorter periods of time. This is an important question because occupants' health may change over time independent of the impacts of weatherization. Household composition may also change several times over the course of even a few years' time.¹⁴ For our monetization calculations, we typically assume that the persistence of health co-benefits lasts only ten years. Well-designed persistence research is needed to support or reject this assumption.

Expand List of Installed Measures – Most of the research to date on the impacts of weatherization has been conducted around comprehensive weatherization programs that install conventional weatherization measures. There is a growing trend around the U.S. to blend weatherization and healthy homes programs such that weatherization, healthy homes, and home repair measures are simultaneously installed in homes. Typical healthy homes measures include carpet removal, air cleaners, and integrated pest management (e.g., to reduce asthma triggers). Frequently, homes need roof and other repairs to make them weatherization ready. Rigorous research is needed to estimate the incremental health impacts from installing a broader set of measures.

Another growing trend relates to the installation of climate resilience measures. Weatherization can provide some respite from the impacts of extreme events such as wildfires and heatwaves by reducing the intrusion of outdoor pollutants and increasing the time that a home is inhabitable, respectively. However, additional resilience measures that may not pass the energy savings cost-effectiveness test described in Section 3 may be able save some energy and also make substantial contributions to home resilience. Here we are thinking about measures like window shades, shade trees, window and door replacement, and improved ventilation systems. Back-up batteries can also be installed in the homes where occupants are dependent upon electrical medical systems.

Dose-Response Modeling – We do not know with much specificity what combinations of weatherization measures (i.e., dose) yields what health co-benefits (i.e., response). For example,

¹⁴ A study of homes that has much more or much less energy savings than estimated from statistical models indicated that a substantial amount of energy savings digressions was attributable to changes in household composition (e.g., grandchildren moving in and out) (Pigg 2014).

we do not know what combination of air sealing, insulation, heating system repair and replacement measures in what type of home with what type of heating fuel with what level of baseline energy efficiency yields what reductions in asthma flare-ups and emergency department visits for occupants who suffer uncontrolled asthma. The research conducted to date has not been able to answer such detailed questions about dose-response relationships for asthma, arthritis, COPD or any of the health issues identified herein. Gaining answers to these questions, at least to some extent, can help physicians have confidence in ‘prescribing’ weatherization for patients and in convincing health insurance organizations that investments in weatherization can help reduce medical bills.¹⁵

An intervening variable in weatherization dose-response modeling is indoor air quality (IAQ). Tightening homes through air sealing and insulation may reduce the infiltration of outdoor air pollutants into indoor environments but also may trap indoor air pollutants that emanate from inside, such as volatile organic compounds from cleaning products. Weatherization could also confine radon indoors post-weatherization than pre-weatherization (Pigg et al., 2014). Peaks levels of indoor air pollutants could increase post-weatherization as well. Future research could focus on measuring changes in IAQ levels and peaks post-weatherization and then develop dose-response relationships that capture IAQ as an intervening variable.

Limits of Weatherization’s Health Impacts – It could be that some occupants in weatherized homes may have too many health problems for weatherization to have an impact. The combined course of their co-morbidities may not be ameliorated in any significant manner by weatherization. Conversely, even homes that have not been deferred for weatherization may have so many environmental problems that weatherization may not be able to impact health (e.g., mold, dust mites). These suppositions, if true, certainly complicate dose-response research. One can argue that insights into current health and personal health histories can be gained through exposome research. The exposome is envisioned to be composed of all environmental pollutants that an individual has been exposed to during their lifetime, including from indoor home environments. Insights about home conditions can be gained through comprehensive home environmental audits.

More Direct Health Measurements – The research described above mainly depends upon self-reports from home occupants collected through surveys. At times, medical diagnostic records may be available to help document changes in health. Still surveys and medical records do not directly measure changes in health. Future research could also encompass more direct measurements of health. For example, spirometer data could be collected to get a more accurate measurement of the impact of weatherization on respiratory conditions. Blood pressure could be measured to explore the impacts of weatherization on personal stress. DNA samples could be collected pre- and post-weatherization to explore whether weatherization has any epigenetic impacts. Direct health measurements can be very expensive but, in some instances, may be able to provide crucial and convincing evidence about the health co-benefits of weatherization.

¹⁵ Fairly precise health impacts and monetary benefits of weatherization could entice energy service organizations to enter into agreements to fund more energy efficiency measures as long as health insurance programs could reimburse them from medical cost savings.

Population Studies – Given how difficult, time consuming, and costly it can be to assess the health co-benefits of weatherization from a dose-response perspective, it might make sense to rely more on population studies. In this context, we are referring to comparisons between samples of weatherized homes versus unweatherized homes. Example research questions could be: are the instances of thermal stress than need medical intervention less in weatherized homes versus unweatherized homes? Are asthma flare-ups suffered by occupants with asthma fewer in weatherized homes than unweatherized homes? The population approach could alter the perspectives of health co-benefits from focusing on individuals and households to the housing stock writ-large. Thus, future research might follow weatherized and unweatherized homes over time instead of tracking individuals and households over time. States such as Wisconsin have over two decades of weatherization records that could be used for this type of retrospective evaluation.

Prevention versus Amelioration – An interesting research question is to what degree can weatherization prevent the onset of chronic health diseases and acute health events. For example, our research suggests that weatherization can reduce thermal stress-related medical encounters for individuals who report thermal stress issues pre-weatherization. Our theory of change suggests that pre-weatherization homes are more often kept at unhealthy temperatures than post-weatherization, and that older HVAC systems may breakdown more frequently. However, our survey approach cannot capture instances where a heating or cooling system was replaced before it broke down or became so inefficient that the health of occupants was at risk during extreme temperature events. Conversely, our survey approach can capture reductions in medical interventions for individuals who already have asthma or arthritis but cannot capture whether weatherization can prevent or substantially delay the onset of these and other chronic diseases.

Big Data – A big data approach may be necessary to produce the requisite evidence to address most of the research questions presented in this section. In this case, medical records would be linked with weatherization records across the country. On the health data side, Medicaid and Medicare records (both diagnostic and cost) need to be included and medical records from leading private health insurance organizations. On the weatherization side, records of weatherization jobs, including addresses and measures installed, would need to be collected from state and local weatherization organizations go back two or more decades. The effort to collect the weatherization data and to link these data to medical records to analyze medical costs for individuals who live in weatherized versus unweatherized homes will be enormous, certainly more than the energy efficiency community in the U.S. could support. The healthcare sector and agencies such as the Centers for Medicare and Medicaid Services, National Institutes of Health, the Census Bureau, the Department of Housing and Urban Development, the Internal Revenue Service, the U.S. Department of Energy, the Environmental Protection Agency, the U.S. Department of Agriculture, and the National Science Foundation would need to pool resources to create the data resource. Data could be made available to researchers through secure Federal Statistical Research Data Centers.

The research opportunities listed above are intellectually challenging. They also present numerous data collection challenges. It is true that tens of thousands of homes are weatherized each year in the U.S. However, weatherization can be stratified by climate zone, home type, and heating fuel, which all may reduce sample sizes available to researchers. Additionally, at times,

local weatherization agencies may not have the staff and other resources to help researchers build valid samples of treatment groups. In the absence of waiting lists for weatherization, it may be difficult to develop valid samples of control homes. In the multifamily space, weatherization agencies may not have access to household contact information. Across the board, social scientists are finding it more difficult to contact potential respondents (e.g., as the world transitions from land-lines to mobile phones and even burner phones) and to get potential respondents to complete surveys. Since the weatherization population skews older demographically, there are additional issues around digital literacy and home internet access.

9.0 Monetization of Co-Benefits of Weatherization

This section addresses the task of placing monetary values on non-energy impacts of weatherization. The first subsection provides an overview of our approach. The second subsection discusses where dollar values that can depict monetary benefits can be found. The third section presents monetization results from the Home Uplift and Low-income Multifamily studies introduced above. The concluding section identified opportunities for future research related to monetization of NEIs.

9.1 Monetization Overview

Our generic approach to monetization is straightforward. A monetary value for a specific NEI has this equation:

$$NEI (\$) = \Delta \text{ in NEI pre-post attributed to weatherization} * \text{monetary value of that } \Delta$$

For example, from the Home Uplift study, a difference -in-differences calculation indicated that utility disconnections fell 1.9% post-weatherization, based on self-reports from our residential survey. The costs to households for a disconnection was \$70.36, based on data provided by utilities in the four cities included in our HUP study. The average NEI per weatherization job is equal to $1.9\% * \$70.36 = \1.34 . The net present value of this benefit over ten years with a 0.1% discount rate is \$13.05. Some form of this equation is used to monetize every NEI listed in Table 8.

In our monetization studies, we typically break the results into household and society benefits. With respect to the former, we consider monetizing any household NEI that can decrease the amount of money that low-income households have to spend to improve their quality of life. For instance, we include the reduction of out-of-pocket medical expenses that may result because of fewer asthma flare-ups or fewer headaches post-weatherization. We also include reductions in deductibles required to be paid by households to insurance plans for emergency department visits and hospitalizations, for example for asthma or injuries incurred during home fires. As another example, we include reductions in predatory interest payments for short-term loans as fewer loans may be taken out to pay energy bills because households benefit from energy cost savings and reductions in other expenditures like those just mentioned.

NEIs whose benefits extend beyond individual households to broad sectors of the economy and society fall into the societal NEI category. In this research, we focus on societal NEIs that are intertwined with household NEIs. NEIs that fall into this category save society money. Here we include reductions in insurance-paid medical costs for doctor's office visits, visits to the emergency department, and hospitalizations. We also include reductions in various government subsidies needed by households for energy or food, which may result if household finances improve.

To be complete, we need to mention that weatherization programs can also positively benefit ratepayers (i.e., utility customers). Weatherization could contribute to grid reliability, utility GHG reduction targets, and meeting utilities' equity goals. With respect to ratepayer NEIs intertwined

with household NEIs, if households are better able to pay their utility bills, then utility costs associated with arrearages, disconnect notices, and disconnections could all be reduced. Finally, costs to utilities through Percentage of Income Payment Programs (PIPPs) could be reduced if unpaid participant generated energy costs are reduced. Ratepayer benefits are not included in this research because our primary data collection and analyses focused on the first two benefit categories.

Monetization of NEIs is vitally important for several reasons. Placing monetary values on NEIs contributes to the systemic and comprehensive evaluation of energy efficiency programs. The values can be used in benefit-cost analyses to provide a fuller picture of how the monetary benefits of a program compare to its costs. Though investments in residential energy efficiency are typically cost-effective (Belaid et al., 2021; Tonn et al., 2018), pressures have built upon utility programs to find more ways to justify the cost effectiveness of their programs in the wake of the high market penetration of highly cost-effective LED lights that utilities install in homes (Niederberger, 2018). Accounting for the monetary values of NEIs can positively contribute to cost-effectiveness tests used by public utility commissions to justify expenditures on energy efficiency programs (ACEEE 2018).

Distinguishing between household and societal NEIs provides clarity about the nature of the NEIs. It can also be important in situations where utilities are limited with regard to the types of NEIs they are allowed to include in their total resource cost (TRC) tests. In Massachusetts, for instance, utilities are only allowed to include monetized household NEIs in their rate applications.

Monetization of health benefits of weatherization can help make the argument that the healthcare and public health sectors should also support these energy efficiency programs. These exercises further reveal that other government subsidy programs may also benefit from weatherization programs, such as energy and food assistance programs.

Two additional points are worth noting here. First, an important characteristic of the NEIs addressed in this research is that the benefits can emerge in the short-term (e.g., impacts on household budgets) or even immediately in the case of thermal stress and asthma (e.g., immediately after weatherization the home is more comfortable and less dusty). Future research should endeavor to encompass potential NEIs that may emerge in the mid- to long-term, such as reductions in cardiovascular disease.

The second point is made with respect to the value of a statistical life (VSL). We believe that weatherization can save lives. For example, more thermally sound homes can prevent deaths from thermal stress (i.e., being too cold or too hot in one's home). Decreasing fires through weatherization supported minor electrical repairs and the repair and replacement of unsafe combustion furnaces can also save lives. The installation of CO monitors, which are part of comprehensive weatherization programs in the U.S., can also have this outcome. Several of our NEI algorithms include estimates of lives saved attributable to weatherization. We use value of life estimates published by federal agencies such as the U.S. Environmental Protection Agency and the U.S. Department of Transportation.⁶

A question very pertinent in the NEI context is whether VSL should be counted as a household or societal benefit. This is because in some instances only the monetized household NEIs estimates can be used in utility program cost effectiveness testing. The two studies presented below count VSL as a household benefit because they were estimated to contribute to utility rate cases. One can argue that this is reasonable because VSL estimations are based upon individual behaviors (i.e., how much more does someone need to be paid to accept a job with a higher risk of death) and individual elicitations about how much they believe their lives are worth. We do not adjust VSL estimates based on the age of individuals whose lives are saved.

9.2 Approaches to Estimating Monetization Values for NEIs

Three approaches to acquiring are these:

Publicly available databases (PADB) – This approach uses cost estimates that have been compiled in publicly available databases by reputable organizations. For example, healthcare costs are collected by the U.S. federal government and are published in publicly accessible databases like the Medical Expenditure Panel Survey¹⁶ and the Health Care Utilization Project.¹⁷ Other federal and state databases contain a wealth of other cost estimates, such as the per hour wages of low-income workers,¹⁸ and average subsidies provided to households that are on food assistance.¹⁹ Also falling into this category are federal government estimates for the value of a statistical life (VSL) that are used to monetize lives saved.

Primary data – It is also possible to collect and/or acquire primary cost data. An example of primary data are direct medical costs that are paid by the U.S. Medicaid program, as reported in claims data. Actual Medicaid costs can be tracked for specific individuals pre- and post-weatherization (e.g., associated with asthma). Typically, it takes a good deal of time and effort to gain approval to access such databases, and very strict procedures are needed to protect the privacy of these data. However, direct cost data for specific individuals could be more informative than data collected from secondary data sources.

Special Studies – Numerous special studies have been conducted to estimate costs associated with specific issues. For example, we have used special studies which have estimated the costs to the U.S. attributable to prescription non-adherence, interest costs associated with short-term predatory loans, and improvements in work productivity that can be gained from improved sleep. In these cases, we pair these costs with answers to survey questions about being able to afford prescriptions, use of predatory loans (e.g., payday, pawn shops), and improvements in sleep, respectively.

It should be noted that we do not use willingness-to-pay, contingent valuation, or other subjective approaches for monetary estimates (e.g., questions that ask respondents to value improvements in home comfort as a percentage of their savings on utility bills). Our point of view is that monetizations based on subjective approaches are harder to defend in public discourse.

¹⁶ Medical Expenditure Panel Survey, n.d. <https://meps.ahrq.gov/mepsweb/>

¹⁷ Healthcare Cost & Utilization Project, n.d. <https://hcup-us.ahrq.gov>

¹⁸ See <https://www.bls.gov/ces/data/employment-and-earnings/2022/home.htm>

¹⁹ See <https://www.fns.usda.gov/SNAP-household-state-averages>

Conversely, basing monetization on real dollar values experienced by households (e.g., savings on interest payments for predatory loans) and society (e.g., reductions in emergency department visits for arthritis) produces estimates that are more salient and actionable for stakeholders.

9.3 Monetization Results²⁰

Table 9 presents the results for monetizing seventeen different NEIs that were addressed in the HUP and LIMF studies. There are several observations to make about the results presented in Table 8. The largest monetizations are related to thermal stress cold. This NEI includes an estimate of the number of lives saved from preventing occupants from getting too cold in their homes. The value for a statistical life (VSL) was \$9.6 million per life saved, per guidance from the U.S. Department of Transportation. Thus, even if the probability that a weatherization job is very small, the monetary value for that job is relatively large. The fire risk reduced NEI is also relatively high because lives save was included in this NEI.

Some of the NEIs have household and societal components, whereas others have only one or the other. For example, thermal stress hot and fire risk reduced have both components because health care costs are broken into household out of pocket and medical insurance. Predatory loan cost savings pertain only to households whereas reductions in food assistance benefit only the government and other programs that provide food assistance subsidies. Improvements in work productivity attributable to improved sleep benefit employers whereas improvements in home productivity obviously benefit households.

The table reveals a difference between evaluations of NEIs for single family versus multifamily weatherization programs. In one case, the MF programs evaluated did not install CO monitors whereas installing CO monitors is mandatory for SF weatherization programs. The monetary value of disconnections was estimated for the single-family HUP but not for LIMF. This is because the sample size for the latter was too small to yield a statistical estimate because so many apartment dwellers pay for their utilities through their rent.

Lastly, there are several NEIs at the bottom of Table 9 that have \$0 NEI values. In the case of asthma, this is because the changes in asthma medical interventions pre- to post-weatherization were not statistically significant but were pointing in the expected direction for the HUP and LIMF evaluations. Conversely, we were able to monetize the asthma NEI in the WAP evaluation and as part of the Washington State Medicaid cost study mentioned in the next subsection. The missed days of work and trips & falls NEIs were monetized for LIMF but not for HUP.

The differences between the monetization for HUP and LIMF are to illustrate the need to conduct independent co-benefit and monetizations for different weatherization programs and for different types of homes.

²⁰ Results presented in this section are drawn from Tonn et al. (2023b)

Table 9. Annual Monetized Estimates with VSL* (per unit weatherized)

Non-Energy Impact/ Study	Home Uplift			LIMF		
	Household Benefit	Societal Benefit	Total Benefit	Household Benefit	Societal Benefit	Total Benefit
CO Poisoning*	\$3.58	\$0.02	\$3.60	--	--	--
Disconnections	\$1.34	N/A	\$1.34	--	N/A	--
Fire Risk Reduced*	\$69.81	\$9.14	\$78.95	\$13.00	\$4.00	\$17.00
Food Spoilage	\$12.33	N/A	\$12.33	\$57.00	N/A	\$57.00
Predatory Loans	\$19.80	N/A	\$19.80	\$2.00	N/A	\$2.00
Home Productivity	\$15.42	N/A	\$15.42	\$49.00	N/A	\$49.00
Work Productivity	N/A	\$5.39	\$5.39	N/A	\$17.00	\$17.00
Prescription Adherence	N/A	\$19.22	\$19.22	N/A	\$59.99	\$59.00
Thermal Stress Hot*	\$2.44	\$12.08	\$14.52	\$0.00	\$0.00	\$0.00
Thermal Stress Cold*	\$1,312.77	\$10.66	\$1,409.23	\$1,426.00	\$38.00	\$1,464.00
Arthritis	--	--	--	\$49.00	\$892.00	\$941.00
Food Assistance	N/A	\$0.00	\$0.00	N/A	\$99.00	\$99.00
Low-birth weight Infants	\$0.00	\$0.00	\$0.00	\$0.00	\$10.00	\$10.00
Missed Days of Work	\$0.00	\$0.00	\$0.00	\$8.00	\$3.00	\$11.00
Trips & Falls	\$0.00	\$0.00	\$0.00	\$3.00	\$46.00	\$49.00
Asthma	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00
COPD	\$0.00	\$0.00	\$0.00	--	--	--

* NEI includes VSL; N/A – Not Applicable, -- unable to include in the analyses.

Table 10 presents aggregated annual and present value results by household and societal NEIs for the two evaluated programs. One can see from the results that VSL plays an important role in these monetizations. The societal monetary value for LIMF is considerably higher than for HUP because the former included the arthritis NEI, which was not included in the latter's resident survey. We should note that the present value of the NEI's over ten years with VSL at a 0.1% discount rate exceed weatherization costs, which were approximately \$10,000 for HUP (Rose et al., 2022) and \$3,000 for LIMF per unit (Carroll et al., 2015b).

Table 10. Aggregate Annual and Societal NEI Monetizations

	HHD Annual	Societal Annual	Total – Annual with VSL	Total -- Annual without VSL	Total – Present Value with VSL**	Total – Present Value without VSL
Home Uplift	\$1,437.49	\$142.31	\$1,579.80	\$208.88	\$15,404.72	\$2,038.84
LIMF	\$1,607.00	\$1,168.99	\$2,775.99	\$413.00	\$27,608.00	\$4107.38

9.4 Future Research – Monetizing the Benefits of Weatherization

Monetization exercises require two major inputs: estimates of change attributable to weatherization; and estimates of the cost implications of those changes. Section 8.5 sets out future research directions related to the former. This section does so for the latter. Unlike the weatherization health centric research, the research efforts laid out in this section are more general in nature. This is, the research may have more general benefits beyond weatherization and much of the work can be described as basic social science research. Four areas of research are presented: special studies, basic social science research, enhanced data collection, and forecasted costs.

Special Studies – As detailed in Tonn et al. (2014b) and Three³, Inc and NMR Group (2021), monetization algorithms are supplemented by results from several special studies. These include papers and reports on loss of productivity due to lack of sleep²¹, the cost to the U.S. from individuals not adhering to prescriptions²², lives saved from the installation of CO monitors (Yoon et al., 1998; Krenzelok et al., 1996). Some studies are particularly on point but are seemingly out-of-date (e.g., hospitalization costs for low birth-weight infants (Rogowski, 1998). Others are more recent but present results that are hard to map to low-income populations. For example, productivity study used focused mainly on manufacturing jobs.

A wish list of special studies that could benefit the monetization of weatherization NEIs, including health co-benefits include:

- Fire – actual data wx and fires.
- Productivity more job categories and home productivity directly
- Short-term, high-interest loans (definitive)
- Low-birth weight babies

Basic Social Science Research – Monetizations of NEIs attributable to weatherization could be enhanced by a better understanding of household financial dynamics. For example, what level of energy cost savings enables households to forego short-term, high-interest loans? How do savings in interest then translate into better affording food and prescriptions? Conversely, if weatherization improves the health of working age adults in the household, how much in wages are not foregone and how much is needed then to impact expenditures on food and prescriptions? How does a one month full or partial payment of utility bills impact needs for loans and ability to afford food and prescriptions, with and without weatherization? Given feedback in household finances, how much cost savings and increases in wages then might be needed for households to end or at least terminate their applications for food and energy assistance? Basic social science research is needed to answer these types of questions.

²¹ <http://www.businessinsider.com/workers-lack-of-sleep-costs-employers-millions-of-dollars-each-year-2011-1> ; <http://www.ncbi.nlm.nih.gov/pubmed/20042880>

²² <http://www.fiercehealthcare.com/story/patients-not-taking-medications-cost-300b/2011-05-27>

Enhanced Data Collection – Monetizations of NEIs attributable to weatherization could benefit from having access to more granular cost data. This is especially pertinent for medical costs. For example, for NEI evaluations for utility weatherization programs, it would be most valuable to have access to medical cost estimates that are drawn from utility’s jurisdictions or even from the states where their programs are being implemented. Unfortunately, in many instances these data are only available at the national level. Because medical costs vary widely by region and state, more granular medical cost data would be valuable, as would data and analyses coming out of the special studies and basic social science research.

Forecasted Costs – An important component in savings-to-investment ratios (SIR) are forecasted energy prices. The U.S. Energy Information Administration (EIA) regularly publishes forecasts for natural gas and electricity prices. If the real prices of energy are expected to increase over time, then the energy cost savings to households from weatherization will increase as will the SIR. When real prices are forecast to fall, then the SIR will decline. EIA energy forecasts can have a major impact on the perceived economic efficiency of WAP.

As health and other co-benefits of weatherization are added to the numerator of the SIR, it may be useful include forecasted medical costs and the price of carbon related GHG emission reductions. Forecasts of real costs for energy assistance, disconnections, and even short-term, high-interest loans could help improve the reliability of monetizations.

10.0 Weatherization and Related Policy Issues

Numerous policy issues face policy makers with respect to weatherization programs in the U.S. and their integration with healthcare and health care policy. This section has four parts. The first presents a high-level policy map that depicts key organizations that impact weatherization funding and how funding flows down to weatherization implementers. The second addresses policy issues that are related to weatherization separate from health and electrification policies. The third section addresses policy issues at the intersection of weatherization and healthcare. The fourth section addresses weatherization and climate change.

10.1 National Weatherization Policy Map

Figure 9 presents a map of the key policy makers and organizations that compose the national weatherization effort in the U.S. At the federal level, the key decision-making organization is the U.S. Congress, which authorizes and appropriates funding for the Weatherization Assistance Program. As discussed above, WAP is administered by the U.S. Department of Energy. WAP funding passes down to state weatherization programs and then to local weatherization programs that are administered by community action agencies. The U.S. Congress also authorizes and appropriates funds for housing programs. The U.S. Department of Housing and Urban Development (HUD) manages several programs that support the installation of healthy homes measures and the home rehabilitation (e.g., lead removal). These funds flow down to local CAAs where they can be blended with weatherization funds to more comprehensively address physical deficiencies in low-income homes.

The U.S. Congress also authorizes and appropriates funds for vitally important social service programs that are administered by the U.S. Department of Health and Human Services (HHS). HHS's Administration for Children and Families administers the Low-Income Home Energy Assistance Program (LIHEAP) that provides funds to help low-income households pay their utility bills. These funds flow down to the states and then to local energy assistance programs that are administered by CAAs. LIHEAP is important to the national weatherization effort in two ways. First and foremost, states are allowed to shift up to 20% of their LIHEAP funding to the state weatherization program. In many years, the amount of LIHEAP funds shifted to weatherization exceeds WAP's budget. Secondly, CAAs direct the highest energy burdened LIHEAP recipient households to their weatherization programs, which helps increase the impact of their WAP dollars.

HHS also is the home of the U.S. Medicaid and Medicare programs. These programs provide health insurance for income eligible households and senior citizens in the U.S. It is hoped that as time passes, these and other health insurance programs will provide funding for weatherization and other housing stock improvements to reduce overall health insurance costs and improve the quality of life for health insurance recipients. Health insurance organizations are specifically identified in this policy map because the financial resources at their disposal dwarf those available to weatherization and housing programs.

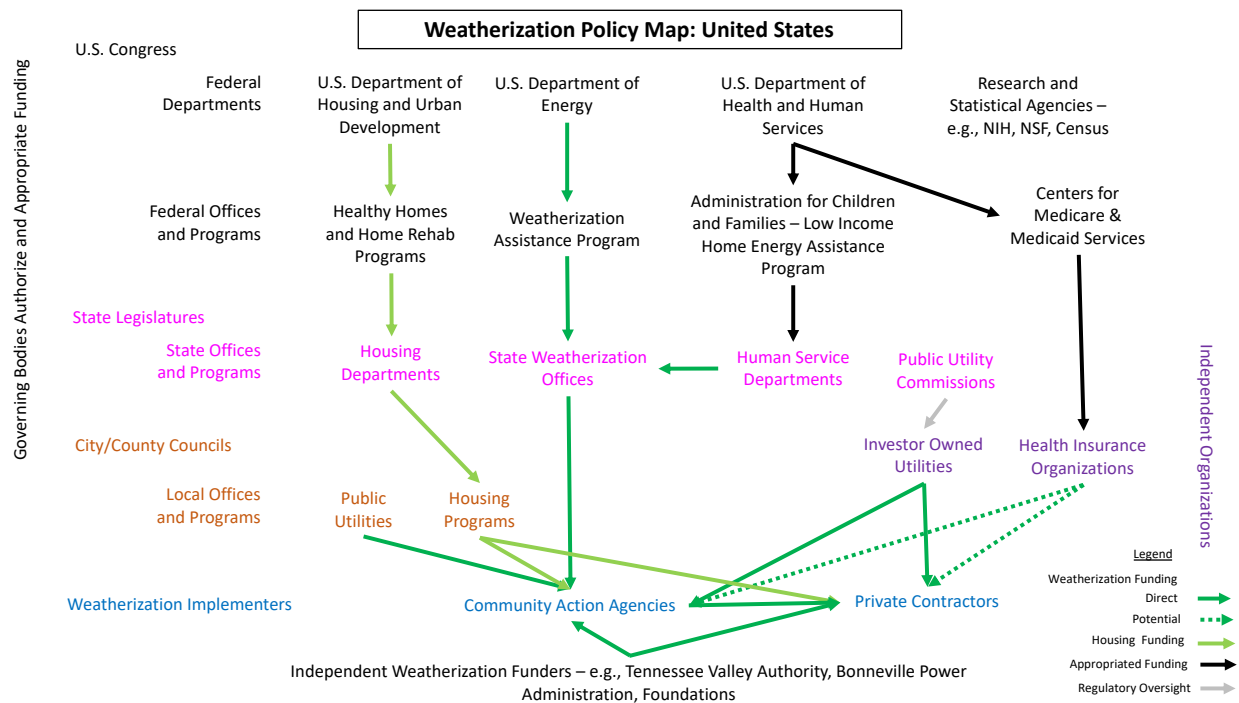


Figure 9: National Weatherization Policy Map

Another funding stream for weatherization involves decisions made by public utility commissions (PUCs) and the investor-owned utilities (IOUs) that they regulate. Through intense negotiations around rate cases, PUCs can direct IOUs to offer weatherization programs. In some states like Massachusetts funds spend by IOUs on weatherization can exceed WAP funding.

As indicated in Fig. 9, city/community councils may oversee their local public utilities. The councils may allow their utilities to fund weatherization programs. For example, the City of Knoxville, Tennessee allows the local utility, the Knoxville Utilities Board (KUB), to spend money on local weatherization from a Round-It-Up fund (i.e., customers can agree to round up their utility bills to the nearest dollar to fund weatherization).

There are various independent weatherization funders. In Tennessee for example, the Tennessee Valley Authority (TVA) co-funds weatherization programs with their local power companies, which includes KUB. Local and national philanthropic foundations also frequently invest in housing programs, including weatherization.

Lastly, weatherization funding flows to weatherization implementers, which include CAAs and private contractors. WAP funding directly flows to CAAs, who may also hire private contractors to install measures. Typically, utility funding for weatherization flows to private contractors, although utilities also ask CAAs to implement their low-income energy efficiency programs.

10.2 Weatherization Policy Issues: Detailed Discussion

As described above, weatherization services in the U.S. are primarily funded federally by the U.S. Department of Energy's Weatherization Assistance Program and investor owned and

municipal utilities across the county. Section 9.1.1 addresses weatherization policy issues from the federal perspective and 9.1.2 from the utility perspective. Section 9.1.3 addresses policy issues related to the renter/multifamily sector that pose challenges to both federally funded and utility supported weatherization programs. Section 9.14 addresses weatherization workforce issues.

10.2.1 Weatherization – Federal Perspective

To have a national weatherization program is a major policy decision. Additionally, how WAP is implemented is a major reason for its success. Federal funds flow first to state weatherization programs and then to local weatherization agencies. The vast majority of local weatherization programs are run by community action agencies (CAAs), which were established as part of President Lyndon Johnson’s Great Society initiative. The CAAs are trusted in their communities and offer many more programs than just weatherization, such as energy assistance and programs for elders. CAAs programs, including weatherization, work synergistically with each other. Because of the trust communities have in their weatherization programs and the fact that CAA’s energy assistance program can refer highly energy burdened households for weatherization, local weatherization programs typically do not have difficulties in recruiting homes for weatherization. In fact, most have long waiting lists for WAP services (Tonn, 2015a).

The national WAP program allows states and local weatherization programs flexibility that may be needed for different climate zones and housing stock. The national WAP program helps set national weatherization standards and provides weatherization training nationwide. This model is one which other countries could consider implementing.

Unfortunately, the most troubling policy issue surrounding the federal WAP program is its very existence. Conservative administrations have tried to zero out funding for WAP for several decades. The argument made is that low-income households do not deserve federal support to weatherize their homes for free because they are themselves responsible for living in poverty.²³ Efforts to eliminate WAP are consistent with other conservative efforts to drastically reduce federal social service programs that support low-income households in the U.S. In contrast, the two most recent Democratic administrations have substantially increased funding for WAP. As a result, large fluctuations in federal funding have made it difficult to attract, train, and retain weatherization workers, especially those that are part of in-house crews (Raissi and Reames, 2020).

Assuming WAP continues to exist, there are several programmatic issues that warrant re-examination by policy makers. Six important policy areas are addressed below.

Savings to Investment Ratio -- As explained in Section 3, all measures eligible for reimbursement must have an SIR of 1.0 or greater. SIR is defined as the energy cost savings divided by measure cost. More measures could be installed in homes if other benefits in addition to energy cost savings could be added to the numerator of this equation. The numerator of the SIR could include a plethora of monetized non-energy impacts, including health co-benefits and

²³ For more on the myth of economic personal responsibility spread by conservatives over the course of many decades, please see Conway and Oreskes (2023).

estimated benefits from reducing emissions of CO₂ and other air pollutants. Given the many research questions still unanswered about weatherization and health co-benefits (and their monetary value) laid out in Section 8.5, it is understandable that progress has not been made in adding health co-benefits to the SIR. However, reductions in GHG emissions attributable are more tractable. Indeed, recently WAP has begun to allow a small number of benefits to be included in the numerator of the SIR to capture the benefits of reducing GHG emissions.²⁴

Another SIR-related issue pertains to only installing measures that pass SIR. More measures can be installed if in combination all of the measures installed in a home pass SIR. WAP has recently begun experimenting with weatherization job or project level SIR tests rather than individual measure SIR tests. This approach also allows measures that may not have SIR ratios above one but are nonetheless important to install. Here we are referencing heat pumps and possibly rooftop solar. This approach will also be important when in the future health co-benefits such as asthma, which rely on combinations of measure to address, are eligible to be added to the numerator of the SIR.

Average Weatherization Job Costs – Another major policy decision relates to the average cost of all weatherization jobs completed by a local weatherization agency. Decisions about SIR and average cost work in tandem to balance benefits received by households and the need to serve as many households as possible. The average amount of funds invested across all homes weatherized by a local weatherization agency must not exceed a predetermined threshold, which is currently just over \$8,000 per home. This policy allows agencies to spend more on some homes and less on others, depending on the needs of the homes entering WAP. If more measures qualify for installation in homes, then in order to fund this expansion the threshold for the average investment in homes probably needs to increase. This could mean that fewer homes will receive WAP services, given no changes to WAP's budget. Policy analyses could reveal whether overall benefits of WAP, including the non-energy impacts, are higher or lower with a higher threshold and fewer homes weatherized.

Weatherization Deferrals – A major issue for WAP is the deferral of weatherization services due to poor and/or unsafe conditions in homes. One frequent source of deferrals is poor roofs. Weatherization cannot effectively save energy if there are holes in roofs. Unfortunately, WAP is unable to pay for extensive roof repair or replacement. In these cases, the weatherization of the home is deferred until the homeowner can remedy the roof issues. Homes with extensive mold issues may also be deferred because these conditions could threaten weatherization worker health. Our research has shown that it is highly unlikely that low-income homeowners will be able have enough money overcome the deferral issues (Rose et al., 2015q).²⁵ An active area of policy making is focusing on ways to blend weatherization funds with housing repair and remediation funds. Blending weatherization with home repair and healthy homes funds is addressed below. In 2022, WAP established a weatherization readiness fund that provides local

²⁴ See WAP Program Notice 22-10 Revised for more details: https://www.energy.gov/sites/default/files/2022-10/WPN-22-10-Revised-NEI_0.pdf

²⁵ Rose, E., Hawkins, B., Ashcraft, L., and Miller, C. 2015. Exploratory Review of Grantee, Subgrantee and Client Experiences with Deferred Services under the Weatherization Assistance Program. ORNL/TM-2014/364, Oak Ridge National Laboratory, Oak Ridge, TN, February.

weatherization agencies funds to fix homes before weatherization.²⁶ These funds do not count towards average job costs, nor do they need to pass any SIR.²⁷

Allowable Measures – In addition to the SIR constraint on measures installed, WAP also has published a list of allowable measures that can be installed in homes. Costs for measures installed in homes to save energy that are not on this list will not be reimbursed to local weatherization agencies. Progress has been made in recent years to designate rooftop solar as an allowable measure. If WAP's remit can be expanded to include improving the resilience of homes to extreme weather events, then measures such as additional insulation and ventilation, backup batteries, window shades and even shade trees might be added to list of allowable measures. Home energy management systems and broadband access could be added to allow low-income homes to become compatible with emerging smart grid technologies.

Affordable Housing Stock Metrics – The current administrative provisions for WAP are decidedly focused on individual homes and households. Measures need to pass SIR and total annual average costs for each unit weatherized must not exceed prescribed levels. This focus on the trees obscures benefits to the forest. In this case, the forest is the energy efficiency of the affordable housing stock in the U.S. Weatherization saves energy. Energy cost savings for one household from their utility bills may not carry over to the next household that occupies the home, but the home will be more energy efficient, nevertheless. Even energy cost savings estimates for the same households may change because composition of households can change radically in short periods of time. Changes in household composition were a major explanatory variable in understanding why households save more less energy than models predict (Pigg, 2014). Affordable weatherized homes may also have non-low-income occupants over time. Lastly, it would be preferable for households moving out of an affordable weatherized home to move into another affordable weatherized home. It is recommended that national housing stock metrics be developed and estimated to communicate how weatherization contributes to the overall energy efficiency of the U.S. housing stock.

Household Health and Financial Sustainability – On one hand, weatherization is supposed to reduce energy burden of low-income and vulnerable households. On the other, the measure of acceptability is SIR, energy cost savings of a measure over the cost of the measure, or of a job over the cost of a job. Energy cost savings are calculated using forecasted energy prices. The question that is not asked is how much energy cost savings could have a substantial impact upon that household. The most recent evaluation of WAP estimated that the average energy cost savings for a home heated with natural gas was just over \$250 per year (Tonn et al., 2018), which translates to around \$5,000 over the 20-year lifetimes of the installed measures. The policy question posed here is whether a savings of \$250 or so per year is life changing. Noted in Section 8, Table 4, is that about 50% of households still find it hard to pay their energy bills post-weatherization. These households still face difficult tradeoffs between paying utility bills and paying for food and prescriptions. What if the life changing energy cost savings is actually \$1000

²⁶ <https://nascsp.org/wp-content/uploads/2023/11/Weatherization-Readiness-Fund-Toolkit-1.pdf>

²⁷ Another reason that a home may not be weatherized is because not enough time has passed since it was last weatherized (e.g., 20-years). It could be that this policy constraint could be relaxed if new weatherization measures are significantly more energy efficient and if homes have undergone physical changes with may have damaged installed weatherization measures.

saving per year? Then the SIR calculus changes. The goal then becomes to find the least cost set of measures that can save that household \$1000. Additionally, the local weatherization agencies could be empowered to blend other programs and resources to hit whatever financial goal is deemed life-changing.²⁸

10.2.2 Weatherization – Utility Perspective

There are over 160 investor-owned utilities (IOUs) in the United States and several hundred additional utilities that are publicly-owned. Each utility is different. Therefore, it is possible that a wide range of weatherization programs can be implemented in the utility sector. Utility weatherization programs may face similar policy issues as does the federal weatherization program. Decisions need to be made about measures to install and how much money can be spent on weatherization. Conversely, utility weatherization programs, especially those offered by IOUs, face a different set of policy challenges. Here are several points for consideration.

All of the IOUs are regulated by state-level public utility commissions (PUCs). This because the utilities have monopolies over providing electricity and natural gas to customers in specifically defined geographic jurisdictions. PUCs approve rates that IOUs can charge different classes of customers (i.e., ratepayers) for electricity and natural gas. The IOUs pay for low-income weatherization programs collected from ratepayers. To secure funding for these programs, the IOUs need to include the costs for these programs in their rate cases (i.e., proposals to the PUCs that explain how revenue collected from ratepayers will be spent).

PUCs apply various tests to assess proposed utility costs. Total resource costs (TRCs) are routinely applied to energy efficiency programs, including low-income weatherization programs. The question that the TRC tests ask is this: does the proposed expenditure save the utility more money than the program costs the utility? Economic savings could take many forms, from reducing the needs for more transmission lines to reducing the costs for fuel for power plants to reducing the needs to build more power plants. For most energy efficiency programs offered by utilities the answer is yes.

For most low-income energy efficiency programs, the answer varies. For several years for programs that installed only LEDs, the answer was yes. Energy efficient lights were very cost effective. Of course, utility programs that only install a couple of measures such as LEDs and low-flow shower heads do not save any one household much money and certainly cannot yield the health co-benefits of comprehensive weatherization programs like WAP.

Unfortunately, comprehensive weatherization programs have trouble meeting TRC tests. This is an issue because regulated utilities also operate with a social contract in mind, where utilities exist for the public good, not only to make profits. As a result, policy decisions have emerged to give regulated utilities a boost to meet TRC tests. These boosts come from allowing utilities to add monetized non-energy impacts of their weatherization programs, such as health co-benefits, to the economic benefits of these programs.

²⁸ Maybe weatherization becomes a requirement to qualify for universal basic income programs.

Utilities do this in two ways. One way is through an adder. In this case, a percentage of the economic cost savings to the utility attributable to a weatherization program can be added. A typical adder is 5%. Therefore, important policy decisions are whether to have an adder and what percentage it should be.

A second way is to identify and monetize specific NEIs, as set out above in Sections 8 and 9. PUCs typically recognize two classes of NEIs; household and societal, with the former being favored over the latter. Thus, if this is the case, economic benefits to households for reductions in out-of-pocket medical costs can be counted for the TRC test but reductions in cost to health insurance programs will not. Thus, one important policy decision for PUCs and utilities is whether to include societal costs in TRC tests. We note the monetizations of individual NEIs typically yield NEI estimates that are substantially above 5% adders. We also note that it takes more time and resources to monetize individual NEIs than it would be to simply specify an adder.

Another important decision is how to account for value of saved lives (VSL). As illustrated in Section 9, VSL estimates can be relatively significant when compared to other elements of NEI monetizations. When utilities are allowed to include VSL and accrue this benefit to households, then low-income weatherization programs get an economic boost.

Theoretically, utilities have much flexibility when designing their weatherization programs. They are not bound by WAP prescriptions, unless they wish to be. They can install more measures or less, as they wish. One constraint on their flexibility, though, is what type of utility they are, an electric power utility, a natural gas utility, or both. The designation may impact what homes they can serve and what measures they can install in homes. For example, a natural gas utility may not be able to provide weatherization for homes that do not use natural gas. Electric utilities may only be able to install measures that directly save electricity in homes that heat with natural gas, though heating costs may dominate low-income households' utility bills. PUCs can render policy decisions to direct utilities to build into their rate bases costs for administering weatherization programs for all homes in their jurisdictions regardless of home heating fuel.

PUCs may allow IOUs to offer percentage of income payment programs (PIPPs). Under these programs, low-income households pay the same amount on their utility bills every month regardless of how much energy they use. PIPP helps low-income households avoid disconnections and help these households plan and budget for paying their utility bills. Months with unexpectedly high and unavoidable utility costs that could push households into arrears if not into an economic crisis are avoided. Weatherizing homes enrolled in PIPPs can reduce the differences between actual energy bills and the payments collected from participating households. Reducing this difference benefits ratepayers. In addition to offering PIPPs, utilities could ask that PIPP recipients have their homes weatherized. The same could be asked of households who have applied for disconnection exemptions for medical reasons.

Utility weatherization programs can hire anyone to implement their programs. If they hire CAAs, then the synergistic societal benefits that CAAs offer can be maintained. The CAAs may be able to blend WAP and utility dollars to bolster the impact of the utility dollars. If they hire private sector implementers, then those synergistic benefits may be lost. This decision is important to fully realize the synergistic co-benefits of weatherization.

10.2.3 Weatherization – Renters and Multifamily Issues

Weatherizing affordable multifamily buildings (MF) is challenging for weatherization programs for several reasons. Most fundamentally, MF buildings are qualitatively different from SF homes to audit and weatherize, especially MF buildings with 5 or more units. For example, energy audits cannot rely on blower door tests to measure and identify air sealing needs. Instead of replacing HVAC systems in homes, it may be required to replace industrial size boilers. It may not be possible to add insulation to the walls of large MF buildings. Computer models of MF buildings that can assist MF energy audits are less reliable than models built for the SF home sector because there are so different types and sizes of MF buildings. Lastly, because of all of these issues, weatherization staff need special training to conduct MF energy audits and to install measures.

MF weatherization is a challenge administratively, too. As noted above, the mandated goal of WAP is to help reduce the energy burdens on households. In the SF sector for owner occupied homes, this is straightforward; the benefits of the energy efficiency measures accrue directly to the owner occupants. In the affordable MF sector, the benefits from the installed energy efficiency measures first accrue to the property owners, presumably from receiving for free tens of thousands of dollars in energy efficiency investments that could increase the values of their properties and allow them to charge higher rents. In some cases, the higher rents could more than offset lower utility bills being directly paid by low-income households. In other cases where rent includes utilities, utility bill savings may not be passed along through rent reductions. To address this issue, property owners typically are asked to co-fund weatherization measures for their buildings and are also asked not to raise rents for a period of time after weatherization. Given these conditions, many property owners may not have an incentive to have their buildings weatherized if their buildings have low vacancy rates.

Another issue is eligibility for weatherization. In the SF sector, if the homeowner's income qualifies, then the home is eligible for weatherization. The situation may be different in the affordable MF sector. Here the issue is that it may be impractical if not impossible to weatherize one apartment in a large MF building. The more economical approach is to weatherize the entire building. However, then the issue becomes what percentage of units must be occupied by low-income households in order for the entire building to be considered income-eligible. Typically, the building is eligible if 66% or more of the units are low-income. Unfortunately, it can be hard to prove what the percentage is because building owners do not keep records of tenants' income and may not allow weatherization programs into the buildings or share contact information to survey occupants. In any case, an important policy decision is to consider reducing the percentage of income eligible homes to allow more MF buildings to qualify for weatherization.

10.2.4 Weatherization Workforce Issues

The federal WAP program is periodically seen as a vehicle for job creation. As mentioned above, WAP received a dramatic increase in funding during the ARRA period to create jobs. Indeed, the ARRA funding tripled the WAP workforce over the course of a few years. The Bipartisan Infrastructure Law (BIL) passed in the early years of the Biden Administration also dramatically increases funding for WAP to create jobs and promote clean energy solutions. While these types

of efforts can increase employment, those jobs are then lost when funding returns to normal levels. The challenges of hiring and training new weatherization workers and then the emotional toll of then having to lay off the workers can impose significant stresses on local weatherization agencies.

Weatherization workforce issues fall into two categories, training and development. With respect to training, weatherization is deceptively technical. A great amount of skill and experience is needed to conduct energy audits, install weatherization measures such as wall and ceiling insulation, and conduct quality assurance of each completed weatherization job. Berger et al. (2014) provide a very detailed summary of the more than 400 measurable skills that weatherization workers need to have.

Weatherization training policy issues include how much training individuals need for each role in the weatherization process, how training needs to be tailored by home type (i.e., single family versus mobile homes versus multifamily) and how best to deliver the training. Presently, energy auditors and QA inspectors need to be certified to hold those jobs. An organization in the U.S. called the Building Performance Institute (BPI) sets certification standards.²⁹ Weatherization training centers across the U.S. then award certifications based on those standards. There are always discussions about whether too much or too little training is required. Of course, training costs time and money, especially if a training center is not nearby. There is also tension between classroom versus apprenticeship training. Weatherization staff, generally speaking, are hands on learners and do not perform well in classroom settings. Movements to prioritize classroom training over apprenticeship training could hamper efforts to build the weatherization workforce in the U.S.

There is a lack of trained weatherization staff in the U.S. Starting out, weatherization can be a hard job, requiring workers to work under homes and in attics in all weather conditions. There is a lot to learn. Weatherization workers also must be respectful of those whose homes they work on and are in. Pay can seem too relatively low. Weatherization is not a well-known field, especially among the younger generations. Like many trade professions in the U.S., the current workforce skews older and is diminishing because of retirements. The U.S. DOE has provided grants to state weatherization offices to support weatherization workforce development (e.g., through new programs at community colleges). Any country interested in widespread implementation of weatherization programs needs to rigorously address training and workforce development issues.

10.2 Weatherization – Healthcare

From the Theory of Change to the results of several surveys to the monetizations of NEIs found in Section 5, 8 and 9, respectively, this report makes a strong case that weatherization can produce health co-benefits. Still, there seems to be a long way to go before the healthcare and weatherization sectors in the U.S. can effectively collaborate. The challenges can be clustered into two categories, evidential and administrative.

²⁹ <https://www.bpi.org>

Section 8.5 lays out many future research questions that when answered can considerably increase the strength and specificity of evidence that weatherization can yield health co-benefits. The healthcare sector needs answers to many of those questions to justify investments in weatherization that would be paid for by medical insurance organizations. For example, for a physician to prescribe weatherization as a treatment for a respiratory problem, the physician needs to be fairly certain that weatherization will prove beneficial given the patient's current condition and characteristics of their home. Conversely, if the benefits to the patient are less certain, the physician may be encouraged to prescribe weatherization if population studies support the contention that longer-term healthcare costs will be reduced and patient health can be improved through weatherizing as many homes as possible. Considerable time and resources are needed to answer the questions set out in Section 8.5.

Efforts to link healthcare and weatherization programs are still needed while the base of evidence continues to grow. This is because there are numerous administrative challenges to overcome. These challenges start with physician interactions with patients. Many physicians may be unaware of the impact of home environments upon specific health issues. Physicians may lack the time to discuss home environments with patients. Physicians would certainly not have the time to visit homes to collect data to justify weatherization prescriptions. Despite these uncertainties, policies could be put into place to allow weatherization prescriptions to accelerate the production of data to answer the many important research questions.

Another policy recommendation would be to have healthcare organizations employ community health workers (CHWs) or others with equivalent training to visit patients' homes. The CHWs could collaborate with local weatherization, healthy homes programs, and housing programs to blend their resources and recommendations with healthcare investments to make the decisions about home investments that co-optimize health benefits and economic efficiency. The CHWs would also be responsible for entering data into diagnostic databases that are related to z-codes, which are ICD-10 diagnostic codes that pertain to social determinants of health (SDOH).³⁰

Additionally, CHWs could help address vexing issues around renters and weatherization. The CHWs could facilitate conversations with property owners about weatherization and agreements to ensure that renters financially benefit or at least are not financially harmed when their homes become more energy efficient. CHWs could have aggregate information on patients that could benefit from weatherization with income eligibility for renters in large multifamily buildings to help qualify those buildings for weatherization.

Another barrier is data sharing between medical, weatherization and other organizations. Health data and accompanying personally identifiable information (PII) need to be kept secure. Healthcare communities are employing secure referral and data sharing systems that link together important social service organizations.³¹ Weatherization programs need to be included in these data sharing systems.³² Ideally, when z-codes are filled out by CHWs or others, this information

³⁰ <https://www.cms.gov/files/document/zcodes-infographic.pdf>

³¹ A leading software solution is offered by Unite Us <https://uniteus.com>.

³² A less formal way that many weatherization programs link to health care is through a program called One Touch. <https://www.tohnenvironmental.com/what-we-do/one-touch/> In this case, weatherization staff provide information

would be shared with various social service providers in these systems to facilitate effective wrap-around services for eligible households.

Healthcare can turn to weatherization agencies to install measures beyond weatherization. These measures could include back-up batteries for power life-maintaining electrical medical equipment during extended power outages and ramps and other home modifications for those with personal mobility challenges.

Lastly, healthcare can benefit from public health benefits of weatherization. Weatherization can reduce energy consumption, when then may reduce the burning of fossil fuels in homes and electric power plants when then may reduce the emissions of harmful air pollutants such as SO₂ and PM_{2.5}. These reductions could then lead to fewer emergency department visits attributable to hazardous outdoor air quality. Weatherized homes may be inhabitable for longer periods of time during extreme weather events and extended power outages, which could help ameliorate the risks to public health and ensuing costs to healthcare from thermal stress.

10.3 Weatherization – Climate Change

Weatherization programs are frequently mentioned in policy discussions about climate change. This is because weatherization can reduce consumption of fossil fuels and also increase the resilience of homes to extreme weather events. In this vein, WAP can be included in the very broad range of policies that can be employed to reduce greenhouse gas emissions. However, WAP should not be judged exclusively on its ability to reduce greenhouse gas emissions. There are many other policies available to governments that may be more cost-effective approaches to reducing greenhouse gas emissions. On the other hand, very few greenhouse gas emission policies and program offer the combination of benefits to households, and society accruable from a dollar spent on weatherization when all of its non-energy impacts are accounted for.

There are several policy issues that pertain to weatherization and electrification of homes. From the WAP perspective, a significant issue relates to the general provision that may discourage fuel switching. Let's say that a single-family home that heats air and water and cooks with natural gas is eligible for weatherization. In an effort to be fuel neutral, the weatherization agency may only be able to replace an inefficient natural gas furnace with an efficient natural gas furnace, not with an electric heat pump for example. Waivers to this rule are given to states in the Northeastern U.S. to replace fuel oil furnaces---which are major sources of air pollutants---with natural gas furnaces. In any case, this policy could be changed to promote electrification of homes. Fuel switching is not an issue for utilities that sell both electricity and natural gas, for example.

Additionally, many electrification measures may not pass the SIR criterion or even appear on the list of approved reimbursable measures. For example, WAP does not replace cooktops or ovens, which eliminates the possibility of installing induction cooktops for example. Rooftop solar systems are typically too expensive for WAP programs to install and do not necessarily reduce energy consumption. Funds are not available to strengthen roofs to handle rooftop solar. Funds

on what conditions they have encountered in homes and with weatherization recipients to healthcare and other organizations.

are not available to retrofit homes' electrical systems to be able to charge electric vehicles. WAP funds cannot be used to install energy storage systems. Of course, policies could be changed to allow WAP programs to effectively contribute to the electrification of homes. It should be noted that utility programs have enormous flexibility compared to WAP and can support home electrification efforts if they so choose and if they get permissions from their PUCs.

11. Summary and Concluding Thoughts

This report takes a very in-depth look at low-income weatherization in the U.S. The federally supported weatherization program has a unique implementation design that features grants to states and then to local weatherization (community action) agencies. A detailed theory of change provides the foundation for the rigorous evaluation of the non-energy impacts of weatherization, including health and household co-benefits. Several comprehensive evaluations support the general hypothesis that weatherization produces health co-benefits. Monetizations of these co-benefits support the conclusion that the health and household co-benefits alone are greater than the weatherization costs. The national policy map of the weatherization and weatherization-related programs and funding streams is rich and intricate. There are numerous policy and supportive research opportunities with respect to weatherization.

Here are three concluding thoughts. One, weatherization is a win-win endeavor. Money spent on weatherization is well worth the investment. Energy is saved. Household energy costs are reduced. The health of occupants is improved, as are household budgets. Reduced energy consumption can lead to reductions in emissions of greenhouse gases and other air pollutants. Investments in homes can increase property values. Weatherization programs support local employment.

Two, weatherization by itself cannot fully fix any one problem facing society. Yes, weatherization can reduce greenhouse gas emissions, but weatherization by itself cannot move a country to net-zero. Weatherization by itself may not even be as cost effective as other greenhouse gas emission alternatives but as just noted, weatherization offers a plethora of other co-benefits which make investing in weatherization a win-win endeavor. Weatherization by itself cannot reduce hardships low-income households face when having to pay energy bills. Yes, many households do find it easier to pay their energy bills but many still have major difficulties. Weatherization can improve health but is not a cure per se for chronic diseases.

Three, weatherization can have powerful synergistic benefits when blended with other health, housing, energy, and climate change resilience programs. Weatherization directly touches households and their homes. Energy auditors and weatherization crews are in homes and can install any number of additional measures to improve health and increase the resilience of homes to climate change. If energy auditors were accompanied by community health workers, then the latter could easily link households to wrap-around social service programs. Conversely, these types of professionals could refer households to weatherization programs. Weatherization programs can be central to these types of collaborations.

Acknowledgments

Preparation of this report was funded by the Wellcome Trust under *Advancing Climate Mitigation Policy Solutions with Health Co-benefits in G7 Countries* (Project No 227151/Z/23/Z).

We wish to thank Paul Francisco, Patricia Fabien, and Carina Gronlund for their insights about weatherization, health, and policy. We also thank Angi Khalil for their help in editing this report.

Appendix A. History of Weatherization NEI Research

The origins of NEI research in the United States can be traced back to the 1970s and 1980s. During these decades, energy efficiency programs were initiated by both the federal government and utility companies. Among the early leaders in this area were the U.S. Department of Energy (USDOE) and the Bonneville Power Administration (BPA). USDOE's low-established “...to increase the energy efficiency of dwellings owned or occupied by low-income persons, reduce their total residential energy expenditures, and improve their health and safety, especially low-income persons who are particularly vulnerable, such as the elderly, the disabled, and children.”¹ From its origin, WAP was intended to address energy poverty, which is problem not only in the U.S. (Power, 2006; EIA, 2018) but also across the globe (Boardman, 2010)

One of BPA's notable projects was called the Hood River Conservation Project (HRCP). This project was designed to assess the maximum penetration of residential energy efficiency that could be attained in a community, this being the town of Hood River, Oregon (Hirst et al., 1989). For this and many other projects and programs, these agencies and utility companies quickly found that they needed rigorous methods to estimate the energy savings attributable to their programs. A new field of study called energy program evaluation emerged to estimate energy savings attributable to energy efficiency programs.²

In the 1990s, energy efficiency program evaluations began to encompass NEIs. One of the seminal pieces of work was conducted by Brown et al. (1993), which estimated a comprehensive set of NEIs attributable to WAP. NEIs addressed by Brown et al. included reductions in outdoor air pollutants, increases in local employment, and increases in property values. Succeeding evaluations expanded the NEIs under consideration to include national security (Riggert et al., 1999) and home comfort (Skumatz and Gardner, 2005). Schweitzer and Tonn (2003) summarize the state of NEI research, including monetizations of NEIs, up to 2003.

Over the past couple of decades, NEI research has expanded its assessment of weatherization benefits accruable to households. Concurrently, research on the positive health impacts of energy efficient housing increased across the world. For example, research supports hypotheses that improving the energy efficiency of homes could improve general health (Willand, 2015), mental health (Phillips 2005) and social well-being (Gibson et al., 2011). Other research has found that weatherization can reduce thermal stress (Howden-Chapman et al., 2008) and improve respiratory conditions (Rose et al., 2015). Outcault et al. (2022) suggest five categories: physiological, psychological, economic, practical, and sociological. We use these five categories: home conditions, general health, acute health events, finances, and life satisfaction (Tonn et al., 2023b).

These strands of research came together in the assessment of health and other household-related NEIs of WAP conducted by Tonn et al. (2014b). This research utilized social science survey and quasi-experimental design techniques to explore new classes of NEIs (e.g., reductions in short-term, high interest loans) and eschewed contingent valuation approaches to monetization to rely on cost estimates published by reputable sources, such as U.S. federal agencies, or actual medical cost records (Rose et al., 2015). Recent publications that describe NEIs of low-income

weatherization programs in the U.S. include (Tonn et al., 2023b, Tonn et al., 2021a; Tonn et al., 2021b; Tonn et al., 2023a).

References

- Ahrentzen, S., Erickson, J., & Fonseca, E. (2016). Thermal and health outcomes of energy efficiency retrofits of homes of older adults. *Indoor Air*, 26(4), 582–593. <https://doi.org/10.1111/ina.12239>
- Alam, M., Rajeev, P., Sanjayan, J., Zou, P. X. W., & Wilson, J. (2018). Mitigation of heat stress risks through building energy efficiency upgrade: a case study of Melbourne, Australia. *Australian Journal of Civil Engineering*, 16(1), 64–78. <https://doi.org/10.1080/14488353.2018.1453331>.
- Alsina-Pagès, R. M., Freixes, M., Orga, F., Foraster, M., & Labairu-Trenchs, A. (2020). Perceptual evaluation of the citizen's acoustic environment from classic noise monitoring. *Cities & Health*, 5(1-2), 145–149. <https://doi.org/10.1080/23748834.2020.1737346>
- American Council for an Energy Efficient Economy. (2018). *Cost-effectiveness tests: overview of state approaches to account for health and environmental benefits of energy efficiency*, <https://www.aceee.org/sites/default/files/he-ce-tests-121318.pdf>.
- Baker, E., Lester, L., Mason, K., & Bentley, R. (2020). Mental health and prolonged exposure to unaffordable housing: a longitudinal analysis. *Social Psychiatry and Psychiatric Epidemiology*, 55(6). <https://doi.org/10.1007/s00127-020-01849-1>
- Barnes, R. (2003). Housing and health uncovered. *Shepherds Bush Housing Association*. https://www.housinglin.org.uk/_assets/resources/housing/housing_advice/housing_health_uncovered.pdf
- Barraclough, K. A., Blashki, G. A., Holt, S. G., & Agar, J. W. M. (2017). Climate change and kidney disease—threats and opportunities. *Kidney International*, 92(3), 526–530. <https://doi.org/10.1016/j.kint.2017.03.047>
- Basner, M., & McGuire, S. (2018). WHO Environmental Noise Guidelines for the European Region: A Systematic Review on Environmental Noise and Effects on Sleep. *International Journal of Environmental Research and Public Health*, 15(3), 519. <https://doi.org/10.3390/ijerph15030519>.
- Belaïd, F. (2018). Exposure and risk to fuel poverty in France: Examining the extent of the fuel precariousness and its salient determinants. *Energy Policy*, 114, 189–200. <https://doi.org/10.1016/j.enpol.2017.12.005>
- Berger, J., Lenahan, T., & Carroll, D. (2014). *National Weatherization Assistance Program Process Field Study: Findings from Observations and Interviews at 19 Local Agencies Across the Country*. https://weatherization.ornl.gov/wp-content/uploads/pdf/WAPRetroEvalFinalReports/ORNL_TM-2014_304.pdf

- Blumenthal, J. A., Sherwood, A., & Babyak, M. A. (2005). Effects of Exercise and Stress Management Training on Markers of Cardiovascular Risk in Patients with Ischemic Heart Disease. A Randomized Controlled Trial. *ACC Current Journal Review*, 14(7), 2–3. <https://doi.org/10.1016/j.accreview.2005.06.008>
- Boch, S. J., Taylor, D. M., Danielson, M. L., Chisolm, D. J., & Kelleher, K. J. (2020). “Home is where the health is”: Housing quality and adult health outcomes in the Survey of Income and Program Participation. *Preventive Medicine*, 132(105990), 105990. <https://doi.org/10.1016/j.ypmed.2020.105990>
- Boardman, B. (2010). *Fixing fuel poverty: challenges and solutions*. Earthscan.
- Boardman, B. (1991). *Fuel poverty: from cold homes to affordable warmth*. Belhaven.
- Braubach, M., Heinen, D., Dame, J. (2007). Preliminary results of the WHO Frankfurt housing intervention project. World Health Organization. Regional Office for Europe. <https://iris.who.int/handle/10665/107908>
- Braveman, P., & Gottlieb, L. (2014). The social determinants of health: It’s time to consider the causes of the causes. *Public Health Reports*, 129(2), 19–31. <https://doi.org/10.1177/00333549141291s206>
- Breyse, J., Dixon, S., Gregory, J., Philby, M., Jacobs, D. E., & Krieger, J. (2013). Effect of Weatherization Combined with Community Health Worker In-Home Education on Asthma Control. *American Journal of Public Health*, 104(1), e57–e64. <https://doi.org/10.2105/ajph.2013.301402>
- T. Bridgeman, D. Croft, P. Guertler, R. Moore, I. Preston. (2012). Improving the Hills approach to measuring fuel poverty. Commissioned by Consumer Focus. Retrieved from <https://www.cse.org.uk/downloads/reports-and-publications/fuel-poverty/policy/energy-justice/improving-Hills-approach-to-measuring-fuel-poverty.pdf>
- Brown, M. A., Berry, L. G., Balzer, R. A., & Faby, E. (1993). National impacts of the Weatherization Assistance Program in single-family and small multifamily dwellings. *OSTI OAI (U.S. Department of Energy Office of Scientific and Technical Information)*. <https://doi.org/10.2172/10179419>
- Brown, M., Soni, A., Lapsa, M., & Southworth, K. (2020). Low-Income Energy Affordability: Conclusions From A Literature Review. *OSTI OAI (U.S. Department of Energy Office of Scientific and Technical Information)*. <https://doi.org/10.2172/1607178>
- Bullen, C., Kearns, R. A., Clinton, J., Laing, P., Mahoney, F., & McDuff, I. (2008). Bringing health home: Householder and provider perspectives on the healthy housing programme in Auckland, New Zealand. *Social Science & Medicine*, 66(5), 1185–1196. <https://doi.org/10.1016/j.socscimed.2007.11.038>

- Burton, M., & Bray, L. (2014). *Macro-Economic Impacts of the Weatherization Assistance Program for Program Year 2008*. https://weatherization.ornl.gov/wp-content/uploads/pdf/WAPRetroEvalFinalReports/ORNL_TM-2014_98.pdf
- Campbell, C., & Pearlman, J. (2018). Access to Social Network Support and Material Hardship. *Social Currents*, 6(3), 284–304. <https://doi.org/10.1177/2329496518820630>
- Campbell, R. (1993). Fuel Poverty and Government Response. *Social Policy & Administration*, 27(1), 58–70. <https://doi.org/10.1111/j.1467-9515.1993.tb00391.x>
- L. Capps, L. Curry, E. Levin. (2019). *Energy-Plus-Health Playbook*. VEIC. Retrieved from <https://www.veic.org/Media/default/documents/resources/manuals/energy-plus-health-playbook.pdf>.
- Carroll, D., & Bausch, D. (2015). *Environmental Emissions Nonenergy Benefits Working Paper: ARRA Period*. <https://doi.org/10.2172/1223074>
- Carroll, D., Daya, G., Johnson, B., Bausch, D., & Blasnik, M. (2015). *Evaluation of the Weatherization Assistance Program During Program Years 2009-2011 (American Recovery and Reinvestment Act Period): Energy Impacts for Multifamily Buildings*. https://weatherization.ornl.gov/wp-content/uploads/pdf/WAPRecoveryActEvalFinalReports/ORNL_TM-2014_583.pdf
- Chard, R., & Walker, G. (2016). Living with fuel poverty in older age: Coping strategies and their problematic implications. *Energy Research & Social Science*, 18, 62–70. <https://doi.org/10.1016/j.erss.2016.03.004>
- Chen, C., Feng, J., Luke, N., Kuo, C.-P., & Fu, J. S. (2022). Localized energy burden, concentrated disadvantage, and the feminization of energy poverty. *IScience*, 104139. <https://doi.org/10.1016/j.isci.2022.104139>
- Chersich, M. F., Pham, M. D., Areal, A., Haghighi, M. M., Manyuchi, A., Swift, C. P., Wernecke, B., Robinson, M., Hetem, R., Boeckmann, M., & Hajat, S. (2020). Associations between high temperatures in pregnancy and risk of preterm birth, low birth weight, and stillbirths: systematic review and meta-analysis. *BMJ*, 371, m3811. <https://doi.org/10.1136/bmj.m3811>
- Colton, M. D., Laurent, J. G. C., MacNaughton, P., Kane, J., Bennett-Fripp, M., Spengler, J., & Adamkiewicz, G. (2015). Health Benefits of Green Public Housing: Associations With Asthma Morbidity and Building-Related Symptoms. *American Journal of Public Health*, 105(12), 2482–2489. <https://doi.org/10.2105/ajph.2015.302793>
- Colton, R. D., Fisher, R. (2011). *Home Energy Affordability in New York: The Affordability Gap (2008 – 2010)*. NYSERDA. <https://accessiblelaw.org/Documents/EnergyAffordabilityGap.pdf>

- Cook, J. J., & Shah, M. (2018). Reducing energy burden with solar: Colorado's strategy and roadmap for states. *National Renewable Energy Laboratory*.
<https://doi.org/10.2172/1431421>
- Daniel, J. , *6 Maps That Show How Bad Energy Poverty Is and Reveal 2 Ways to Make it Better*. (2019, July 9). The Equation. <https://blog.ucsusa.org/joseph-daniel/6-maps-that-show-how-bad-energy-poverty-is>.
- Daniel, L., Baker, E., Beer, A., & Pham, N. T. A. (2019). Cold housing: evidence, risk and vulnerability. *Housing Studies*, 36(1), 1–21.
<https://doi.org/10.1080/02673037.2019.1686130>
- Demski, C., Thomas, G., Becker, S., Evensen, D., & Pidgeon, N. (2019). Acceptance of energy transitions and policies: Public conceptualisations of energy as a need and basic right in the United Kingdom. *Energy Research & Social Science*, 48, 33–45.
<https://doi.org/10.1016/j.erss.2018.09.018>
- De Souza, R., Evans-Agnew, R., & Espina, C. (2019). Federal Weatherization and Health Education Team up. *Journal of Public Health Management and Practice*, 25(1), E21–E24. <https://doi.org/10.1097/phh.0000000000000786>
- Doll, S. C., Davison, E. L., & Painting, B. R. (2016). Weatherization impacts and baseline indoor environmental quality in low-income single-family homes. *Building and Environment*, 107, 181–190. <https://doi.org/10.1016/j.buildenv.2016.06.021>
- Dotsikas, K., Osborn, D., Walters, K., & Dykxhoorn, J. (2022). Trajectories of housing affordability and mental health problems: a population-based cohort study. *Social Psychiatry and Psychiatric Epidemiology*, 58(5). <https://doi.org/10.1007/s00127-022-02314-x>
- Drehobl, A., & Ross, L. (2016). *Lifting the high energy burden in America's largest cities: How energy*. American Council for an Energy Efficient Economy.
<https://www.aceee.org/research-report/u1602>
- Dubois, U., & Meier, H. (2016). Energy affordability and energy inequality in Europe: Implications for policymaking. *Energy Research & Social Science*, 18, 21–35.
<https://doi.org/10.1016/j.erss.2016.04.015>
- Ellaway, A., Macintyre, S., & Fairley, A. (2000). Mums on Prozac, kids on inhalers: the need for research on the potential for improving health through housing interventions. *Health bulletin*, 58(4), 336–339.
- Engvall, K., Norrby, C., & Norbäck, D. (2003). Ocular, nasal, dermal and respiratory symptoms in relation to heating, ventilation, energy conservation, and reconstruction of older multi-family houses. *Indoor Air*, 13(3), 206–211. <https://doi.org/10.1034/j.1600-0668.2003.00174.x>

- Ezratty, V., Duburcq, A., Emery, C., Lambrozo, J. (2009). Residential energy systems: links with health in European LARES study, *Environnement Risques & Sante*. 8 (6) (2009) 497–506, <https://doi.org/10.1684/ers.2009.0303>.
- Filcak, R., Zivcic, L. (2017). Energy poverty and multi-dimensional perspectives of social inequities and policy challenges, *International Issues & Slovak Foreign Policy Affairs* 26 (1–2) 40–61.
- Fisk, W. J., Eliseeva, E. A., & Mendell, M. J. (2010). Association of residential dampness and mold with respiratory tract infections and bronchitis: a meta-analysis. *Environmental Health*, 9(72), 72. <https://doi.org/10.1186/1476-069X-9-72>
- Forster, M., Tarcali, G., Till, M. (2014) Income and non-income poverty in Europe: What is the minimum acceptable standard in an enlarged European Union? *27th General Conference of the International Association for Research in Income and Wealth*, Djurhamn, Sweden.
- Francisco, P. W., Jacobs, D. E., Targos, L., Dixon, S. L., Breysse, J., Rose, W., & Cali, S. (2016). Ventilation, indoor air quality, and health in homes undergoing weatherization. *Indoor Air*, 27(2), 463–477. <https://doi.org/10.1111/ina.12325>
- Frank, D. A., Neault, N. B., Skalicky, A., Cook, J. T., Wilson, J. D., Levenson, S., Meyers, A. F., Heeren, T., Cutts, D. B., Casey, P. H., Black, M. M., & Berkowitz, C. (2006). Heat or Eat: The Low-Income Home Energy Assistance Program and Nutritional and Health Risks Among Children Less Than 3 Years of Age. *Pediatrics*, 118(5), e1293–e1302. <https://doi.org/10.1542/peds.2005-2943>
- Fujii, H., Fukuda, S., Narumi, D., Ihara, T., & Watanabe, Y. (2015). Fatigue and sleep under large summer temperature differences. *Environmental Research*, 138, 17–21. <https://doi.org/10.1016/j.envres.2015.02.006>
- Garnham, L., Rolfe, S., Anderson, I., Seaman, P., Godwin, J., & Donaldson, C. (2021). Intervening in the cycle of poverty, poor housing and poor health: the role of housing providers in enhancing tenants' mental wellbeing. *Journal of Housing and the Built Environment*, 37(1), 1–21. <https://doi.org/10.1007/s10901-021-09852-x>
- Gerin, W., Zawadzki, M. J., Brosschot, J. F., Thayer, J. F., Christenfeld, N. J. S., Campbell, T. S., & Smyth, J. M. (2012). Rumination as a Mediator of Chronic Stress Effects on Hypertension: A Causal Model. *International Journal of Hypertension*, 2012. <https://doi.org/10.1155/2012/453465>
- Giannakopoulos, G., Dimitrakaki, C., Pedeli, X., Kolaitis, G., Rotsika, V., Ravens-Sieberer, U., & Tountas, Y. (2009). Adolescents' wellbeing and functioning: relationships with parents' subjective general physical and mental health. *Health and Quality of Life Outcomes*, 7(1), 100. <https://doi.org/10.1186/1477-7525-7-100>

- Gibson, M., Thomson, H., Kearns, A., & Petticrew, M. (2011). Understanding the Psychosocial Impacts of Housing Type: Qualitative Evidence from a Housing and Regeneration Intervention. *Housing Studies*, 26(04), 555–573. <https://doi.org/10.1080/02673037.2011.559724>
- Gillard, R., Snell, C., & Bevan, M. (2017). Advancing an energy justice perspective of fuel poverty: Household vulnerability and domestic retrofit policy in the United Kingdom. *Energy Research & Social Science*, 29, 53–61. <https://doi.org/10.1016/j.erss.2017.05.012>
- Glasgow Center for Population Health (2013), *The built environment and health: an evidence review*. Briefing Paper 11, Glasgow, Scotland, November, https://www.gcph.co.uk/assets/000/000/250/BP_11_-_Built_environment_and_health_-_updated_original.pdf?1700036397
- Glassman, B. (2017). *A multidimensional poverty measure using the American Community Survey*. Social, Economic, and Housing Statistics Division, U.S. Census Bureau. <https://www.census.gov/content/dam/Census/library/working-papers/2017/demo/SEHSD-WP2017-47.pdf>
- Grey, C. N. B., Schmieder-Gaite, T., Jiang, S., Nascimento, C., & Poortinga, W. (2017). Cold homes, fuel poverty and energy efficiency improvements: A longitudinal focus group approach. *Indoor and Built Environment*, 26(7), 902–913. <https://doi.org/10.1177/1420326x17703450>
- Guo, Y., Gasparrini, A., Armstrong, B. G., Tawatsupa, B., Tobias, A., Lavigne, E., Coelho, M. de S. Z. S., Pan, X., Kim, H., Hashizume, M., Honda, Y., Guo, Y. L., Wu, C.-F., Zanobetti, A., Schwartz, J. D., Bell, M. L., Overcenco, A., Punnasiri, K., Li, S., & Tian, L. (2016). Temperature Variability and Mortality: A Multi-Country Study. *Environmental Health Perspectives*, 124(10), 1554–1559. <https://doi.org/10.1289/ehp149>
- Halonen, J. I., Vahtera, J., Stansfeld, S., Yli-Tuomi, T., Salo, P., Pentti, J., Kivimäki, M., & Lanki, T. (2012). Associations between Nighttime Traffic Noise and Sleep: The Finnish Public Sector Study. *Environmental Health Perspectives*, 120(10), 1391–1396. <https://doi.org/10.1289/ehp.1205026>
- Haverinen-Shaughnessy, U., Pekkonen, M., Leivo, V., Prasauskas, T., Turunen, M., Kiviste, M., Aaltonen, A., & Martuzevicius, D. (2018). Occupant satisfaction with indoor environmental quality and health after energy retrofits of multi-family buildings: Results from INSULAtE-project. *International Journal of Hygiene and Environmental Health*, 221(6), 921–928. <https://doi.org/10.1016/j.ijheh.2018.05.009>
- Hayes, S., Kubes, C., & Verbose, C. (2020). *Making health count: Monetizing the health benefits of in-home service*. American Council for an Energy-Efficient Economy. <https://www.aceee.org/research-report/h2001>

- Jian-Rong He, Yu Liu, Xiao-Yan Xia, Wen-Jun Ma, Hua-Liang Lin, Hai-Dong Kan, Jin-Hua Lu, Qiong Feng, Wei-Jian Mo, Ping Wang, Hui-Min Xia, Xiu Qiu, & Muglia, L. J. (2016). Ambient Temperature and the Risk of Preterm Birth in Guangzhou, China (2001-2011). *Environmental Health Perspectives*, 124(7), 1100–1106. <https://doi.org/10.1289/ehp.1509778>
- Hernández, D. (2013). Energy Insecurity: A Framework for Understanding Energy, the Built Environment, and Health Among Vulnerable Populations in the Context of Climate Change. *American Journal of Public Health*, 103(4), e32–e34. <https://doi.org/10.2105/ajph.2012.301179>
- Hernández, D., Phillips, D., & Siegel, E. (2016). Exploring the Housing and Household Energy Pathways to Stress: A Mixed Methods Study. *International Journal of Environmental Research and Public Health*, 13(9), 916. <https://doi.org/10.3390/ijerph13090916>
- Hills, J. (2012). *Getting the measure of fuel poverty - STICERD*. Hills Fuel Poverty Review. <https://sticerd.lse.ac.uk/dps/case/cr/CASEREport72.pdf>
- Hirst, E., Goeltz, R., & Trumble, D. (1989). Effects of the Hood River Conservation Project on electricity use. *Energy and Buildings*, 13(1), 19–30. [https://doi.org/10.1016/0378-7788\(89\)90014-5](https://doi.org/10.1016/0378-7788(89)90014-5)
- Howden-Chapman, P., Viggers, H., Chapman, R., O’Sullivan, K., Telfar Barnard, L., & Lloyd, B. (2012). Tackling cold housing and fuel poverty in New Zealand: A review of policies, research, and health impacts. *Energy Policy*, 49, 134–142. <https://doi.org/10.1016/j.enpol.2011.09.044>
- Howden-Chapman, P., Matheson, A., Crane, J., Viggers, H., Cunningham, M., Blakely, T., Cunningham, C., Woodward, A., Saville-Smith, K., O’Dea, D., Kennedy, M., Baker, M., Waipara, N., Chapman, R., & Davie, G. (2007). Effect of insulating existing houses on health inequality: cluster randomised study in the community. *BMJ*, 334(7591), 460. <https://doi.org/10.1136/bmj.39070.573032.80>
- Howden-Chapman, P., Pierse, N., Nicholls, S., Gillespie-Bennett, J., Viggers, H., Cunningham, M., Phipps, R., Boulic, M., Fjallstrom, P., Free, S., Chapman, R., Lloyd, B., Wickens, K., Shields, D., Baker, M., Cunningham, C., Woodward, A., Bullen, C., & Crane, J. (2008). Effects of improved home heating on asthma in community dwelling children: randomised controlled trial. *BMJ*, 337(sep23 1), a1411–a1411. <https://doi.org/10.1136/bmj.a1411>
- International Energy Agency (2014). *Capturing the multiple benefits of energy efficiency, a guide to quantifying the value added*. OECD/IEA, Paris, France. Accessed, https://www.oecd.org/content/dam/oecd/en/publications/reports/2014/09/capturing-the-multiple-benefits-of-energy-efficiency_g1g476b1/9789264220720-en.pdf

- Jabre, N. A., Keet, C. A., McCormack, M., Peng, R., Balcer-Whaley, S., & Matsui, E. C. (2020). Material Hardship and Indoor Allergen Exposure among Low-Income, Urban, Minority Children with Persistent Asthma. *Journal of Community Health*, 45(5), 1017–1026. <https://doi.org/10.1007/s10900-020-00822-y>
- Jensen, E. (2009). *Teaching with Poverty in Mind*. ASCD.
- Johnson, Richard J., Sánchez-Lozada, Laura G., Newman, Lee S., Lanaspa, Miguel A., Diaz, Henry F., Lemery, J., Rodriguez-Iturbe, B., Tolan, Dean R., Butler-Dawson, J., Sato, Y., Garcia, G., Hernando, A., & Roncal-Jimenez, Carlos A. (2019). Climate Change and the Kidney. *Annals of Nutrition and Metabolism*, 74(Suppl. 3), 38–44. <https://doi.org/10.1159/000500344>
- Kanakri, S. M., Shepley, M., Varni, J. W., & Tassinari, L. G. (2017). Noise and autism spectrum disorder in children: An exploratory survey. *Research in Developmental Disabilities*, 63(1), 85–94. <https://doi.org/10.1016/j.ridd.2017.02.004>
- Kang, S. (2021). The Cumulative Relationship between Housing Instability and Mental Health: Findings from the Panel Study of Income Dynamics. *Journal of Social Distress and Homelessness*, 31(2), 1–13. <https://doi.org/10.1080/10530789.2021.1925038>
- Kenny, G. P., Flouris, A. D., Yagouti, A., & Notley, S. R. (2018). Towards establishing evidence-based guidelines on maximum indoor temperatures during hot weather in temperate continental climates. *Temperature*, 6(1), 11–36. <https://doi.org/10.1080/23328940.2018.1456257>
- Khubchandani, J., Kleinfelder, J., Tietjen, G., Bhattacharjee, S., & Ghosh, S. (2012). Headache symptoms and indoor environmental parameters: Results from the EPA BASE study. *Annals of Indian Academy of Neurology*, 15(5), 95. <https://doi.org/10.4103/0972-2327.100029>
- Kim, Y. S., Park, Y. S., Allegrante, J. P., Marks, R., Ok, H., Ok Cho, K., & Garber, C. E. (2012). Relationship between physical activity and general mental health. *Preventive Medicine*, 55(5), 458–463. <https://doi.org/10.1016/j.ypmed.2012.08.021>
- Kratz M. (2005). Dietary cholesterol, atherosclerosis and coronary heart disease. *Handbook of experimental pharmacology*, (170), 195–213. https://doi.org/10.1007/3-540-27661-0_6
- Kravatz, M., Belliveau, E., Tonn, B. & Clendenning, G. (2018). Co-funded Health-Focused Housing Intervention Measure Benefits: Establishing a Co-funded Low-Income Residential Program Model, ACEEE, Asilomar, CA, 2018.
- Krenzelok, E. P., Roth, R., & Full, R. (1996). Carbon monoxide ... the silent killer with an audible solution. *The American Journal of Emergency Medicine*, 14(5), 484–486. [https://doi.org/10.1016/s0735-6757\(96\)90159-x](https://doi.org/10.1016/s0735-6757(96)90159-x)

- Kyle, S. D., Crawford, M. R., Morgan, K., Spiegelhalter, K., Clark, A. A., & Espie, C. A. (2013). The Glasgow Sleep Impact Index (GSII): A novel patient-centred measure for assessing sleep-related quality of life impairment in Insomnia Disorder. *Sleep Medicine*, 14(6), 493–501. <https://doi.org/10.1016/j.sleep.2012.10.023>
- Kyle, S. D., Crawford, M. R., Morgan, K., Spiegelhalter, K., Clark, A. A., & Espie, C. A. (2013). The Glasgow Sleep Impact Index (GSII): A novel patient-centred measure for assessing sleep-related quality of life impairment in Insomnia Disorder. *Sleep Medicine*, 14(6), 493–501. <https://doi.org/10.1016/j.sleep.2012.10.023>
- Landon, J., Shepherd, D., & Lodhia, V. (2016). A qualitative study of noise sensitivity in adults with autism spectrum disorder. *Research in Autism Spectrum Disorders*, 32(1), 43–52. <https://doi.org/10.1016/j.rasd.2016.08.005>
- W, L., H, C., M, C., J, J., C, K., & H, K. (2019). Projections of diurnal temperature range and diurnal temperature range-related excess mortality under climate change. *Environmental Epidemiology*, 3, 226–227. <https://doi.org/10.1097/01.ee9.0000608348.49159.75>
- Lees, C. (2020, December 9). *Online scams and mental health - Money and Mental Health*. Money and Mental Health Policy Institute. <https://www.moneyandmentalhealth.org/online-scams-mental-health/>
- Léger, D., Scheuermaier, K., Philip, P., Paillard, M., & Guilleminault, C. (2001). SF-36: Evaluation of Quality of Life in Severe and Mild Insomniacs Compared With Good Sleepers. *Psychosomatic Medicine*, 63(1), 49–55. <https://doi.org/10.1097/00006842-200101000-00006>
- Liddell, C., & Morris, C. (2010). Fuel poverty and human health: A review of recent evidence. *Energy Policy*, 38(6), 2987–2997. <https://doi.org/10.1016/j.enpol.2010.01.037>
- Lindemann, U., Stotz, A., Beyer, N., Oksa, J., Skelton, D. A., Becker, C., Rapp, K., & Klenk, J. (2017). Effect of Indoor Temperature on Physical Performance in Older Adults during Days with Normal Temperature and Heat Waves. *International Journal of Environmental Research and Public Health*, 14(2), 186. <https://doi.org/10.3390/ijerph14020186>
- Llorca, M., Rodriguez-Alvarez, A., & Jamasb, T. (2020). Objective vs. subjective fuel poverty and self-assessed health. *Energy Economics*, 87, 104736. <https://doi.org/10.1016/j.eneco.2020.104736>
- City of London Warmer Homes Program. (2020). <https://www.london.gov.uk/what-we-do/housing-and-land/improving-quality/warmer-homes>
- Longhurst, N., & Hargreaves, T. (2019). Emotions and fuel poverty: The lived experience of social housing tenants in the United Kingdom. *Energy Research & Social Science*, 56, 101207. <https://doi.org/10.1016/j.erss.2019.05.017>

- Lukeso, D., Guidotti, T. L., Franklin, D. E., & Burt, A. (2014). Indoor environmental and air quality characteristics, building-related health symptoms, and worker productivity in a federal government building complex. *Archives of Environmental & Occupational Health*, 71(2), 85–101. <https://doi.org/10.1080/19338244.2014.965246>
- Mamashli, F., Khan, S., Bharadwaj, H., Michmizos, K., Ganesan, S., Garel, K.-L. A., Hashmi, J. A., Herbert, M. R., Hämäläinen, M., & Kenet, T. (2017). Auditory Processing in Noise Is Associated with Complex Patterns of Disrupted Functional Connectivity in Autism Spectrum Disorder. *Autism Research: Official Journal of the International Society for Autism Research*, 10(4), 631–647. <https://doi.org/10.1002/aur.1714>
- Marć, M., Śmiełowska, M., Namieśnik, J., & Zabiegała, B. (2017). Indoor air quality of everyday use spaces dedicated to specific purposes—a review. *Environmental Science and Pollution Research*, 25(3), 2065–2082. <https://doi.org/10.1007/s11356-017-0839-8>
- Marchand, R., Genovese, A., Koh, S. C. L., & Brennan, A. (2019). Examining the relationship between energy poverty and measures of deprivation. *Energy Policy*, 130, 206–217. <https://doi.org/10.1016/j.enpol.2019.03.026>
- Mayes, J. L. (2019). Urban noise levels are high enough to damage auditory sensorineural health. *Cities & Health*, 1–7. <https://doi.org/10.1080/23748834.2019.1577204>
- Medline Plus. (2019). *How to lower cholesterol with diet*. Medlineplus.gov; National Library of Medicine. <https://medlineplus.gov/howtolowercholesterolwithdiet.html>
- Meyer, S., Laurence, H., Bart, D., Middlemiss, L., & Maréchal, K. (2018). Capturing the multifaceted nature of energy poverty: Lessons from Belgium. *Energy Research & Social Science*, 40, 273–283. <https://doi.org/10.1016/j.erss.2018.01.017>
- Misra, R., & Fitch, C. (2020). A model exploring the relationship between nutrition knowledge, behavior, diabetes self-management and outcomes from the dining with diabetes program. *Preventive Medicine*, 106296. <https://doi.org/10.1016/j.ypmed.2020.106296>
- Mould, R., & Baker, K. J. (2017). Documenting fuel poverty from the householders' perspective. *Energy Research & Social Science*, 31, 21–31. <https://doi.org/10.1016/j.erss.2017.06.004>
- Mullin, P., and Tonn, B. (1993). *Factors associated with proxy knowledge of employment related information*. Proceedings of the American Statistical Association, Winter Conference, Fort Lauderdale, Florida. https://www.asasrms.org/Proceedings/papers/1993_161.pdf
- Neckerman, K. M., Garfinkel, I., Teitler, J. O., Waldfogel, J., & Wimer, C. (2016). Beyond Income Poverty: Measuring Disadvantage in Terms of Material Hardship and Health. *Academic Pediatrics*, 16(3), S52–S59. <https://doi.org/10.1016/j.acap.2016.01.015>

- Nerbass, Fabiana B., Moist, L., Clark, William F., Vieira, Marcos A., & Pecoits-Filho, R. (2019). Hydration Status and Kidney Health of Factory Workers Exposed to Heat Stress: A Pilot Feasibility Study. *Annals of Nutrition and Metabolism*, 74(Suppl. 3), 30–37. <https://doi.org/10.1159/000500373>
- Nicol, F. (2019). Temperature and sleep. *Energy and Buildings*, 204, 109516. <https://doi.org/10.1016/j.enbuild.2019.109516>
- Nicholls, L., & Strengers, Y. (2018). Heatwaves, cooling and young children at home: Integrating energy and health objectives. *Energy Research & Social Science*, 39, 1–9. <https://doi.org/10.1016/j.erss.2017.10.002>
- Niederberger, A. (2018). *LEDs are the new norm: are utility programs obsolete?* Medium. <https://medium.com/enervue/leds-are-the-new-normal-are-utility-programs-obsolete-8aa5dcf90f1f>.
- Norback, D., Michel, I., & Widstrom, J. (1990). Indoor air quality and personal factors related to the sick building syndrome. *Scandinavian Journal of Work, Environment & Health*, 16(2), 121–128. <https://doi.org/10.5271/sjweh.1808>
- Noris, F., Adamkiewicz, G., Delp, W. W., Hotchi, T., Russell, M., Singer, B. C., Spears, M., Vermeer, K., & Fisk, W. J. (2013). Indoor environmental quality benefits of apartment energy retrofits. *Building and Environment*, 68, 170–178. <https://doi.org/10.1016/j.buildenv.2013.07.003>
- Nuamah, J., Rodriguez-Paras, C., & Sasangohar, F. (2020). Veteran-Centered Investigation of Architectural and Space Design Considerations for Post-Traumatic Stress Disorder (PTSD). *HERD: Health Environments Research & Design Journal*, 14(1), 193758672092555. <https://doi.org/10.1177/1937586720925554>
- Ohrnberger, J., Fichera, E., & Sutton, M. (2017). The relationship between physical and mental health: A mediation analysis. *Social Science & Medicine*, 195(195), 42–49. <https://doi.org/10.1016/j.socscimed.2017.11.008>
- Öhrström, E., Hadzibajramovic, E., Holmes, M., & Svensson, H. (2006). Effects of road traffic noise on sleep: Studies on children and adults. *Journal of Environmental Psychology*, 26(2), 116–126. <https://doi.org/10.1016/j.jenvp.2006.06.004>
- Okamoto-Mizuno, K., & Mizuno, K. (2012). Effects of thermal environment on sleep and circadian rhythm. *Journal of Physiological Anthropology*, 31(1). <https://doi.org/10.1186/1880-6805-31-14>
- OECD. (2008). *Non-income poverty: What can we learn from indicators of material deprivation?* | read online. OECD Publishing. https://read.oecd-ilibrary.org/social-issues-migration-health/growing-unequal/non-income-poverty_9789264044197-9-en

- Oreskes, N., & Conway, E. M. (2023). *The Big Myth*. Bloomsbury Publishing USA.
- Osman, L. M., Ayres, J. G., Garden, C., Reglitz, K., Lyon, J., & Douglas, J. G. (2009). A randomised trial of home energy efficiency improvement in the homes of elderly COPD patients. *European Respiratory Journal*, 35(2), 303–309. <https://doi.org/10.1183/09031936.00187708>
- Outcault, S., Sanguinetti, A., Dessouky, N., & Magaña, C. (2022). Occupant Non-Energy Impact Identification Framework: A human-centered approach to understanding residential energy retrofits. *Energy and Buildings*, 263, 112054. <https://doi.org/10.1016/j.enbuild.2022.112054>
- Papada, L., & Kaliampakos, D. (2020). Being forced to skimp on energy needs: A new look at energy poverty in Greece. *Energy Research & Social Science*, 64, 101450. <https://doi.org/10.1016/j.erss.2020.101450>
- Park, G.-R., & Kim, J. (2022). Cumulative exposure to poor housing conditions and psychological well-being: Does the relationship differ for young and middle-aged adults and older adults? *Aging & Mental Health*, 1–8. <https://doi.org/10.1080/13607863.2022.2102145>
- Perreault, K., Lapalme, J., Potvin, L., & Riva, M. (2022). “We’re Home Now”: How a Rehousing Intervention Shapes the Mental Well-Being of Inuit Adults in Nunavut, Canada. *International Journal of Environmental Research and Public Health*, 19(11), 6432. <https://doi.org/10.3390/ijerph19116432>
- Philip, P., Leger, D., Taillard, J., Quera-Salva, M., Niedhammer, I., JG Jasso Mosqueda, Bioulac, B., & Gérard, D. (2006). *Insomniac complaints interfere with quality of life but not with absenteeism: Respective role of depressive and organic comorbidity*. 7(7), 585–591. <https://doi.org/10.1016/j.sleep.2006.04.006>
- Phillips, D. R., Siu, O., Yeh, A. G. O., & Cheng, K. H. C. (2005). The impacts of dwelling conditions on older persons’ psychological well-being in Hong Kong: the mediating role of residential satisfaction. *Social Science & Medicine*, 60(12), 2785–2797. <https://doi.org/10.1016/j.socscimed.2004.11.027>
- Pigg, S., Cautley, D., Francisco, P., Hawkins, B., & Brennan, T. (2014). *Weatherization and Indoor Air Quality: Measured Impacts in Single-family Homes under the Weatherization Assistance Program*. https://weatherization.ornl.gov/wp-content/uploads/pdf/WAPRetroEvalFinalReports/ORNL_TM-2014_170.pdf
- Pigg, S. (2014). *National Weatherization Assistance Program Impact Evaluation: A Field Investigation of Apparent Low and High Savers*. https://weatherization.ornl.gov/wp-content/uploads/pdf/WAPRetroEvalFinalReports/ORNL_Tm-2014_308.pdf

- Platt, S., Mitchell, R., Petticrew, M., Walker, J.J., Hopton, J.L., Martin, C., Corbett, J., & Hope, S. (2007). The Scottish executive central heating programme: assessing impacts on health.
- Poortinga, W., Jiang, S., Grey, C., & Tweed, C. (2017). Impacts of energy-efficiency investments on internal conditions in low-income households. *Building Research & Information*, 46(6), 653–667. <https://doi.org/10.1080/09613218.2017.1314641>
- Power, M. (2006). Fuel poverty in the USA: the overview and outlook. *Energy Action*. 2006; 98. <http://www.opportunitystudies.org/wp-content/uploads/2011/11/fuel-poverty.pdf>. Accessed March 20, 2019.
- Raissi, S., & Reames, T. G. (2020). “If we had a little more flexibility.” perceptions of programmatic challenges and opportunities implementing government-funded low-income energy efficiency programs. *Energy Policy*, 147, 111880. <https://doi.org/10.1016/j.enpol.2020.111880>
- Riggert, J., Oh, A., Hall, N., & Reed, J. (1999). An Evaluation of the Energy and Non-Energy Impacts of Vermont’s Weatherization Assistance Program, TecMRKT Works, Oregon, Wisconsin.
- Rogowski J. (1998). Cost-effectiveness of care for very low birth weight infants. *Pediatrics*, 102(1 Pt 1), 35–43. <https://doi.org/10.1542/peds.102.1.35>
- Rolfe, S., Garnham, L., Godwin, J., Anderson, I., Seaman, P., & Donaldson, C. (2020). Housing as a social determinant of health and wellbeing: developing an empirically-informed realist theoretical framework. *BMC Public Health*, 20(1), 1–19. <https://doi.org/10.1186/s12889-020-09224-0>
- Rose, E., Marincic, M., Tonn, B., & Hawkins, B. (2022). *TVA Home uplift metro areas final report on non-energy impacts*. Three3, Inc. http://www.threecubed.org/uploads/2/9/1/9/29191267/tva_home_uplift_metro_areas_nei_report_march_2022.pdf
- Rose, E., Hawkins, B., Tonn, B., Paton, D., & Shah, L. (2015). *Exploring Potential Impacts of Weatherization and Healthy Homes Interventions on Asthma-related Medicaid Claims and Costs in a Small Cohort in Washington State*. https://weatherization.ornl.gov/wp-content/uploads/pdf/WAPRecoveryActEvalFinalReports/ORNL_TM-2015_213.pdf
- Rose, E., Hawkins, B., Ashcraft, L., & Miller, C. (2015). *Exploratory Review of Grantee, Subgrantee and Client Experiences with Deferred Services under the Weatherization Assistance Program*. https://weatherization.ornl.gov/wp-content/uploads/pdf/WAPRecoveryActEvalFinalReports/ORNL_TM-2014_364.pdf

- Rowe, S. (2017, May 10). *4 Tips to Cope with Debt Depression*. Psych Central. <https://psychcentral.com/blog/stressed-about-money-tips-to-cope-with-debt-depression#next-steps>
- Sanchis-Soler, G., Tortosa-Martínez, J., Manchado-Lopez, C., & Cortell-Tormo, J. M. (2020). The effects of stress on cardiovascular disease and Alzheimer's disease: Physical exercise as a counteract measure. *Stress and Brain Health: In Clinical Conditions*, 157–193. <https://doi.org/10.1016/bs.irn.2020.01.002>
- Sawyer, A., Sherriff, N., Bishop, D., Darking, M., & Huber, J. W. (2022). “It’s changed my life not to have the continual worry of being warm” – health and wellbeing impacts of a local fuel poverty programme: a mixed-methods evaluation. *BMC Public Health*, 22(1). <https://doi.org/10.1186/s12889-022-12994-4>
- Schenck-Fontaine, A., & Panico, L. (2019). Many Kinds of Poverty: Three Dimensions of Economic Hardship, Their Combinations, and Children’s Behavior Problems. *Demography*, 56(6), 2279–2305. <https://doi.org/10.1007/s13524-019-00833-y>
- Schubert, C. R., Cruickshanks, K. J., Dalton, D. S., Klein, B. E., Klein, R., & Nondahl, D. M. (2002). Prevalence of sleep problems and quality of life in an older population. *Sleep*, 25(8), 889–893.
- Schweitzer, M., & Tonn, B. (2003). Non-energy benefits of the US Weatherization Assistance Program: a summary of their scope and magnitude. *Applied Energy*, 76(4), 321–335. [https://doi.org/10.1016/s0306-2619\(03\)00003-5](https://doi.org/10.1016/s0306-2619(03)00003-5)
- Sen, A. (1983). Poor, Relatively Speaking. *Oxford Economic Papers*, 35(2), 153–169. <http://www.jstor.org/stable/2662642>
- Shah, P., Misra, A., Gupta, N., Hazra, D. K., Gupta, R., Seth, P., Agarwal, A., Gupta, A. K., Jain, A., Kulshreshta, A., Hazra, N., Khanna, P., Gangwar, P. K., Bansal, S., Tallikoti, P., Mohan, I., Bhargava, R., Sharma, R., Gulati, S., & Bharadwaj, S. (2010). Improvement in nutrition-related knowledge and behaviour of urban Asian Indian school children: findings from the “Medical education for children/Adolescents for Realistic prevention of obesity and diabetes and for healthy ageing” (MARG) intervention study. *British Journal of Nutrition*, 104(3), 427–436. <https://doi.org/10.1017/s0007114510000681>
- Shen, Z., Chen, C., Zhou, H., Fefferman, N., & Shrestha, S. (2023). Community vulnerability is the key determinant of diverse energy burdens in the United States. *Energy Research & Social Science*, 97, 102949. <https://doi.org/10.1016/j.erss.2023.102949>
- Shortt, N., & Rugkåsa, J. (2007). “The walls were so damp and cold” fuel poverty and ill health in Northern Ireland: Results from a housing intervention. *Health & Place*, 13(1), 99–110. <https://doi.org/10.1016/j.healthplace.2005.10.004>

- Shrubsole, C., Macmillan, A., Davies, M., & May, N. (2014). 100 Unintended consequences of policies to improve the energy efficiency of the UK housing stock. *Indoor and Built Environment*, 23(3), 340–352. <https://doi.org/10.1177/1420326x14524586>
- Silva, L. B., de Souza, E. L., de Oliveira, P. A., & Andrade, B. J. (2019). Implications of indoor air temperature variation on the health and performance of Brazilian students. *Indoor and Built Environment*, 29(10), 1374–1385. <https://doi.org/10.1177/1420326x19878228>
- Simon, J., Levensohn, A., Metzger, B. B., Hardman, S., & Klein, S. J. (1993). Burn prevention through Weatherization Assistance Programs. *American journal of public health*, 83(12), 1787–1788. <https://doi.org/10.2105/ajph.83.12.1787>
- Singh, A., Aitken, Z., Baker, E., & Bentley, R. (2019). Do financial hardship and social support mediate the effect of unaffordable housing on mental health? *Social Psychiatry and Psychiatric Epidemiology*, 55(6), 705–713. <https://doi.org/10.1007/s00127-019-01773-z>.
- Singh, A., Daniel, L., Baker, E., & Bentley, R. (2019). Housing Disadvantage and Poor Mental Health: A Systematic Review. *American Journal of Preventive Medicine*, 57(2), 262–272. <https://doi.org/10.1016/j.amepre.2019.03.018>
- Skumatz, L., Gardner, J., Schauer, L., Rathbun, P. (2005). The non-energy benefits of Wisconsin's low-income weatherization assistance program: revised report. *Wisconsin Department of Administration*, Division of Energy. Canada. COI: 20.500.12592/6ywppxf
- Skumatz, L. (2014). *Non-energy benefits / non-energy impacts (NEBs/NEIs) and their role & values in cost-effectiveness tests: state of Maryland final report*. https://assets.ctfassets.net/ntcn17sslow9/7hSd2GZVRtPoZKuKs9WxDc/99f7076a6561f1fb44644756e6bf5707/2014__NEBs_report_for_Maryland.pdf
- Somerville, M., Mackenzie, I., Owen, P., & Miles, D. (2000). Housing and health: does installing heating in their homes improve the health of children with asthma? *Public Health*, 114(6), 434–439. <https://doi.org/10.1038/sj.ph.1900687>
- Step toe, A., Rosengren, A., & Hjemdahl, P. (2011). Introduction to cardiovascular disease, stress and adaptation. *Stress and Cardiovascular Disease*, 1–14. https://doi.org/10.1007/978-1-84882-419-5_1
- Sun, S., Spangler, K. R., Weinberger, K. R., Yanosky, J. D., Braun, J. M., & Wellenius, G. A. (2019). Ambient Temperature and Markers of Fetal Growth: A Retrospective Observational Study of 29 Million U.S. Singleton Births. *Environmental Health Perspectives*, 127(6), 067005. <https://doi.org/10.1289/ehp4648>
- Tang, J., Chen, N., Liang, H., & Gao, X. (2022). The Effect of Built Environment on Physical Health and Mental Health of Adults: A Nationwide Cross-Sectional Study in China.

- International Journal of Environmental Research and Public Health*, 19(11), 6492.
<https://doi.org/10.3390/ijerph19116492>
- Thomson, H., Bouzarovski, S., & Snell, C. (2017). Rethinking the measurement of energy poverty in Europe: A critical analysis of indicators and data. *Indoor and Built Environment*, 26(7), 879–901. <https://doi.org/10.1177/1420326x17699260>
- Thomson, H., Snell, C., & Bouzarovski, S. (2017). Health, Well-Being and Energy Poverty in Europe: A Comparative Study of 32 European Countries. *International Journal of Environmental Research and Public Health*, 14(6), 584.
<https://doi.org/10.3390/ijerph14060584>
- Thomson, H., & Thomas, S. (2015). Developing empirically supported theories of change for housing investment and health. *Social Science & Medicine*, 124, 205–214.
<https://doi.org/10.1016/j.socscimed.2014.11.043>
- Three³, Inc and NMR Group, Inc. (2021). *Low-Income Multifamily Health-and Safety-Related NEIs Study (TXC50)*. https://www.threecubed.org/uploads/2/9/1/9/29191267/low-income_multifamily_health-_and_safety-related_non-energy_impact_study.pdf
- Tohn Environmental Strategies, Three3, Inc., and the National Center for Healthy Housing, 2016. *Occupant Health Benefits of Residential Energy Efficiency*. (2016).
<https://e4thefuture.org/wp-content/uploads/2016/11/Occupant-Health-Benefits-Residential-EE.pdf>
- Tonn, B., Rose, E., Hawkins, B., & Conlon, B. (2014b). *Health and Household-Related Benefits Attributable to the Weatherization Assistance Program*.
https://weatherization.ornl.gov/wp-content/uploads/pdf/WAPRetroEvalFinalReports/ORNL_TM-2014_345.pdf
- Tonn, B., Carroll, D., Pigg, S., Blasnik, M., Dalhoff, G., Berger, J., Rose, E., Hawkins, B., Eisenberg, J., Ucar, F., Bensch, I., & Cowan, C. (2014a). *Weatherization Works - Summary of Findings from the Retrospective Evaluation of the U.S. Department of Energy's Weatherization Assistance Program*.
<https://www.energy.gov/sites/prod/files/2015/09/f26/weatherization-works-retrospective-evaluation.pdf>
- Tonn, B., Rose, E., & Hawkins, B. (2014c). *Weatherization Beyond the Numbers: Case Studies of Fifteen High-Performing Weatherization Agencies -Conducted*.
https://weatherization.ornl.gov/wp-content/uploads/pdf/WAPRetroEvalFinalReports/ORNL_TM-2014_317.pdf
- Tonn, B., Rose, E., & Hawkins, B. (2015a). *National Weatherization Assistance Program Characterization Describing the Recovery Act Period*.
https://weatherization.ornl.gov/wp-content/uploads/pdf/WAPRecoveryActEvalFinalReports/ORNL_TM-2015_195.pdf

- Tonn, B., Carroll, D., Rose, E., Hawkins, B., Pigg, S., Bausch, D., Dalhoff, G., Blasnik, M., Eisenberg, J., Cowan, C., & Conlon, B. (2015b). *Weatherization Works II -Summary of Findings from the ARRA Period Evaluation of the U.S. Department of Energy's Weatherization Assistance Program*. https://weatherization.ornl.gov/wp-content/uploads/pdf/WAPRecoveryActEvalFinalReports/ORNL_TM-2015_139.pdf
- Tonn, B., Rose, E., & Hawkins, B. (2018). Evaluation of the U.S. department of energy's weatherization assistance program: Impact results. *Energy Policy*, 118, 279–290. <https://doi.org/10.1016/j.enpol.2018.03.051>
- Tonn, B., Hawkins, B., Rose, E., & Marincic, M. (2021a). Income, housing and health: Poverty in the United States through the prism of residential energy efficiency programs. *Energy Research & Social Science*, 73, 101945. <https://doi.org/10.1016/j.erss.2021.101945>
- Tonn, B., Rose, E., Hawkins, B., & Marincic, M. (2021b). Health and financial benefits of weatherizing low-income homes in the southeastern United States. *Building and Environment*, 197, 107847. <https://doi.org/10.1016/j.buildenv.2021.107847>
- Tonn, B., Hawkins, B., Rose, E., Marincic, M., Pigg, S., & Cowan, C. (2023a). Saving lives by saving energy? Examining the health benefits of energy efficiency in multifamily buildings in the United States. *Building and Environment*, 109716. <https://doi.org/10.1016/j.buildenv.2022.109716>
- Tonn, B., Rose, E., & Marincic, M. (2023b). Cascading benefits of low-income weatherization upon health and household well-being. *Building and Environment*, 242, 110470–110470. <https://doi.org/10.1016/j.buildenv.2023.110470>
- Tonn, B., Marincic, M., & Rose, E. (2023c). A dollar well spent: Monetizing the societal benefits of low-income weatherization programs in the United States. *Energy Research & Social Science (Print)*, 107, 103341–103341. <https://doi.org/10.1016/j.erss.2023.103341>
- Townsend, P., Phillimore, P., & Alastair Beattie. (1989). *Health and deprivation: inequality and the North*. Routledge.
- Townsend, P. (1979). *Poverty in the United Kingdom a survey of household resources and standards of living* (p. 31). Berkeley [U.A.] Univ. Of Calif. Press.
- Underhill, L. J., Milando, C. W., Levy, J. I., Dols, W. S., Lee, S. K., & Fabian, M. P. (2020). Simulation of indoor and outdoor air quality and health impacts following installation of energy-efficient retrofits in a multifamily housing unit. *Building and Environment*, 170, 106507. <https://doi.org/10.1016/j.buildenv.2019.106507>

- Unger, T., Borghi, C., Charchar, F., Khan, N. A., Poulter, N. R., Prabhakaran, D., Ramirez, A., Schlaich, M., Stergiou, G. S., Tomaszewski, M., Wainford, R. D., Williams, B., & Schutte, A. E. (2020). 2020 international society of hypertension global hypertension practice guidelines. *Hypertension*, 75(6), 1334–1357. <https://doi.org/10.1161/hypertensionaha.120.15026>
- U.S. Energy Information Administration, (2018). *One in three U.S. households faces a challenge in meeting energy needs - Today in Energy - U.S. Energy Information Administration (EIA)*. (n.d.). www.eia.gov. <https://www.eia.gov/todayinenergy/detail.php?id=37072>
- van Kasteren, Y., Champion, S., & Perimal-Lewis, L. (2019). Thermal comfort and physical activity in an office setting. *Proceedings of the Australasian Computer Science Week Multiconference*. <https://doi.org/10.1145/3290688.3290733>
- Vermeulen, M. C. M., Heijden, K. B., Kocевska, D., Treur, J. L., Huppertz, C., Beijsterveldt, C. E. M., Boomsma, D. I., Swaab, H., Someren, E. J. W., & Bartels, M. (2020). Associations of sleep with psychological problems and well-being in adolescence: causality or common genetic predispositions? *Journal of Child Psychology and Psychiatry*, 62(1), 28–39. <https://doi.org/10.1111/jcpp.13238>
- Walker, G., Simcock, N., & Day, R. (2016). Necessary energy uses and a minimum standard of living in the United Kingdom: Energy justice or escalating expectations? *Energy Research & Social Science*, 18, 129–138. <https://doi.org/10.1016/j.erss.2016.02.007>
- Wang, N., & Rotondo, J. (2020). *Energy and Health Nexus: Making the case for building energy efficiency considerations of occupant health and productivity*. https://www.pnnl.gov/sites/default/files/media/file/EED_0831_BROCH_HealthyBuildings_v4.pdf
- Wilby, R. L., Kasei, R., Gough, K. V., Amankwaa, E. F., Abarike, M., Anderson, N. J., Codjoe, S. N. A., Griffiths, P., Kaba, C., Abdullah, K., Kayaga, S., Matthews, T., Mensah, P., Murphy, C., & Yankson, P. W. K. (2021). Monitoring and moderating extreme indoor temperatures in low-income urban communities. *Environmental Research Letters*, 16(2), 024033. <https://doi.org/10.1088/1748-9326/abdbf2>
- Willand, N., Ridley, I., & Maller, C. (2015). Towards explaining the health impacts of residential energy efficiency interventions – A realist review. Part 1: Pathways. *Social Science & Medicine*, 133, 191–201. <https://doi.org/10.1016/j.socscimed.2015.02.005>
- Wilson, J., Dixon, S. L., Zuluaga, M., Jacobs, D. E., Breysse, J., & Berger, D. (2020). Venting for health: indoor air quality improvements from upgraded ventilation systems in multifamily high-rise housing. *Energy Efficiency*, 13(8), 1727–1735. <https://doi.org/10.1007/s12053-020-09902-3>

- Wu, S., Fu, Y., & Yang, Z. (2022). Housing condition, health status, and age-friendly housing modification in Europe: The last resort? *Building and Environment*, 108956. <https://doi.org/10.1016/j.buildenv.2022.108956>
- Yano, Y., Ning, H., Reis, J. P., Lewis, C. E., Launer, L. J., Bryan, R. N., Yaffe, K., Sidney, S., Albanese, E., Greenland, P., Lloyd-Jones, D., & Liu, K. (2016). Blood Pressure Reactivity to Psychological Stress in Young Adults and Cognition in Midlife: The Coronary Artery Risk Development in Young Adults (CARDIA) Study. *Journal of the American Heart Association*, 5(1). <https://doi.org/10.1161/jaha.115.002718>
- Yoon, S. S. (1998). Deaths From Unintentional Carbon Monoxide Poisoning and Potential for Prevention with Carbon Monoxide Detectors. *JAMA*, 279(9), 685. <https://doi.org/10.1001/jama.279.9.685>